A line-of-sight Einstein–Boltzmann solver for the CMB and the matter power spectrum

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We describe a new C++ Einstein–Boltzmann code that computes the angular power spectra of the cosmic microwave background (CMB) temperature and E-mode polarization, and the linear matter power spectrum, for a flat ΛCDM model. The calculation is split into four modules: the background expansion, the recombination history, the linear perturbations, and the line-of-sight integration. The code includes photon polarization, massless neutrinos, helium and a tanh model of reionization. For the fiducial model ($h = 0.67$, $\Omega_b = 0.05$, $\Omega_{\rm CDM} = 0.267$) we find radiation–matter equality at $z = 3400$, an age of 13.85 Gyr and last scattering at $z_* = 1085$ (with reionization; 1090 without) with sound horizon $r_s(z_*) = 144.9\,\mathrm{Mpc}$. The first acoustic peak lies at $\ell = 221$ and $P(k)$ peaks at $k = 0.0167\,h\,\mathrm{Mpc}^{-1}$. Every module agrees with the CLASS code run with the same physics, the TT spectrum to better than 1.1% for $2 \le \ell \le 2000$ and $P(k)$ to 0.11%. The unlensed TT spectrum matches the shape of the Planck 2018 data, and the residuals at $\ell \gtrsim 1000$ follow the pattern expected from gravitational lensing. Including lensing with the full-sky correlation-function method lowers the χ^2 against the 83 binned Planck high-ℓ points from 781 to 220; the lensed TT agrees with CLASS to 10^{-3} for $\ell \le 2000$. We also show that supernova distances alone prefer $h \simeq 0.70$ when the density parameters are held fixed. Finally, we extend the code to open and closed universes using hyperspherical Bessel functions; for $|\Omega_K| \le 0.2$ the spectra agree with CLASS as well as in the flat case. We also add a massive neutrino species; the suppression of the matter power spectrum (16% at $k = 0.5\,h\,\mathrm{Mpc}^{-1}$ for $m_\nu = 0.3\,\mathrm{eV}$) and of the lensing potential agree with CLASS to 5×10^{-4}. Finally, we compute the real-space two-point functions: the CMB angular correlation functions, the matter correlation function with its BAO peak at $100\,h^{-1}\,\mathrm{Mpc}$, and the angular power spectrum and correlation function of the matter projected over a redshift distribution, which agree with CLASS to 2×10^{-3}. Replacing the Peebles recombination model by RECFAST reduces the rms differences from CLASS to 0.048% in TT and 0.32% in EE for $30 \le \ell \le 1000$. Lastly, we describe a library of eleven dark-energy and modified-gravity models: quintessence, coupled quintessence, dark scattering, early dark energy, interacting dark energy, μ–Σ, DGP, Brans–Dicke, $f(R)$, the cubic Galileon, and Horndeski gravity in the α basis. They are validated with constraint equations, independent growth calculations and reconstructions of one theory from another, and against CLASS, hi_class and CLASS_EDE where these apply.

I. INTRODUCTION

The anisotropies of the CMB and the clustering of matter on large scales are the two cleanest probes of the early Universe. Both are set by linear physics. Small primordial curvature fluctuations, generated during inflation, evolve under gravity and under photon–baryon interactions. The result is a set of transfer functions that map the primordial spectrum onto the observed spectra. Computing these transfer functions is the job of an Einstein–Boltzmann solver. Public codes such as CMBFAST [1], CAMB [2] and CLASS [3, 4] do this to subpercent precision and are the basis of all modern parameter estimates, including those of Planck [5].

This paper describes the four milestones of writing such a code from scratch. The structure follows the physics:

1. **Background cosmology** (Sec. II): the expansion history $H(x)$, conformal time and distances. We test it against supernova data.

2. **Recombination history** (Sec. III): the free electron fraction X_e, the optical depth τ and the visibility function $\tilde{g}$, which decide when and where CMB photons last scatter.

3. **Perturbations** (Sec. IV): the linearized Einstein and Boltzmann equations for dark matter, baryons, photons (temperature and polarization) and massless neutrinos, solved for each Fourier mode k.

4. **Power spectra** (Sec. V): the line-of-sight integration [1] that gives the photon transfer functions $\Theta_\ell(k)$ and the angular spectra C_ℓ^{TT}, C_ℓ^{EE}, C_ℓ^{TE}, together with the matter power spectrum $P(k)$.

Each module is a separate C++ class that takes the previous one as input. Each section below gives the theory, the implementation and tests, and the results for one milestone. Sec. VI compares all modules with CLASS, and Sec. XII concludes. The course project also includes Markov chain Monte Carlo (MCMC) fits of supernova data; as instructed, these are omitted here.

Throughout we use the time variable $x = \ln a$, the conformal Hubble factor $\mathcal{H} = aH$, and primes for derivatives with respect to x. The perturbations use the conformal

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Newtonian gauge and the conventions of Callin [7] and Dodelson [8]. Internally the code works in SI units.

The fiducial model is $h = 0.67$, $\Omega_{b0} = 0.05$, $\Omega_{\rm CDM0} = 0.267$, $\Omega_{K0} = 0$, $T_{\rm CMB} = 2.7255\,{\rm K}$, $N_{\rm eff} = 3.046$, $Y_p = 0.245$, reionization at $z_{\rm re} = 8$ ($\Delta z = 0.5$) and helium reionization at $z = 3.5$ ($\Delta z = 0.5$), with primordial amplitude $A_s = 2.1 \times 10^{-9}$ and tilt $n_s = 0.965$ at $k_{\rm pivot} = 0.05\,{\rm Mpc}^{-1}$. $\Omega_{\Lambda 0}$ is fixed by flatness.

II. MILESTONE I: THE BACKGROUND COSMOLOGY

The first milestone computes the expansion history of a homogeneous and isotropic Universe. Every later module needs it: the recombination rates are set by the competition between atomic reaction rates and the expansion rate H, the perturbation equations contain $\mathcal{H}$ and its derivatives, and the line-of-sight integral needs the conformal distance $\eta_0 - \eta$ to the last-scattering surface. The background also contains the first test of the model against data, through the distances to type Ia supernovae.

A. Theory

The metric and the Friedmann equations. The cosmological principle states that on large scales the Universe is homogeneous and isotropic. The most general metric with these symmetries is the Friedmann–Lemaître–Robertson–Walker (FLRW) metric,

$$ {\rm d}s^2 = -c^2{\rm d}t^2 + a^2(t)\Big[\frac{{\rm d}r^2}{1-kr^2} + r^2{\rm d}\Omega^2\Big], \qquad (1) $$

where r is a comoving radial coordinate, k is the spatial curvature and $a(t)$, normalized to $a(t_0) = 1$ today, is the scale factor. All the dynamics is in $a(t)$. Because of the symmetry, the matter content must be a set of perfect fluids with energy density $\rho_i c^2$ and pressure $P_i = w_i \rho_i c^2$. Inserting the metric and fluids in Einstein's equations gives two independent equations. The 00 component is the Friedmann equation,

$$ H^2 \equiv \Big(\frac{\dot a}{a}\Big)^2 = \frac{8\pi G}{3}\sum_i \rho_i - \frac{kc^2}{a^2}, \qquad (2) $$

and the trace of the ij components gives the acceleration equation,

$$ \frac{\ddot a}{a} = -\frac{4\pi G}{3}\sum_i \Big(\rho_i + \frac{3P_i}{c^2}\Big). \qquad (3) $$

Combining them gives energy conservation, $\dot\rho_i + 3H(\rho_i + P_i/c^2) = 0$, which for constant w_i integrates to $\rho_i \propto a^{-3(1+w_i)}$. The relevant cases are:

- Non-relativistic matter (baryons and cold dark matter), $w = 0$. The number of particles in a comoving volume is conserved, so $\rho_M \propto a^{-3}$.
- Radiation (photons and massless neutrinos), $w = 1/3$. The number density falls as a^{-3} and, in addition, every particle loses energy to the redshift, $E \propto a^{-1}$, so $\rho_r \propto a^{-4}$.
- A cosmological constant, $w = -1$, with ρ_Λ constant: the vacuum energy per volume does not dilute.

Curvature enters Eq. (2) like a fluid with $\rho \propto a^{-2}$. Dividing by H_0^2 and defining the density parameters relative to the critical density $\rho_{c0} = 3H_0^2/8\pi G$ (the density of a flat universe), $\Omega_{i0} = \rho_{i0}/\rho_{c0}$ and $\Omega_{K0} = -kc^2/H_0^2$, gives

$$ H(x) = H_0\sqrt{\Omega_{M0}e^{-3x} + \Omega_{r0}e^{-4x} + \Omega_{K0}e^{-2x} + \Omega_{\Lambda 0}}, \qquad (4) $$

where $\Omega_M = \Omega_b + \Omega_{\rm CDM}$ and $\Omega_r = \Omega_\gamma + \Omega_\nu$. We use $x = \ln a$ as time variable. The interesting history spans some twenty e-folds of expansion, and most quantities vary as powers of a, which are simple exponentials in x. Evaluating Eq. (4) today gives the constraint $\sum_i \Omega_{i0} = 1$, which fixes $\Omega_{\Lambda 0}$ in a flat universe. The density parameters at other times are $\Omega_i(x) = \Omega_{i0}e^{-n_i x}H_0^2/H^2(x)$, with $n_i = 3, 4, 2, 0$ for matter, radiation, curvature and Λ; they measure which component drives the expansion at a given time.

The radiation density. The CMB is a blackbody with temperature $T_{\rm CMB} = 2.7255\,{\rm K}$, so its density follows from the Stefan–Boltzmann law: integrating the Bose–Einstein distribution over momenta, with two polarization states, gives $\rho_\gamma c^2 = 2\cdot\frac{\pi^2}{30}(k_B T)^4/(\hbar c)^3$ and

$$ \Omega_{\gamma 0} = 2\cdot\frac{\pi^2}{30}\frac{(k_B T_{\rm CMB})^4}{\hbar^3 c^5}\cdot\frac{8\pi G}{3H_0^2}. \qquad (5) $$

Neutrinos decoupled from the plasma at $T \sim 1\,{\rm MeV}$, just before electrons and positrons annihilated. The annihilation heated the photons but not the neutrinos, and entropy conservation gives $T_\nu = (4/11)^{1/3}T_\gamma$ afterwards. Fermions also carry a factor 7/8 relative to bosons in the energy density. With $N_{\rm eff}$ species, $\Omega_{\nu 0} = N_{\rm eff}\,\frac{7}{8}(4/11)^{4/3}\Omega_{\gamma 0}$. $N_{\rm eff} = 3.046$ rather than 3 because the neutrinos were still slightly coupled when the annihilation began and shared a little of its energy.

Clocks and horizons. Cosmic time t is the time measured by an observer at rest with respect to the expansion. Conformal time η is defined by $c\,{\rm d}t = a\,{\rm d}\eta$, so that light moves on straight lines in (η, χ) coordinates with ${\rm d}\chi = c\,{\rm d}\eta$. In terms of x,

$$ \frac{{\rm d}t}{{\rm d}x} = \frac{1}{H}, \qquad \frac{{\rm d}\eta}{{\rm d}x} = \frac{c}{\mathcal{H}}, \qquad \mathcal{H} \equiv aH = \dot a. \qquad (6) $$

$\eta(x)$ is the comoving distance light has travelled since the Big Bang, the *particle horizon*. Two regions farther apart than η have never been in causal contact. The quantity $c/\mathcal{H}$ is the comoving Hubble radius, the distance over which the expansion velocity reaches c, and over which causal physics can act within one expansion time. A perturbation of comoving wavenumber k is outside the

horizon when $ck < \mathcal{H}$ and inside when $ck > \mathcal{H}$. In a decelerating universe ($\ddot{a} < 0$) $\mathcal{H} = \dot{a}$ decreases, so ever larger scales enter the horizon. This is central for the perturbations of Milestone III.

Single-component limits. If one component with $\rho \propto a^{-n}$ dominates, then $\mathcal{H} \propto a^{1-n/2} = e^{-(n-2)x/2}$, and Eq. (6) gives $\eta = 2c/[(n-2)\mathcal{H}]$ for $n > 2$. Hence

$$\frac{\mathcal{H}'}{\mathcal{H}} = 1 - \frac{n}{2}, \qquad \frac{\mathcal{H}''}{\mathcal{H}} = \left(1 - \frac{n}{2}\right)^2, \qquad \frac{\eta\mathcal{H}}{c} = \frac{2}{n-2}, \tag{7}$$

that is $(-1, 1, 1)$ in radiation domination (RD) and $(-\frac{1}{2}, \frac{1}{4}, 2)$ in matter domination (MD). In Λ domination H is constant, $\mathcal{H} \propto a$, $\mathcal{H}'/\mathcal{H} = \mathcal{H}''/\mathcal{H} = 1$, and η approaches a finite value: there is an event horizon, and we will never see beyond it. The corresponding laws for cosmic time are $t \propto a^2$ (RD), $t \propto a^{3/2}$ (MD) and $t \propto \ln a$ (Λ). These limits are our main tests.

Distances. Light from an object at x has travelled a comoving distance $\chi = \eta_0 - \eta(x)$. In a curved universe the comoving *transverse* distance is $r = \chi\,\mathrm{sinn}(\sqrt{|\Omega_{K0}|}H_0\chi/c)/(\sqrt{|\Omega_{K0}|}H_0\chi/c)$, with sinn = sinh for $\Omega_{K0} > 0$ and sin for $\Omega_{K0} < 0$; for flat space $r = \chi$. An object of physical size D at scale factor a subtends the angle $\theta = D/(ar)$, so the angular diameter distance is $d_A = ar$. A source of luminosity L spreads its light over a sphere of comoving area $4\pi r^2$, and each photon arrives with its energy reduced by $(1+z)$ and its arrival rate reduced by another $(1+z)$ because of time dilation. The flux is then $F = L/(4\pi d_L^2)$ with the luminosity distance $d_L = r/a = (1+z)^2 d_A$.

At low redshift $d_L \simeq cz/H_0$, Hubble's law. The next term, $d_L \simeq (cz/H_0)[1 + \frac{1}{2}(1-q_0)z]$, depends on the deceleration parameter $q_0 = -\ddot{a}/(aH^2)|_0 = \frac{1}{2}\Omega_{M0} - \Omega_{\Lambda 0}$ (neglecting radiation). In an accelerating universe ($q_0 < 0$) distant objects are farther away, and therefore fainter, than in a decelerating one. This is how supernovae revealed the accelerated expansion.

Key epochs. Three epochs are of special interest:

- Radiation–matter equality, $\rho_r = \rho_M$, at $a_{\rm eq} = \Omega_{r0}/\Omega_{M0}$.
- Matter–Λ equality at $a^3 = \Omega_{M0}/\Omega_{\Lambda 0}$.
- The onset of acceleration, $\ddot{a} = 0$. Equation (3) shows that a component accelerates the expansion only if $w < -1/3$. Matter contributes ρ_M and Λ contributes $\rho_\Lambda + 3P_\Lambda/c^2 = -2\rho_\Lambda$, so acceleration starts when $\rho_M = 2\rho_\Lambda$, i.e. $\Omega_{M0}a^{-3} = 2\Omega_{\Lambda 0}$. Since $\ddot{a} = \dot{\mathcal{H}}$, this is also where $\mathcal{H}' = 0$.

The wavenumber that enters the horizon at equality, $k_{\rm eq} = \mathcal{H}(x_{\rm eq})/c$, separates modes that entered during RD from those that entered during MD. It sets the turnover scale of $P(k)$ (Sec. V).

B. Implementation and tests

`BackgroundCosmology` writes $E^2 = H^2/H_0^2$ as a sum of virtual component densities, each with analytic first and second x-derivatives. $\mathcal{H}$, $\mathcal{H}'$ and $\mathcal{H}''$ are therefore analytic. A new model, for example dynamical dark energy, only overrides the density of one component; the code includes a CPL $w_0 w_a$ example. We solve Eq. (6) on $x \in [-20, 5]$ with an adaptive Runge–Kutta (RK8PD) solver and store η and t in cubic splines. The initial values come from the exact solution for a universe with radiation and matter only,

$$\eta = \frac{2ca}{H_0\left(\sqrt{\Omega_{r0} + \Omega_{M0}a} + \sqrt{\Omega_{r0}}\right)}, \tag{8}$$

which is exact up to the negligible Λ term at $x = -20$; the analogous expression is used for t. The epochs are found by root finding on the analytic expressions.

Figure 1 shows the two standard tests. $\mathcal{H}'/\mathcal{H}$ and $\mathcal{H}''/\mathcal{H}$ reach the plateaus of Eq. (7) in each era, with smooth transitions at the equality epochs. $\eta\mathcal{H}/c$ goes from 1 in RD towards 2 in MD. It never quite reaches 2: matter domination is too short, and η still remembers the radiation era when Λ takes over. In the Λ era η freezes while $\mathcal{H}$ grows, so $\eta\mathcal{H}/c$ increases. The derivative of the η spline satisfies $\eta'\mathcal{H}/c = 1$ to $\max|\eta'\mathcal{H}/c - 1| = 9.8\times10^{-6}$, which checks both the ODE and the spline. The unit tests also check t against the analytic Einstein–de Sitter result, the analytic derivatives against finite differences, and η against an independent quadrature. Section VI shows that H, t and η agree with CLASS to $\sim 10^{-7}$.

C. Results

The expansion history is shown in Figs. 2–4 and the key epochs are listed in Table I.

The three eras. Figure 4 shows the succession of eras that follows from the different dilution laws. Going back in time, radiation ($\propto a^{-4}$) eventually wins over matter ($\propto a^{-3}$), which in turn wins over the constant ρ_Λ. Radiation dominates until $z = 3400$, when the Universe was only 51 000 years old. Matter then dominates for most of cosmic history, and Λ only takes over at $z = 0.29$. Before $z \sim 2$ the cosmological constant is completely negligible; that we happen to live just when Ω_M and Ω_Λ are comparable is the so-called coincidence problem. Today $\Omega_{\Lambda 0} = 0.6829$, $\Omega_{\gamma 0} = 5.51\times10^{-5}$ and $\Omega_{\nu 0} = 3.81\times10^{-5}$.

The conformal Hubble factor. $\mathcal{H}$ (Fig. 2) falls as e^{-x} in RD and as $e^{-x/2}$ in MD, exactly as Eq. (7) predicts. Because $\mathcal{H} = \dot{a}$, its slope is the sign of the acceleration: $\mathcal{H}$ has a minimum exactly at the onset of acceleration, $z = 0.627$, and then grows as e^x. Acceleration starts well before matter–Λ equality at $z = 0.292$, because the pressure of the cosmological constant counts twice in Eq. (3): Λ only needs half the matter density to make $\ddot{a} > 0$. A falling $\mathcal{H}$ means the comoving Hubble radius $c/\mathcal{H}$ grows

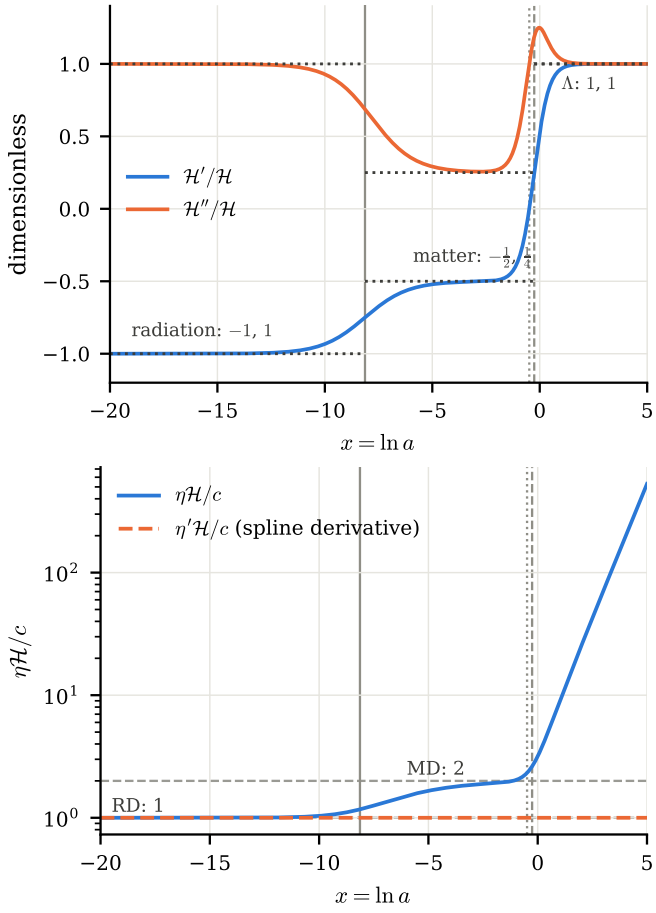


FIG. 1. Tests of the background. *Top:* $\mathcal{H}'/\mathcal{H}$ and $\mathcal{H}''/\mathcal{H}$. The short dotted lines are the analytic values of Eq. (7) in the radiation, matter and Λ eras. *Bottom:* $\eta\mathcal{H}/c$, which is 1 in RD and approaches 2 in MD, and $\eta'\mathcal{H}/c$ from the derivative of the spline, which should be exactly 1. In all Milestone I figures the vertical lines mark radiation–matter equality (solid), matter–Λ equality (dashed) and the onset of acceleration (dotted).

TABLE I. Key epochs of the fiducial background. The ages are cosmic times since the Big Bang.

Event	x	z	t
Radiation–matter equality	−8.1319	3400.3	51 029 yr
Onset of acceleration	−0.4865	0.6267	7.750 Gyr
Matter–Λ equality	−0.2558	0.2915	10.371 Gyr
Today	0	0	13.848 Gyr

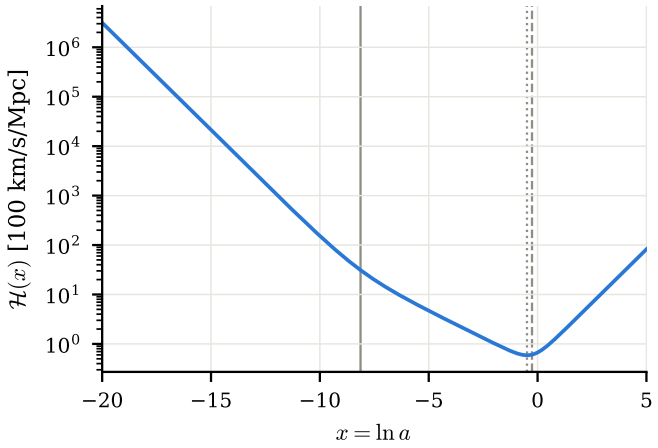


FIG. 2. The conformal Hubble factor $\mathcal{H} = aH = \dot{a}$ in units of 100 km s^{-1} Mpc^{-1}. It decreases during RD and MD, when the expansion decelerates, and increases after the onset of acceleration (dotted line).

and new scales enter the horizon; after $z = 0.63$ this has reversed, and scales are now leaving the horizon again.

Age and horizon. The age of the Universe is $t_0 = 13.85$ Gyr (Fig. 3). This is $0.949\,H_0^{-1}$, with $H_0^{-1} = 14.59$ Gyr. A flat matter-only (Einstein–de Sitter) universe with the same H_0 would be only $\frac{2}{3}H_0^{-1} = 9.7$ Gyr old, younger than the oldest stars. The cosmological constant slowed the expansion less in the past, so it took longer to reach today's expansion rate, and the Universe is older. The conformal time today is $\eta_0/c = 46.29$ Gyr ($\eta_0 = 14\,192$ Mpc $= 3.17\,c/H_0$). This is the comoving radius of the observable Universe: light has travelled for 13.85 Gyr, but the space it crossed has expanded in the meantime.

The radiation era is short in cosmic time but long in conformal time, because $\mathrm{d}\eta = c\,\mathrm{d}t/a$ weighs early times by $1/a$. About 40% of the conformal time at recombination accumulates before equality. The comoving horizon at recombination is $\eta_* \approx 280$ Mpc, only 2% of the distance $\eta_0 - \eta_*$ to the last-scattering surface. It subtends an angle of about 1° on the sky. Regions of the CMB separated by more than this were causally disconnected at recombination, which is the horizon problem that inflation is invoked to solve. It is also why the acoustic features of the CMB appear at $\ell \sim 200$ (Sec. V).

Supernovae. Type Ia supernovae are standardizable candles: after empirical corrections for light-curve width and colour, their peak luminosities are nearly the same. Their fluxes therefore measure $d_L(z)$. Figure 5 compares our luminosity distance with the binned JLA distances of Betoule et al. [9], plotted as d_L/z. At low redshift $d_L/z \to c/H_0$, and the rise with z measures $1 - q_0$. Our model has $q_0 = -0.52$.

The fiducial model reproduces the shape of $d_L(z)$ but is about 4% too far away at all redshifts, giving $\chi^2 = 141.7$ for $N = 31$ points. With the density parameters fixed, $d_L \propto 1/h$ and h only rescales the curve. The best fit is then $h = 0.699$ with $\chi^2 = 30.7$, a good fit. The data clearly require acceleration. For comparison, the best fit of a flat matter-only universe (EdS, $q_0 = +\frac{1}{2}$) gives $\chi^2 = 317$, and even an empty, coasting universe ($q_0 = 0$) gives $\chi^2 = 49$.

The preference for $h \approx 0.70$ should not be read as a measurement of H_0. Supernovae measure *relative* distances. The absolute scale of the data comes from the ab-

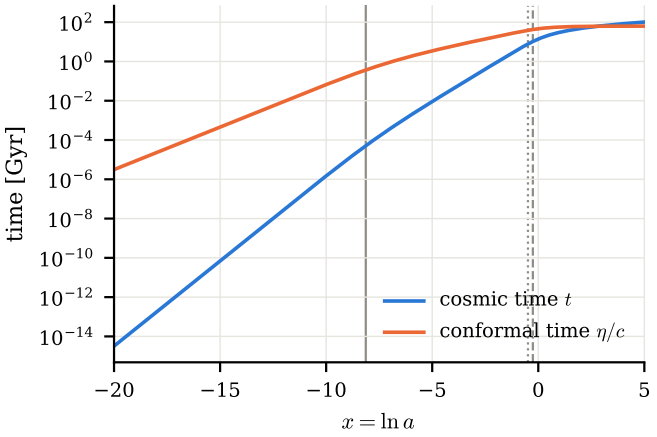


FIG. 3. Cosmic time t and conformal time η/c in Gyr. In RD $t \propto a^2$ and $\eta \propto a$; in MD $t \propto a^{3/2}$ and $\eta \propto a^{1/2}$. η freezes at $\eta_0/c = 46.3\,\mathrm{Gyr}$ in the Λ era, while t grows logarithmically in a.

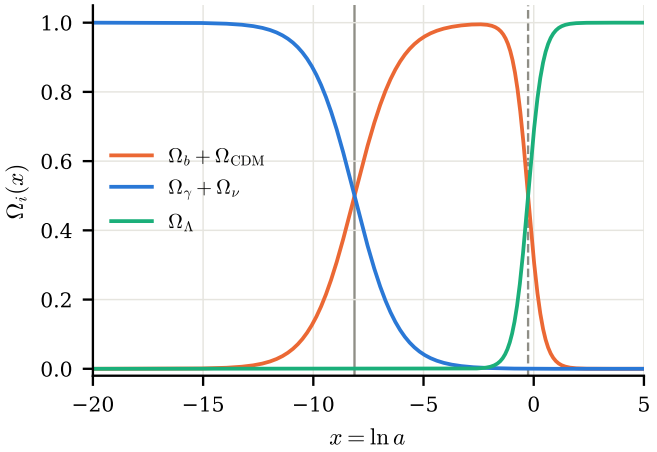


FIG. 4. Density parameters of relativistic species ($\Omega_\gamma + \Omega_\nu$), matter ($\Omega_b + \Omega_{\mathrm{CDM}}$) and dark energy ($\Omega_\Lambda$) as functions of x. They always sum to one in a flat universe.

solute magnitude adopted when converting the distance moduli to d_L, which is degenerate with H_0. A joint fit of h, Ω_{M0} and Ω_{K0} by MCMC is part of the course project but omitted here.

III. MILESTONE II: THE RECOMBINATION HISTORY

Before recombination the Universe is filled with a plasma of photons, free electrons and nuclei. Thomson scattering off the free electrons is so frequent that a photon travels only a tiny distance between collisions, and photons and baryons move together as a single fluid. When the Universe has cooled enough for electrons to bind into neutral atoms, the scattering stops: the photons decouple and stream freely towards us as the CMB.

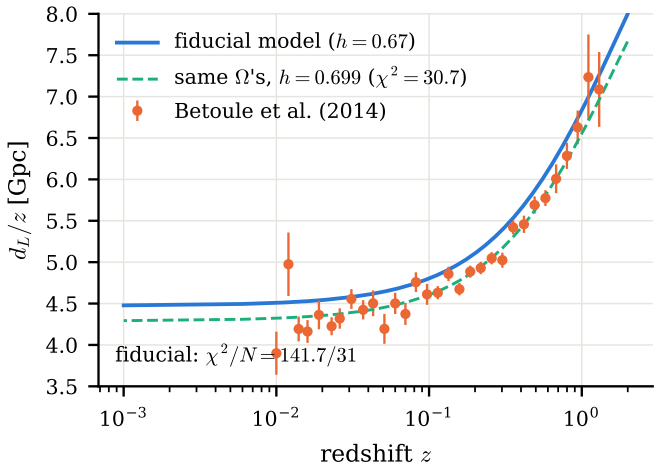


FIG. 5. Luminosity distance divided by redshift for the fiducial model (solid), and for the same density parameters with the best-fit h (dashed), compared with the binned JLA supernova data [9]. The intercept at $z \to 0$ is c/H_0; the slope measures the acceleration of the expansion.

This milestone computes the free electron fraction $X_e(x)$, and from it the optical depth τ and the visibility function $\tilde{g}$, which tells us where the CMB photons last scattered.

A. Theory

We define $X_e = n_e/n_H$, where $n_H = (1-Y_p)\rho_b/m_H \propto a^{-3}$ is the number density of hydrogen nuclei, free or bound, and Y_p is the primordial helium mass fraction. Helium contributes $f_{\mathrm{He}} = n_{\mathrm{He}}/n_H = Y_p/[4(1-Y_p)]$ nuclei per hydrogen nucleus, so fully ionized hydrogen and helium give $X_e = 1 + 2f_{\mathrm{He}}$.

Saha equilibrium. At early times the reaction $e^- + p \leftrightarrow \mathrm{H} + \gamma$ runs much faster than the expansion and is in chemical equilibrium, $\mu_e + \mu_p = \mu_{\mathrm{H}}$ (photons have $\mu_\gamma = 0$). The particles are non-relativistic and non-degenerate, so each species has the Maxwell–Boltzmann density $n_i = g_i(m_i k_B T/2\pi\hbar^2)^{3/2} e^{(\mu_i - m_i c^2)/k_B T}$. In the ratio $n_e n_p/n_{\mathrm{H}}$ the chemical potentials cancel, and the mass difference $m_e + m_p - m_{\mathrm{H}} = \epsilon_0/c^2$ leaves the binding energy $\epsilon_0 = 13.6\,\mathrm{eV}$. With $F = (m_e k_B T_b/2\pi\hbar^2)^{3/2}$ this gives the Saha equation. The same argument applies to the two ionization stages of helium:

$$\frac{n_e x_{\mathrm{H}^+}}{1 - x_{\mathrm{H}^+}} = F e^{-\epsilon_0/k_B T_b}, \qquad (9)$$

$$\frac{n_e x_{\mathrm{He}^+}}{x_{\mathrm{He}^0}} = 4F e^{-\chi_0/k_B T_b}, \qquad \frac{n_e x_{\mathrm{He}^{++}}}{x_{\mathrm{He}^+}} = F e^{-\chi_1/k_B T_b}.$$

Here $\chi_0 = 24.59\,\mathrm{eV}$ and $\chi_1 = 54.4\,\mathrm{eV}$ are the helium ionization energies, and the factors 4 and 1 are the ratios of the spin degeneracies. The electron density $n_e = n_H[x_{\mathrm{H}^+} + f_{\mathrm{He}}(x_{\mathrm{He}^+} + 2x_{\mathrm{He}^{++}})]$ closes the system. For hydrogen alone, Eq. (9) is a quadratic in X_e.

Equation (9) explains why recombination happens so late. Writing $n_e = X_e n_H$ and $n_H \propto \eta_b n_\gamma$, with $\eta_b \approx 6\times10^{-10}$ baryons per photon, the right-hand side divided by n_H is of order $\eta_b^{-1}(k_BT/m_ec^2)^{3/2}e^{-\epsilon_0/k_BT}$. The huge factor η_b^{-1} keeps hydrogen ionized until the exponential has become very small, i.e. until $\epsilon_0/k_BT \approx 40$–$50$. Physically, there are so many photons per atom that even the far Wien tail of the Planck spectrum, above 13.6 eV, contains enough photons to ionize every atom that forms.

Peebles equation. As X_e drops, the rates fall below the expansion rate and equilibrium can no longer be maintained. Recombination is moreover not a single-step process. A direct recombination to the ground state emits a photon above 13.6 eV, which is immediately absorbed by a neighbouring atom and ionizes it again; such recombinations achieve nothing. Recombination to an excited state cascades down to $n = 2$. From $2p$ the atom emits a Lyman-α photon, but the Universe is extremely optically thick to Lyman-α, so the photon is reabsorbed and the atom is re-excited. Only two routes lead permanently to the ground state:

- the slow two-photon decay $2s \to 1s$, with rate $\Lambda_{2s\to1s} = 8.227\,\mathrm{s}^{-1}$;
- the redshifting of a Lyman-α photon out of the line by the expansion, with rate Λ_α.

Meanwhile the $n = 2$ state, bound by only $\epsilon_0/4 = 3.4\,\mathrm{eV}$, is easily photoionized again. Peebles' effective three-level atom [10] captures this physics:

$$\frac{\mathrm{d}X_e}{\mathrm{d}x} = \frac{C_r(T_b)}{H}\Big[\beta(T_b)(1-X_e) - n_H\alpha^{(2)}(T_b)X_e^2\Big]. \quad (10)$$

The terms are as follows:

- The second term in the bracket is recombination. Its rate per hydrogen atom is $n_e\alpha^{(2)}$, where $\alpha^{(2)}$ is the case-B recombination coefficient. It counts only captures to $n \geq 2$, the ones that can eventually lead to a neutral atom.
- The first term is photoionization. Its rate β follows from $\alpha^{(2)}$ by detailed balance, $\beta = \alpha^{(2)}Fe^{-\epsilon_0/k_BT_b}$, so that the bracket vanishes in Saha equilibrium. The rate out of $n = 2$ is $\beta^{(2)} = \beta e^{3\epsilon_0/4k_BT_b}$.
- The factor

$$C_r = \frac{\Lambda_{2s\to1s}+\Lambda_\alpha}{\Lambda_{2s\to1s}+\Lambda_\alpha+\beta^{(2)}}, \quad \Lambda_\alpha = \frac{H(3\epsilon_0)^3}{(8\pi)^2(\hbar c)^3 n_{1s}}, \quad (11)$$

  with $n_{1s} = (1-X_e)n_H$, is the probability that an atom in $n = 2$ reaches the ground state before it is ionized again.

Early on $\beta^{(2)} \gg \Lambda$'s and $C_r \ll 1$: this bottleneck makes recombination slower than in equilibrium. Late on, the recombination rate $n_e\alpha^{(2)}$ drops below H because both n_e and $\alpha^{(2)}$ decrease. The few remaining electrons can no longer find a proton within a Hubble time, and X_e freezes out at a finite value. The rate coefficients we use are those of Callin [7].

Baryon temperature. The gas is kept at the photon temperature by Compton scattering of the abundant CMB photons off the free electrons. A non-relativistic gas expanding adiabatically would instead cool as $T_b \propto a^{-2}$, faster than $T_\gamma \propto a^{-1}$. The two effects combine to

$$\frac{\mathrm{d}T_b}{\mathrm{d}x} = -2T_b + \frac{8\sigma_T a_R T_\gamma^4}{3m_ecH}\frac{X_e}{1+f_{\mathrm{He}}+X_e}(T_\gamma - T_b), \quad (12)$$

where the second term is the Compton heating rate in units of H; the fraction counts the energy transferred per particle in the gas. This rate is $\sim 10^6$ at recombination, so $T_b = T_\gamma$ until the residual electrons become too few. The baryon temperature enters Eq. (10) and matters mainly for the freeze-out value of X_e.

Reionization. The intergalactic medium is ionized today. Ultraviolet radiation from the first stars and galaxies reionized hydrogen (and helium once) at $z \sim 6$–10, and quasars reionized the second helium electron at $z \sim 3$. We do not model these sources. Instead we use the phenomenological CAMB/CLASS parametrization [11]:

$$X_e = X_e^{\mathrm{rec}} + (1+f_{\mathrm{He}}-X_e^{\mathrm{rec}})\frac{1+\tanh u}{2} + f_{\mathrm{He}}\frac{1+\tanh u_{\mathrm{He}}}{2}, \quad (13)$$

with $u = (y_{\mathrm{re}}-y)/\Delta y$, $y = (1+z)^{3/2}$, $\Delta y = \frac{3}{2}(1+z_{\mathrm{re}})^{1/2}\Delta z$, and $u_{\mathrm{He}} = (z_{\mathrm{He}}-z)/\Delta z_{\mathrm{He}}$. The variable y is proportional to the cosmic time during MD, so the transition is symmetric in time. Hydrogen and the first helium electron are reionized at $z_{\mathrm{re}} = 8$, and the second helium electron at $z_{\mathrm{He}} = 3.5$.

Optical depth and visibility. A photon travelling through a gas of free electrons scatters at the rate $n_e\sigma_T c$, where σ_T is the Thomson cross section. The probability that a photon emitted at x reaches us without scattering is $e^{-\tau(x)}$, where the optical depth $\tau(x) = \int_x^0 n_e\sigma_T c\,\mathrm{d}t$ obeys

$$\tau'(x) = -\frac{c\,n_e\sigma_T}{H}, \qquad \tilde{g}(x) = -\tau' e^{-\tau}. \quad (14)$$

$|\tau'|$ is the number of scatterings per Hubble time. The visibility function $\tilde{g}$ is the probability density, per unit x, that a photon observed today scattered for the last time at x, and $\int \tilde{g}\,\mathrm{d}x = 1$. It peaks where the falling scattering rate and the growing transparency balance.

The sound horizon and the drag epoch. Before decoupling, photons and baryons form one fluid. The photons provide the pressure, $P = \rho_\gamma c^2/3$, while both components provide the inertia. The sound speed is $c_s^2 = \delta P/\delta\rho = \frac{c^2}{3}(1+R_b)^{-1}$, with $R_b = 3\rho_b/4\rho_\gamma$. Following Ref. [7] we use the inverse, $R = 1/R_b = 4\Omega_{\gamma0}/(3\Omega_{b0}a)$, so

$$r_s(x) = \int_{-\infty}^{x}\frac{c_s}{\mathcal{H}}\mathrm{d}x', \quad c_s = c\sqrt{\frac{R}{3(1+R)}}. \quad (15)$$

r_s is the comoving distance a sound wave travels before x. It is the length scale imprinted in both the CMB peaks and the clustering of galaxies, the baryon acoustic oscillation (BAO) scale. The photons decouple at $\tau = 1$. The baryons are released later. There are $\sim 10^9$ photons per baryon, so each baryon keeps being pushed by scattering photons after a typical photon has stopped scattering. The rate at which the photons transfer momentum to the baryons is $|\tau'|$ times the ratio of photon to baryon momentum density, $|\tau'|/R_b = |\tau'|R$. The drag epoch is where the drag optical depth $\tau_d(x) = -\int_x^0 \tau' R\,\mathrm{d}x'$ reaches 1. Since $R > 1$ at recombination, it comes after decoupling.

B. Implementation and tests

`RecombinationHistory` solves on 40 000 points in $x \in [-20, 0]$. We use Saha for hydrogen and helium while the hydrogen ionization fraction is above 0.99, solving the coupled Saha system by root finding in $\ln n_e$. After that, x_{H^+} follows the Peebles equation (10) together with T_b. Helium stays in Saha equilibrium, given the electron density. The coupled (X_e, T_b) system is stiff, because the Compton rate is $\sim 10^6 H$, so we use the implicit multistep BDF method of GSL. X_e is splined in $\ln X_e$, which varies over many decades. We integrate τ backwards from $\tau(0) = 0$, and take τ' analytically from n_e. The code can switch off helium, reionization, T_b evolution and the Peebles stage (pure Saha), and includes the RECFAST fudge factor 1.14 on $\alpha^{(2)}$ as an option [12].

Tests (Figs. 6–8) include:

- $X_e = 1+2f_{\mathrm{He}} = 1.162$ before helium recombination and $1 + f_{\mathrm{He}}$ on the He^+ plateau.
- With $Y_p = 0$ the full helium solver reproduces the hydrogen-only code exactly (maximum relative difference 0).
- $\int \tilde{g}\,\mathrm{d}x = 1$ to machine precision.
- τ', τ'', $\tilde{g}'$ and $\tilde{g}''$ agree with finite differences of the splines.
- r_s agrees with an independent quadrature.
- $T_b = T_\gamma$ at high z and $T_b \propto a^{-2}$ at low z.
- The optical depth to reionization agrees with the analytic estimate for instantaneous reionization in a matter+Λ background (see below).

C. Results

Helium and hydrogen recombination. Figure 6 shows the free electron fraction. Helium recombines first, in two steps, because its electrons are more tightly bound:

- $\mathrm{He}^{++} \to \mathrm{He}^+$ (54.4 eV) at $z \approx 6000$, the drop from 1.162 to 1.081;
- $\mathrm{He}^+ \to \mathrm{He}$ (24.6 eV) at $z \approx 2500$, the drop from 1.081 to 1.

Both happen when $\chi/k_BT \approx 40$, the same "η_b^{-1} delay" as for hydrogen. Helium is fully neutral long before hydrogen recombination starts. It therefore has no direct effect on the last-scattering surface; it matters only by removing electrons at $z \gtrsim 2000$, which affects the diffusion damping of small-scale anisotropies. The hydrogen-only run gives the same numbers as the run with helium to better than 10^{-8} (both runs use the same n_H).

Hydrogen recombines between $z \approx 1500$ and $z \approx 900$. With the Peebles equation $X_e = 0.5$ at $z = 1274$ ($T = 3475\,\mathrm{K}$, $k_BT = 0.30\,\mathrm{eV}$), so indeed $\epsilon_0/k_BT \approx 45$. Table II lists the key times. Here we switch reionization off, so that τ measures the recombination surface alone.

Saha versus Peebles. The Saha approximation recombines far too early and too fast: $X_e = 0.5$ at $z = 1370$, decoupling at $z = 1292$, and r_s is 11% smaller. Saha assumes that the reactions are instantaneous. In reality the $n = 2$ bottleneck limits the net recombination rate: at $z = 1200$ we find $C_r \approx 0.007$, so most atoms that reach $n = 2$ are ionized again before they reach the ground state. Recombination therefore falls behind equilibrium. The difference grows with time. In Saha equilibrium X_e drops exponentially and reaches the numerical floor of 10^{-30} in our code. In the Peebles solution the recombination rate per electron, $n_H\alpha^{(2)}X_e$, falls below H at $z \approx 400$, and the residual ionization freezes at $X_e = 2.0 \times 10^{-4}$ today. These residual electrons keep T_b coupled to T_γ until $z \approx 500$ and are important for the formation of the first molecules.

The RECFAST fudge factor 1.14 on $\alpha^{(2)}$ mimics the effect of a full multi-level atom. It speeds up recombination slightly: z_* changes by +0.8, r_s by 0.05%, and the freeze-out value is lowered to 1.8×10^{-4}. These small changes nevertheless matter at the precision of Planck.

Decoupling. The photons decouple ($\tau = 1$) at $z = 1090$, when the Universe is 372 000 years old and $T = 2970\,\mathrm{K}$. At that moment $X_e = 0.13$: decoupling does not require the gas to be fully neutral, only that the scattering rate drops below the expansion rate. Indeed $|\tau'| = 13$ at $\tau = 1$, so a photon still scatters a few more times per Hubble time, but X_e is dropping so fast that these scatterings stop within $\Delta x \sim 0.1$. The comoving mean free path at that time is 16 Mpc, compared with the horizon $\eta_* = 280\,\mathrm{Mpc}$. Decoupling is at lower z than $X_e = 0.5$ and higher z than $X_e = 0.1$ ($z = 1063$). Within a few percent, "recombination" and "last scattering" are the same event for the purpose of the CMB.

Optical depth and visibility. Figure 7 shows the optical depth. Before recombination $\tau \propto$ a high power of a^{-1}, because $n_e \propto a^{-3}$ and $H \propto a^{-2}$ in RD. Across recombination $|\tau'|$ drops by more than three orders of magnitude, so the Universe goes from opaque to transparent within a small fraction of a Hubble time. Fig-

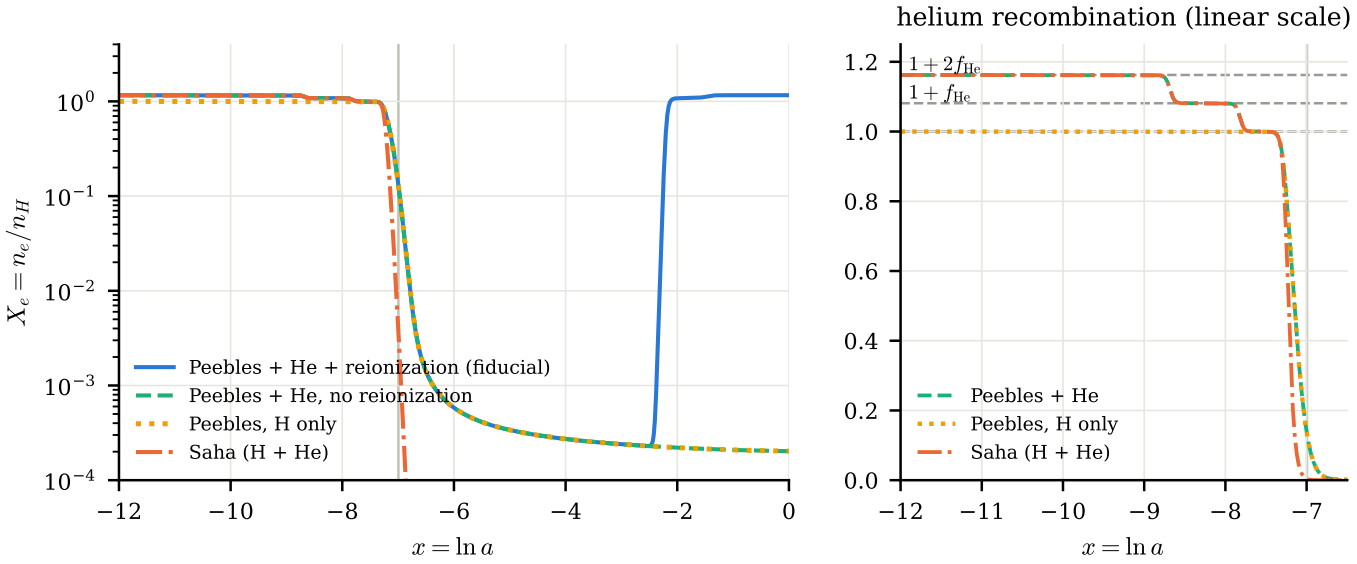


FIG. 6. Free electron fraction. *Left:* the fiducial model (Peebles hydrogen, Saha helium and reionization), the same without reionization, the hydrogen-only Peebles solution, and the Saha solution for H and He. The vertical line marks the last-scattering surface ($\tau = 1$). *Right:* the same on a linear scale, showing the two helium recombinations at $z \approx 6000$ and $z \approx 2500$ before hydrogen recombination.

TABLE II. Recombination and decoupling without reionization for the Peebles (with Saha helium) and pure Saha solutions. Times are cosmic times; r_s is the comoving sound horizon.

	Peebles	Saha
$X_e = 0.5$: z	1273.6	1369.9
Recombination, $X_e = 0.1$: x	−6.9701	−7.1340
z	1063.4	1252.9
t [kyr]	387.6	293.8
Decoupling, $\tau = 1$: x	−6.9948	−7.1645
z	1089.9	1291.7
t [kyr]	371.8	279.0
Peak of $\tilde{g}$: z	1085.7	1286.8
$r_s(\tau = 1)$ [Mpc]	144.41	128.90
Drag epoch: z_d	1060.3	1260.1
$r_s(z_d)$ [Mpc]	147.03	131.09
X_e today	2.02×10^{-4}	$\to 0$

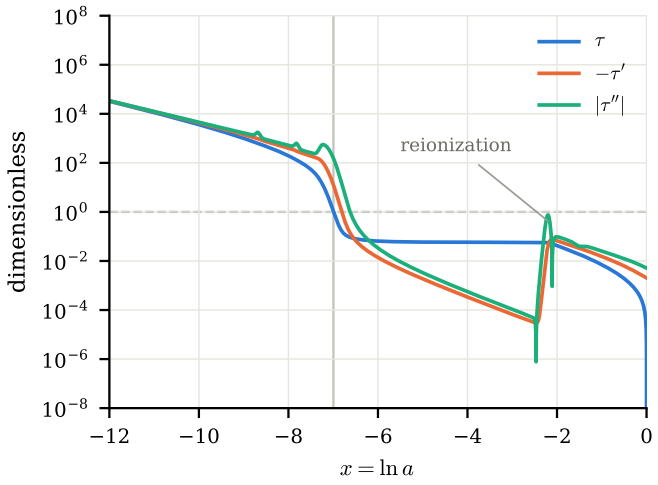


FIG. 7. Optical depth τ and its derivatives for the fiducial model. τ drops by many orders of magnitude during recombination. Reionization gives the shoulder at $x \approx -2$ and the total $\tau_{\rm reion} = 0.058$.

ure 8 shows the visibility function. It is sharply peaked around $x \approx -7$, with a full width at half maximum of $\Delta x \approx 0.18$ ($\Delta z \approx 200$). This corresponds to a comoving thickness of ≈ 39 Mpc, about a quarter of the sound horizon. About two thirds of the CMB photons last scattered between $z = 1000$ and $z = 1200$. The finite thickness of the last-scattering surface has two observable consequences: anisotropies on scales smaller than its thickness are washed out, and polarization is generated only within it (Sec. IV). The derivatives $\tilde{g}'$ and $\tilde{g}''$, which appear in the source function, are smooth and well resolved by our grid.

Sound horizon and drag epoch. At decoupling $R_b = 0.62$, so the sound speed is $c_s = 0.45c$, below the relativistic value $c/\sqrt{3} = 0.58c$. The sound horizon $r_s(z_*) = 144.4$ Mpc is therefore about half the particle horizon. The baryons are released at the drag epoch $z_d = 1060$, when the sound horizon has grown to 147.0 Mpc. This is the BAO scale measured in galaxy surveys.

Reionization. Reionization adds a second, broad and much lower bump in $\tilde{g}$ at $z \lesssim 10$ (Fig. 8, right). The optical depth to reionization is $\tau_{\rm reion} = 0.0580$. For an

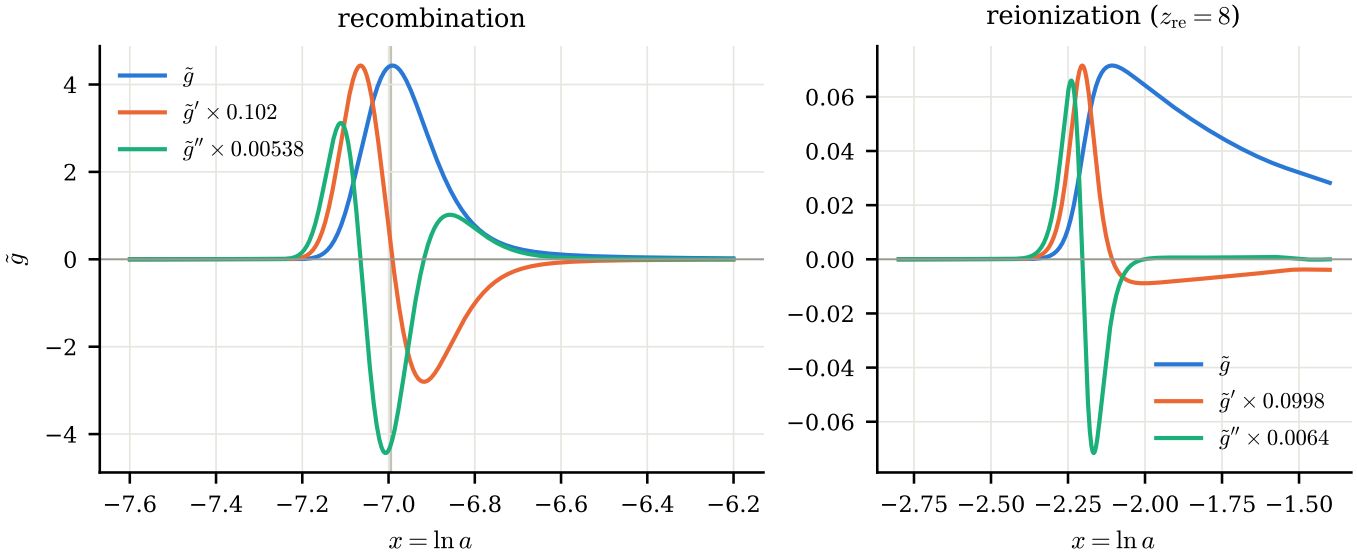


FIG. 8. Visibility function $\tilde{g}$ and its first two derivatives (rescaled to the same amplitude) around recombination (*left*) and reionization (*right*). The vertical line in the left panel marks $\tau = 1$.

instantaneous reionization at $z_{\rm re} = 8$ in a matter+Λ universe, $\tau = \int_0^{z_{\rm re}} c\sigma_T n_e(z)\,\mathrm{d}z/[(1+z)H(z)]$ gives 0.0578 with our parameters, in agreement. This is close to the Planck value 0.054 ± 0.007 [5]. About $1 - e^{-\tau} \approx 6\%$ of the CMB photons scatter again after reionization. On scales smaller than the horizon at reionization, rescattering mixes photons from different directions, which damps the primary anisotropies by $e^{-\tau_{\rm reion}}$ in amplitude. On larger scales it creates new polarization: the low-ℓ "reionization bump" in C_ℓ^{EE} (Sec. V). With reionization the electron fraction today is $X_e = 1.162 = 1 + 2f_{\rm He}$.

With reionization the $\tau = 1$ surface also moves slightly: $\tau_{\rm reion}$ is added to the total, so $\tau = 1$ is reached at $z_* = 1084.9$ with $r_s(z_*) = 144.85\,$Mpc. These are the numbers to compare with CLASS, which gives $z_* = 1084.70$ and $r_s = 144.87\,$Mpc (Sec. VI).

IV. MILESTONE III: EVOLUTION OF THE PERTURBATIONS

The Universe is not perfectly homogeneous. The CMB temperature varies across the sky at the level of 10^{-5}, and these tiny primordial fluctuations have grown into the galaxies and clusters we see today. The third milestone follows the fluctuations from the radiation era until today. Since they are small, we expand Einstein's equations and the Boltzmann equations for every species to first order around the FLRW background of Milestone I. The background is translation invariant, so in Fourier space each wavenumber evolves independently, and isotropy means that only the magnitude k matters. The output is a set of transfer functions and source functions that the line-of-sight integration of Milestone IV turns into the observed spectra.

A. Theory

The perturbed metric. We work in the conformal Newtonian gauge,

$$\mathrm{d}s^2 = a^2(\eta)\big[-(1+2\Psi)c^2\mathrm{d}\eta^2 + (1+2\Phi)\delta_{ij}\mathrm{d}x^i\mathrm{d}x^j\big], \quad (16)$$

with the variables and sign conventions of Callin [7], which follow Dodelson [8]. Ψ plays the role of the Newtonian gravitational potential: it governs the motion of non-relativistic particles and the redshift of light climbing out of a potential well. Φ is a perturbation of the spatial curvature, so a changing Φ stretches space locally and redshifts photons in the same way the background expansion does. In the absence of anisotropic stress the two are equal and opposite, $\Psi = -\Phi$.

We write $\kappa \equiv ck/\mathcal{H}$, the wavenumber in units of the comoving Hubble scale. A mode is outside the horizon when $\kappa \ll 1$ and inside when $\kappa \gg 1$; most of the interesting physics happens at horizon entry, $\kappa \approx 1$.

The Boltzmann equation for photons. The photon distribution function is a Planck spectrum with a direction-dependent temperature, $f = [\exp(p/k_B T(1+\Theta)) - 1]^{-1}$, where the fractional temperature perturbation $\Theta(\boldsymbol{x}, \hat{\boldsymbol{p}}, \eta)$ depends on position and on the direction of the photon momentum. In Fourier space Θ depends on k and on $\mu = \hat{\boldsymbol{k}} \cdot \hat{\boldsymbol{p}}$. The linearized Boltzmann equation $\mathrm{d}f/\mathrm{d}\eta = C[f]$ becomes

$$\Theta' + ik\mu\Theta = -\Phi' - ik\mu\Psi - \tau'\Big[\Theta_0 - \Theta + i\mu v_b - \tfrac{1}{2}P_2(\mu)\Pi\Big]. \quad (17)$$

Each term has a clear physical meaning:

- $ik\mu\Theta$ is free streaming. Photons move in direction $\hat{\boldsymbol{p}}$, so an inhomogeneity in space turns into an anisotropy in direction.

- $-\Phi'$ is the redshift caused by the local change of spatial curvature.
- $-i\kappa\mu\Psi$ is the gravitational redshift (or blueshift) of photons that climb out of (or fall into) a potential gradient.
- The last term is Thomson scattering, at rate $-\tau'$ per unit x. Scattering drives the distribution towards isotropy, $\Theta \to \Theta_0$. Electrons moving with velocity v_b Doppler shift the scattered photons, which gives the μv_b term. Thomson scattering is not isotropic but goes as $1+\cos^2$ of the scattering angle, which gives the quadrupole term $P_2\Pi$.

We expand in Legendre polynomials, $\Theta_\ell = \frac{1}{(-i)^\ell}\int_{-1}^{1}\frac{\mathrm{d}\mu}{2}P_\ell(\mu)\Theta(\mu)$, and the low multipoles are the familiar fluid quantities:

- the monopole Θ_0 gives the density, $\delta_\gamma = 4\Theta_0$ since $\rho_\gamma \propto T^4$;
- the dipole gives the velocity, $v_\gamma = -3\Theta_1$;
- the quadrupole Θ_2 is the anisotropic stress.

Multiplying Eq. (17) by P_ℓ and integrating, the $\mu\Theta$ term couples each ℓ to $\ell\pm1$, because μP_ℓ is a combination of $P_{\ell-1}$ and $P_{\ell+1}$. This gives the hierarchy

$$\Theta_0' = -\kappa\Theta_1 - \Phi', \tag{18}$$

$$\Theta_1' = \frac{\kappa}{3}(\Theta_0 - 2\Theta_2 + \Psi) + \tau'\Big(\Theta_1 + \frac{v_b}{3}\Big), \tag{19}$$

$$\Theta_\ell' = \frac{\kappa\big[\ell\Theta_{\ell-1} - (\ell+1)\Theta_{\ell+1}\big]}{2\ell+1} + \tau'\Big(\Theta_\ell - \frac{\Pi}{10}\delta_{\ell2}\Big), \tag{20}$$

where the factors of i have been absorbed into the definition of the velocities. The monopole equation is the continuity equation of the photon fluid, and the dipole equation is its Euler equation. Free streaming moves power from low to high ℓ: an inhomogeneity on scale $1/k$ viewed from a distance d appears as an anisotropy at $\ell \sim kd$. Scattering instead damps all $\ell \geq 2$ multipoles and locks Θ_1 to the baryon velocity.

Polarization. Thomson scattering of radiation with a quadrupole anisotropy produces linearly polarized light: the scattered wave is polarized perpendicular to the plane of scattering, and an incoming quadrupole makes the polarization from different directions not cancel. Its multipoles Θ_ℓ^P obey a similar hierarchy [28], sourced by $\Pi = \Theta_2 + \Theta_0^P + \Theta_2^P$:

$$\Theta_\ell^{P\prime} = \frac{\kappa\big[\ell\Theta_{\ell-1}^P - (\ell+1)\Theta_{\ell+1}^P\big]}{2\ell+1} + \tau'\Big[\Theta_\ell^P - \frac{\Pi}{2}\Big(\delta_{\ell0} + \frac{\delta_{\ell2}}{5}\Big)\Big]. \tag{21}$$

Polarization has no gravitational source: it is created only by scattering of a quadrupole.

Neutrinos. Neutrinos decoupled long before the earliest time we consider, so their Boltzmann equation is Eq. (17) without the collision term. Their multipoles $\mathcal{N}_\ell$ obey the photon hierarchy with $\tau' = 0$. Massless neutrinos therefore free-stream from the start. Their quadrupole $\mathcal{N}_2$ is a large anisotropic stress during the radiation era, when they make up 41% of the radiation density.

Dark matter and baryons. Cold dark matter is collisionless and non-relativistic. The first two moments of its Boltzmann equation (equivalently, energy–momentum conservation) are the continuity and Euler equations,

$$\delta_{\rm CDM}' = \kappa v_{\rm CDM} - 3\Phi', \qquad v_{\rm CDM}' = -v_{\rm CDM} - \kappa\Psi. \tag{22}$$

In the continuity equation κv is the inflow of matter, and $-3\Phi'$ the change of the local volume when the spatial metric is perturbed. In the Euler equation $-v$ is the Hubble drag, which redshifts peculiar momenta as a^{-1}, and $-\kappa\Psi$ is the gravitational force. Baryons obey the same equations plus the momentum they exchange with the photons through Thomson scattering:

$$\delta_b' = \kappa v_b - 3\Phi', \tag{23}$$

$$v_b' = -v_b - \kappa\Psi + \tau' R(3\Theta_1 + v_b). \tag{24}$$

The factor $R = 4\rho_\gamma/3\rho_b$ ensures momentum conservation. The momentum the photons lose (the τ' term in the Θ_1 equation) is gained by the baryons, whose momentum density is smaller by $\rho_b/(\frac{4}{3}\rho_\gamma)$.

Einstein equations. The perturbed Einstein equations relate the potentials to the matter. The 00 component, together with the $0i$ component, gives

$$\Phi' = \Psi - \frac{\kappa^2}{3}\Phi + \frac{H_0^2}{2\mathcal{H}^2}\Big[\frac{\Omega_{\rm CDM}\delta_{\rm CDM} + \Omega_b\delta_b}{a} + \frac{4(\Omega_\gamma\Theta_0 + \Omega_\nu\mathcal{N}_0)}{a^2}\Big], \tag{25}$$

where Ω_i are today's values. Deep inside the horizon ($\kappa \gg 1$) this reduces to the Poisson equation, $k^2\Phi \propto a^2\delta\rho$. The traceless part of the ij components gives

$$\Psi = -\Phi - \frac{12H_0^2}{c^2k^2a^2}\big[\Omega_\gamma\Theta_2 + \Omega_\nu\mathcal{N}_2\big], \tag{26}$$

so $\Psi \neq -\Phi$ only if there is anisotropic stress: the quadrupoles of the photons and neutrinos.

Truncation. We truncate the hierarchies at $\ell_{\rm max}$ with the closure of Ma & Bertschinger [13]. It assumes the free-streaming solution $\Theta_\ell \propto j_\ell(k\eta)$ for the highest multipole, to avoid an artificial reflection of power back down to low ℓ:

$$\Theta_{\ell_{\rm max}}' = \kappa\Theta_{\ell_{\rm max}-1} - \frac{c(\ell_{\rm max}+1)}{\mathcal{H}\eta}\Theta_{\ell_{\rm max}} + \tau'\Theta_{\ell_{\rm max}}. \tag{27}$$

Initial conditions. At $x = -18$ every mode is far outside the horizon ($\kappa \ll 1$) and the Universe is radiation dominated. Inflation produces *adiabatic* fluctuations:

the whole Universe is locally the same, but slightly ahead of or behind in time. Then the number densities of all species are perturbed equally, $\delta_{\rm CDM} = \delta_b = \frac{3}{4}\delta_\gamma = \frac{3}{4}\delta_\nu$. Solving the equations to lowest order in κ gives $\Phi' = 0$ and $\Theta_0 = -\Psi/2$. The neutrino quadrupole in Eq. (26) gives $\Phi = -(1 + 2f_\nu/5)\Psi$, with $f_\nu = \Omega_\nu/(\Omega_\gamma + \Omega_\nu)$. We normalize to a primordial comoving curvature perturbation $\mathcal{R} = 1$, which is conserved outside the horizon and is the quantity inflation predicts. Then

$$\begin{aligned}
\Psi &= -\frac{1}{\frac{3}{2} + \frac{2}{5}f_\nu}, \quad \Phi = -\Big(1 + \frac{2f_\nu}{5}\Big)\Psi, \\
\delta_{\rm CDM} &= \delta_b = -\frac{3}{2}\Psi, \quad v_{\rm CDM} = v_b = -\frac{\kappa}{2}\Psi, \\
\Theta_0 &= \mathcal{N}_0 = -\frac{\Psi}{2}, \quad \Theta_1 = \mathcal{N}_1 = \frac{\kappa}{6}\Psi, \\
\mathcal{N}_2 &= -\frac{(cka)^2\Phi}{12H_0^2\Omega_\nu}\Big(1 + \frac{5}{2f_\nu}\Big)^{-1},
\end{aligned} \tag{28}$$

and $\mathcal{N}_\ell = \kappa\mathcal{N}_{\ell-1}/(2\ell{+}1)$ for $\ell \geq 3$. The actual amplitude of each mode is restored in Milestone IV through the primordial power spectrum $\mathcal{P}_\mathcal{R}(k)$.

Tight coupling and acoustic oscillations. At early times $|\tau'| \gg 1$: a photon scatters thousands of times per Hubble time. The scattering terms force $\Theta_1 \simeq -v_b/3$, and the photon multipoles with $\ell \geq 2$ are suppressed by powers of κ/τ'. The photons and baryons then behave as one fluid. Combining the Θ_0, Θ_1 and v_b equations at leading order in $1/\tau'$ gives, in conformal time,

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}\eta}\Big[(1 + R_b)\dot\Theta_0\Big] + \frac{c^2k^2}{3}\Theta_0 = -\frac{c^2k^2}{3}(1 + R_b)\Psi \\
- \frac{\mathrm{d}}{\mathrm{d}\eta}\Big[(1 + R_b)\dot\Phi\Big],
\end{aligned} \tag{29}$$

with $R_b = 1/R = 3\rho_b/4\rho_\gamma$. This is a driven harmonic oscillator. Photon pressure provides the restoring force, baryons add inertia $(1 + R_b)$ and lower the sound speed to $c_s = c/\sqrt{3(1 + R_b)}$, and gravity drives it. For slowly varying potentials and R_b the adiabatic solution is

$$\Theta_0 + (1 + R_b)\Psi \simeq A\cos(kr_s), \tag{30}$$

where r_s is the sound horizon of Milestone II. These are the acoustic oscillations of the photon–baryon fluid. Two features are important for the CMB:

- The observed temperature includes the gravitational redshift, $\Theta_0 + \Psi = A\cos(kr_s) - R_b\Psi$. Baryons shift the zero point of the oscillation, so compressions in potential wells are enhanced relative to rarefactions.

- When a mode enters the horizon in RD, the potentials decay just as the fluid is first compressed. The decay removes the gravitational pull that would otherwise slow the rebound, which boosts the oscillation amplitude ("radiation driving").

At next order in $1/\tau'$ photons diffuse over their mean free path and smooth out the oscillations (Silk damping): $\Theta_0 \propto e^{-k^2/k_D^2}$ with [27]

$$k_D^{-2} = \int_0^\eta \frac{c\,\mathrm{d}\eta'}{6(1 + R_b)\,n_e\sigma_T a}\Big[\frac{R_b^2}{1 + R_b} + \frac{16}{15}\Big]. \tag{31}$$

This is a random walk: the diffusion length is the geometric mean of the mean free path and the horizon.

Numerically, the tight-coupling regime is too stiff to integrate the full hierarchy. We evolve only $\delta_{\rm CDM}, v_{\rm CDM}, \delta_b, v_b, \Phi, \Theta_0, \Theta_1$ (and the neutrinos), and set

$$\Theta_2 = -\frac{8}{15}\frac{\kappa}{\tau'}\Theta_1, \quad \Theta_0^P = \frac{5}{4}\Theta_2, \quad \Theta_2^P = \frac{1}{4}\Theta_2, \tag{32}$$

with $\Theta_\ell = -\frac{\ell}{2\ell+1}\frac{\kappa}{\tau'}\Theta_{\ell-1}$ for $\ell \geq 3$ (and the same for Θ^P). These are the quasi-static solutions of the hierarchy when the τ' terms dominate. Without polarization, 8/15 becomes 20/45. Polarization feeds back on Θ_2 through Π, which raises the quadrupole by a factor 6/5. The baryon velocity and Θ_1 follow from the second-order expansion of Ref. [7], which eliminates the large τ' terms in favour of the slip $q \equiv 3\Theta_1' + v_b'$:

$$\begin{aligned}
q = \Big\{ &-\big[(1 - R)\tau' + (1 + R)\tau''\big](3\Theta_1 + v_b) - \kappa\Psi \\
&+ \Big(1 - \frac{\mathcal{H}'}{\mathcal{H}}\Big)\kappa(-\Theta_0 + 2\Theta_2) - \kappa\Theta_0'\Big\} \\
&\Big/\Big[(1 + R)\tau' + \frac{\mathcal{H}'}{\mathcal{H}} - 1\Big], \\
v_b' = &\frac{-v_b - \kappa\Psi + R\big[q + \kappa(-\Theta_0 + 2\Theta_2) - \kappa\Psi\big]}{1 + R}, \\
\Theta_1' = &\frac{1}{3}(q - v_b').
\end{aligned} \tag{33}$$

Tight coupling ends when $|\tau'| < 10$, when $|\kappa/\tau'| > 0.1$, or at $z = 1700$ (before recombination starts), whichever comes first.

Analytic limits. A few exact results guide the interpretation and serve as tests:

- Outside the horizon Φ is constant within each era. Through radiation–matter equality it drops by the factor 9/10 (without neutrinos). With $\mathcal{R} = 1$, $\Phi = 3/5$ in MD. The full superhorizon solution through equality is known analytically [14].

- Inside the horizon in RD, the potential obeys $\Phi = 3\Phi_0(\sin y - y\cos y)/y^3$ with $y = k\eta/\sqrt{3}$. It oscillates and decays as a^{-2}, because radiation pressure prevents the density contrast from growing while the background density falls as a^{-4}.

- In MD Φ is constant on all scales: $\delta \propto a$ and $\rho_M a^2 \propto a^{-1}$ cancel in the Poisson equation. In the Λ era $\Phi \propto D(a)/a$ decays, where D is the linear growth factor.

- Sub-horizon dark matter obeys $\ddot{\delta}+2H\dot{\delta} = 4\pi G\rho_M\delta$ (ignoring the radiation perturbations). In RD the source is negligible and $\delta \propto \ln a$ (the Mészáros effect). In MD $\delta \propto a$. In general the growth rate is $f = \mathrm{d}\ln\delta/\mathrm{d}\ln a \approx \Omega_M(x)^{0.55}$.

Source functions. After the evolution we build the temperature and E-mode source functions:

$$S_T(k,x) = \tilde{g}\Big[\Theta_0 + \Psi + \frac{\Pi}{4}\Big] + e^{-\tau}(\Psi' - \Phi') - \frac{1}{ck}\frac{\mathrm{d}}{\mathrm{d}x}(\mathcal{H}\tilde{g}v_b) + \frac{3}{4c^2k^2}\frac{\mathrm{d}}{\mathrm{d}x}\Big[\mathcal{H}\frac{\mathrm{d}}{\mathrm{d}x}(\mathcal{H}\tilde{g}\Pi)\Big], \tag{34}$$

$$S_E(k,x) = \frac{3\tilde{g}\,\Pi}{4k^2(\eta_0-\eta)^2}. \tag{35}$$

They come from integrating Eq. (17) along the line of sight (Sec. V); the derivatives arise from integrating the μ-dependent terms by parts. They express the temperature we see as a sum over all places where a photon last scattered (weighted by $\tilde{g}$) or was redshifted on its way (weighted by $e^{-\tau}$). The four terms are, in order:

- the Sachs–Wolfe (SW) term: the intrinsic temperature at last scattering plus the gravitational redshift out of the potential well there;
- the integrated Sachs–Wolfe (ISW) term: the net energy change of a photon crossing a time-varying potential;
- the Doppler term: from the velocity of the electrons that last scattered the photon;
- a small correction from the anisotropic, polarization-dependent nature of Thomson scattering.

B. Implementation and tests

`Perturbations` solves 400 modes, logarithmically spaced in $k \in [5\times10^{-5}, 0.3]\,\mathrm{Mpc}^{-1}$, on a non-uniform grid of 2100 points in $x \in [-18, 0]$ that is dense around recombination. The modes run in parallel with OpenMP. We keep $\ell_{\max} = 8$ for Θ_ℓ and Θ_ℓ^P, and $\ell_{\max,\nu} = 22$ for the neutrinos; going to $\ell_{\max,\nu} = 30$ changes the spectra by less than 5×10^{-4}. Only the low photon multipoles enter the sources, so a small $\ell_{\max}$ is enough. The tight-coupling system is solved with an explicit Runge–Kutta method (RK8PD). The full system is stiff, because $|\tau'|$ is still large when tight coupling ends, so we use the implicit BDF method of GSL with a finite-difference Jacobian. The relative tolerance is 10^{-8}. At the switch we initialize the full hierarchy from the tight-coupling expressions above. The derivatives in S_T come from splines in x, and all quantities are stored as 2D bicubic splines in (x, k).

The tests (Fig. 9 and the unit tests) check the analytic limits listed above:

- On superhorizon scales the solution matches the exact Kodama–Sasaki solution through equality, and $\Phi \to 3/5$ in MD.
- For modes that enter in RD, Φ follows the analytic radiation-era solution until equality approaches (middle panel).
- Before decoupling, $\Theta_0 + \Psi$ oscillates as $\cos(kr_s)$, with the phase given by the sound horizon of Milestone II (left panel). This tests the sound speed, and with it R.
- In matter and Λ domination the growth rate $\mathrm{d}\ln\delta_{\mathrm{CDM}}/\mathrm{d}x$ of small-scale modes follows $\Omega_M(x)^{0.55}$ (right panel).

In addition, the Poisson equation holds on subhorizon scales today, the tight-coupling and full solutions are continuous at the switch, and the neutrino multipoles free-stream as expected. All perturbations agree with CLASS to $\sim 0.1\%$.

C. Results

We show four modes, chosen to probe the different regimes:

- $k = 0.001\,\mathrm{Mpc}^{-1}$ enters the horizon ($\kappa = 1$) at $x = -4.13$ ($z \approx 61$), well into MD. It is outside the horizon at recombination and probes the largest scales in the CMB.
- $k = 0.01\,\mathrm{Mpc}^{-1}$ enters at $x = -8.08$, close to equality.
- $k = 0.1\,\mathrm{Mpc}^{-1}$ enters at $x = -10.71$ and $k = 0.3\,\mathrm{Mpc}^{-1}$ at $x = -11.83$, both deep in RD. They oscillate many times before recombination.

Tight coupling ends at $x = -7.44$ ($z = 1700$) for the first three modes, and at $x = -7.89$ for $k = 0.3\,\mathrm{Mpc}^{-1}$, where κ/τ' becomes large first. Horizon entry is marked by coloured dotted lines in the figures, and grey lines mark radiation–matter equality and decoupling.

Dark matter. Figure 10 shows the density contrasts. Outside the horizon all δ's are nearly constant at $\delta = -\frac{3}{2}\Psi$. There is no causal mechanism to change them, and in our gauge the superhorizon density contrast only changes through the changing potential at equality. After horizon entry the behaviour of δ_{CDM} depends on when the mode entered:

- Modes that enter in RD ($k = 0.1,\ 0.3\,\mathrm{Mpc}^{-1}$) receive a kick from the decaying potential at horizon entry. After that they grow only slowly, because the dark matter's self-gravity is negligible compared with the expansion driven by the smooth radiation. The dark matter coasts on the velocity it got at entry, and δ_{CDM} grows logarithmically (the Mészáros effect).

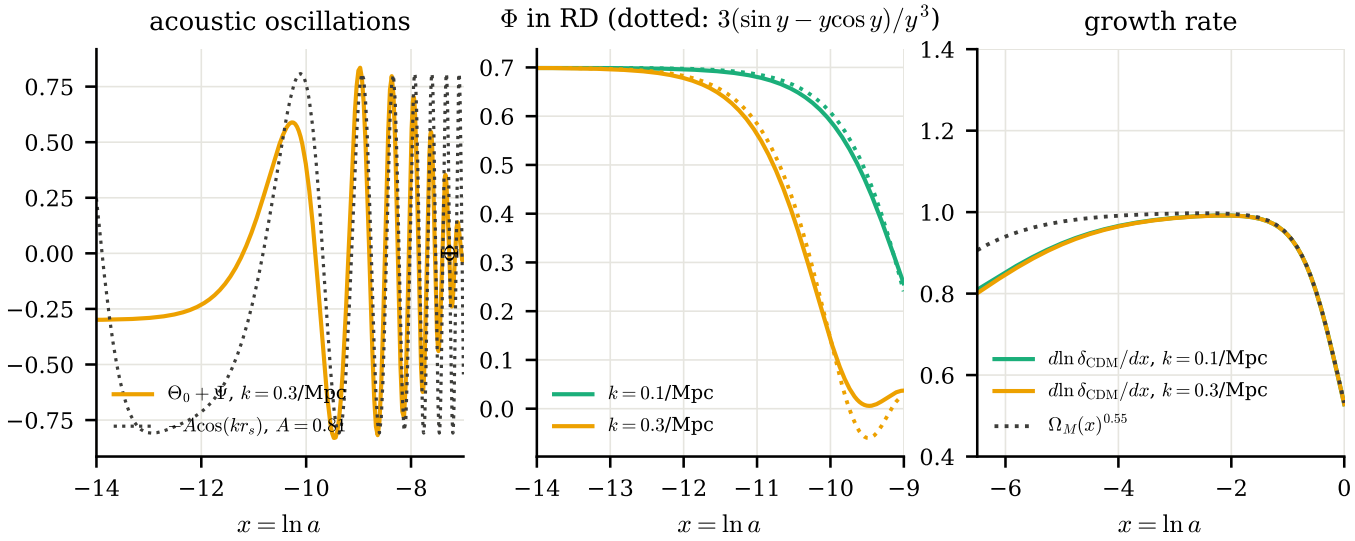


FIG. 9. Tests of the perturbations. *Left:* $\Theta_0 + \Psi$ for $k = 0.3\,\mathrm{Mpc}^{-1}$ during the radiation era compared with $-A\cos(kr_s)$, using the sound horizon r_s from Milestone II. After horizon entry the decaying potential drives the oscillator, so the amplitude A is fitted; the test is the phase. *Middle:* Φ for modes that enter in RD, compared with the analytic radiation-era solution (dotted). *Right:* the growth rate of $\delta_{\rm CDM}$ compared with $\Omega_M(x)^{0.55}$.

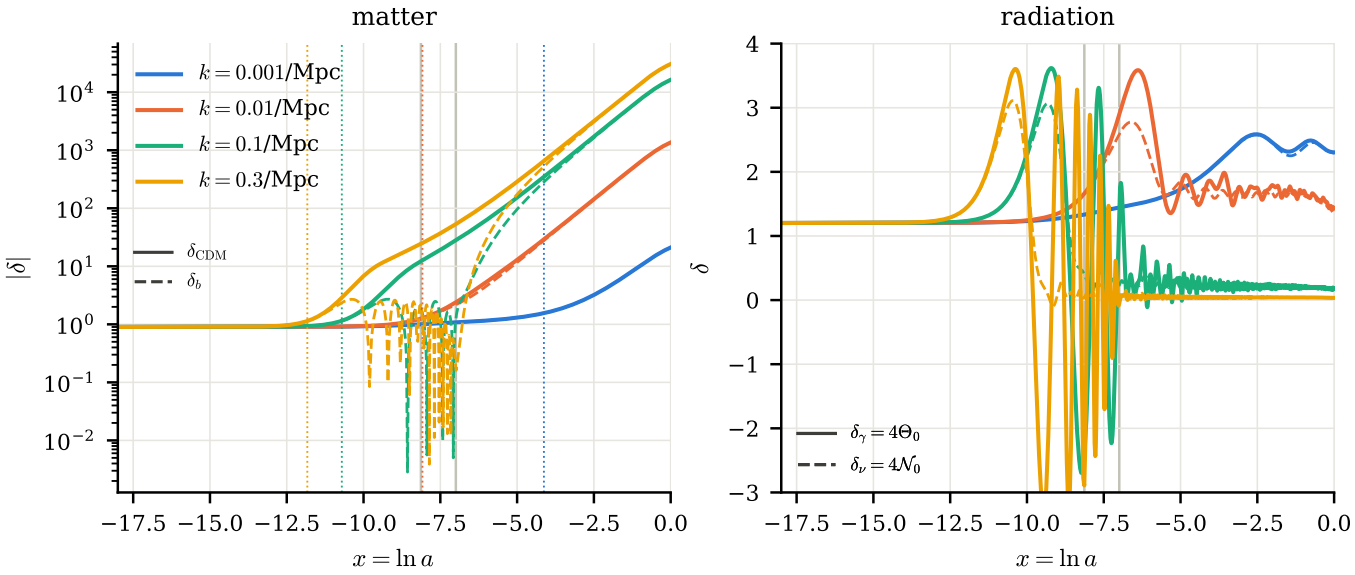


FIG. 10. Density perturbations. *Left:* $|\delta_{\rm CDM}|$ (solid) and $|\delta_b|$ (dashed). The coloured dotted lines mark horizon entry of each mode. *Right:* photon $\delta_\gamma = 4\Theta_0$ (solid) and neutrino $\delta_\nu = 4\mathcal{N}_0$ (dashed). Grey lines mark radiation–matter equality and decoupling ($\tau = 1$).

- After equality, matter dominates gravity and all modes grow as $\delta_{\rm CDM} \propto a$.
- In the Λ era growth slows down (Fig. 9, right). For $k = 0.1\,\mathrm{Mpc}^{-1}$, $\delta_{\rm CDM}$ grows by a factor 107 from $x = -5$ to today, instead of $e^5 = 148$ for $\delta \propto a$.
- The mode $k = 0.001\,\mathrm{Mpc}^{-1}$ starts to grow as it approaches horizon entry in MD and has not reached the $\delta \propto a$ regime until late.

Small scales enter earlier and spend more time inside the horizon, but the time they spend in RD is lost to the Mészáros stagnation. This suppression of modes with $k > k_{\rm eq}$ is the origin of the turnover in $P(k)$ (Sec. V).

Baryons. As long as they are tightly coupled to the photons, the baryons share the photon acoustic oscillations, and adiabaticity gives $\delta_b \simeq \frac{3}{4}\delta_\gamma$. Their density contrast is stuck at the photon level, while $\delta_{\rm CDM}$ already grows. After decoupling the baryons lose the photon pressure and fall into the potential wells that

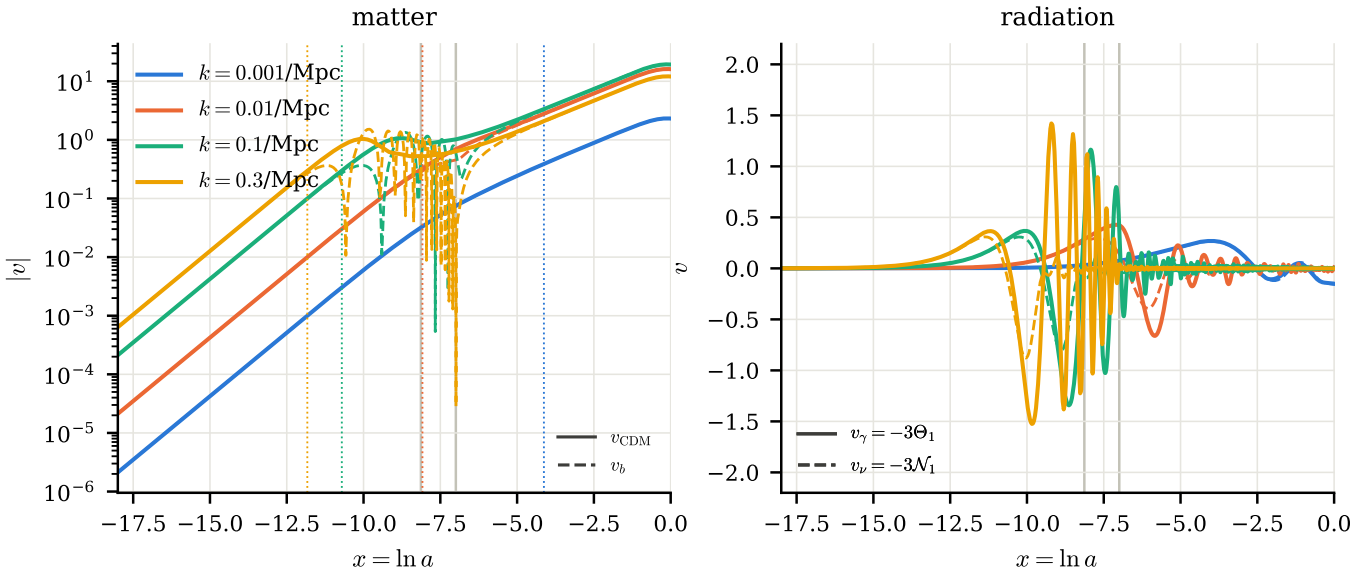


FIG. 11. Velocities. *Left:* $|v_{\rm CDM}|$ (solid) and $|v_b|$ (dashed). *Right:* photon $v_\gamma = -3\Theta_1$ (solid) and neutrino $v_\nu = -3\mathcal{N}_1$ (dashed).

the dark matter has built. In MD the solution is $\delta_b \simeq \delta_{\rm CDM}(1 - a_d/a)$, so within a few e-folds δ_b catches up. For $k = 0.1\,{\rm Mpc}^{-1}$, $\delta_b/\delta_{\rm CDM}$ goes from 0.04 at $x = -7$ to 0.29 at $x = -6$ and 0.93 at $x = -3$; today they are equal to within 0.5%. The remaining difference is the acoustic imprint, the BAO. Without dark matter the baryon perturbations would have had to grow from their value at decoupling, $\delta_b \sim 10^{-5}\times$ (a few), by only the factor $\sim 10^3$ available since recombination. That is not enough to form galaxies. Dark matter is needed to give structure a head start.

Photons and neutrinos. The photons (Fig. 10, right) oscillate after horizon entry: these are the acoustic oscillations of Eq. (30). The amplitude grows at horizon entry because of radiation driving, and the oscillation is offset from zero by the baryon loading term $-R_b\Psi$. Diffusion then damps the oscillation. At decoupling $k_D \approx 0.14\,{\rm Mpc}^{-1}$ (Eq. 31), so the $k = 0.3\,{\rm Mpc}^{-1}$ mode has lost most of its amplitude: $|\Theta_0 + \Psi|$ drops from 0.84 at $x \approx -9$ to 0.2 at decoupling. The $k = 0.1\,{\rm Mpc}^{-1}$ mode is only mildly damped. After decoupling the photons free-stream, Θ_0 decays, and the power moves to higher multipoles; at late times $\Theta_0 \approx -\Psi$, so that the observed $\Theta_0 + \Psi$ averages to zero inside the horizon.

The neutrinos have been free-streaming all along. They have no pressure that can restore an acoustic oscillation, so δ_ν oscillates with decaying amplitude like a Bessel function as soon as the mode enters the horizon.

Velocities. The velocities (Fig. 11) tell the same story. Outside the horizon $v \propto \kappa$ grows as the mode approaches the horizon. In MD the continuity equation gives $v = \delta'/\kappa \propto a/a^{1/2} = a^{1/2}$: $v_{\rm CDM}$ grows by a factor 2.65 between $x = -3$ and $x = -1$ for $k = 0.1\,{\rm Mpc}^{-1}$, close to $e^1 = 2.72$. Growth slows in the Λ era. v_b oscillates with the photons until decoupling (as Θ_1, out of phase with Θ_0) and then quickly joins $v_{\rm CDM}$. The photon velocity $-3\Theta_1$ oscillates in quadrature with the density, which is the origin of the Doppler term in the CMB.

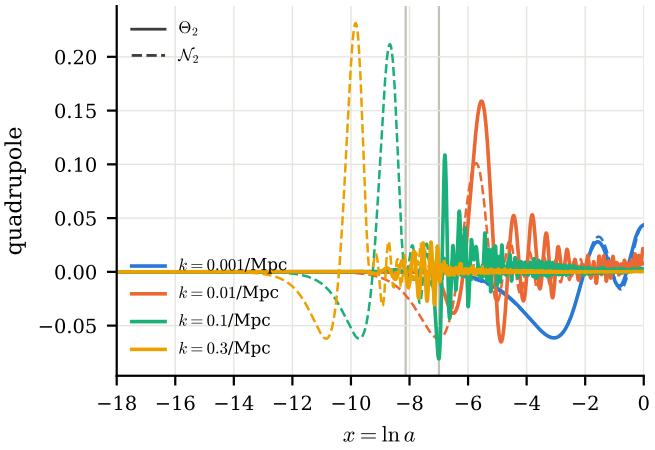


FIG. 12. Photon quadrupole Θ_2 (solid) and neutrino quadrupole $\mathcal{N}_2$ (dashed).

Quadrupoles. The photon quadrupole (Fig. 12) is suppressed by scattering until recombination, $\Theta_2 \propto (\kappa/\tau')\Theta_1$: every scattering erases the anisotropy. As $|\tau'|$ falls during recombination it grows rapidly, and after decoupling it reaches about a quarter of Θ_0 as the photons free-stream. The neutrino quadrupole is present from the start, since nothing isotropizes the neutrinos.

Potentials. Figure 13 shows the potentials.

- Φ is constant outside the horizon. The neutrino anisotropic stress makes $\Phi + \Psi \neq 0$: in RD $\Phi/\Psi = -(1 + 2f_\nu/5)$, a 16% effect. As matter takes over, the neutrinos' share of the energy, and with it $\Phi+\Psi$,

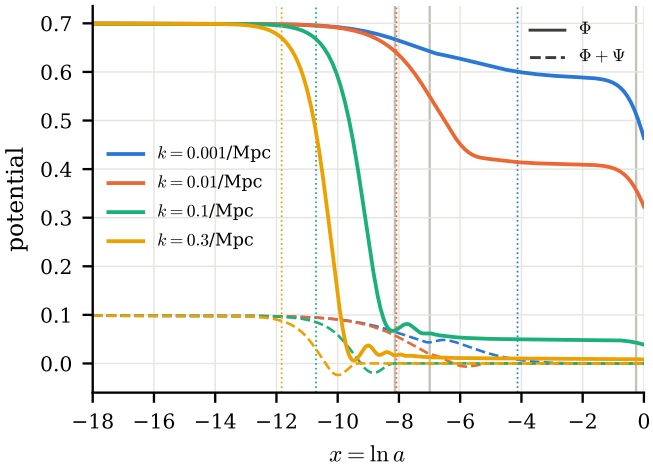


FIG. 13. Potentials Φ (solid) and $\Phi + \Psi$ (dashed). $\Phi + \Psi \neq 0$ is the anisotropic stress of the neutrinos, which is significant only during RD.

falls; it also decays once a mode is inside the horizon and the neutrinos have dispersed.

- For modes that enter in RD, Φ decays and oscillates after horizon entry, as in the analytic solution. Radiation pressure keeps the photon density perturbation from growing while the background radiation density falls as a^{-4}, so the potential it sources decays as a^{-2}.
- Once matter dominates, Φ is constant on all scales: the growing matter perturbation exactly compensates the dilution of the background. This is why Φ for $k = 0.1$ and $0.3\,\mathrm{Mpc}^{-1}$ freezes at a small value after equality.
- For $k = 0.001\,\mathrm{Mpc}^{-1}$ we see the superhorizon transition from 0.699 in RD to 0.586 at $x = -1.5$, close to the MD value 3/5; the ratio 0.84 differs from 9/10 because of the neutrinos.
- When Λ takes over, growth slows down and all potentials decay, by about 20% today. This late decay is the source of the late ISW effect in the CMB.

Polarization. The polarization multipoles (Fig. 14) are generated only by Thomson scattering of a quadrupole. Polarization therefore needs two things at once: scattering, and a quadrupole. Before recombination the scattering is so efficient that the quadrupole is tiny, $\Theta_2 \sim (\kappa/\tau')\Theta_1$. After decoupling the quadrupole is large, but there is no more scattering. Polarization is produced only during the short interval when the photons are starting to free-stream but still scattering occasionally, i.e. during the width of the visibility function. Since $\Theta_2 \propto \kappa\Theta_1$, the polarization traces the photon *velocity*, not the density, at last scattering. As a result:

- the E-mode signal is about a factor 10 weaker than the temperature signal;
- it is very sensitive to the duration of recombination;
- its acoustic peaks are out of phase with those of the temperature (Sec. V).

After decoupling the multipoles free-stream and decay. Reionization gives a small second burst at $x \approx -2$, which is responsible for the low-ℓ E-mode bump.

V. MILESTONE IV: THE CMB AND MATTER POWER SPECTRA

The last milestone turns the source functions of Milestone III into observables: the angular power spectra of the CMB temperature and polarization, and the matter power spectrum today. We also interpret the shapes of these spectra in terms of the physics of the earlier milestones.

A. Theory

Statistics of the CMB sky. The temperature fluctuation in direction $\hat{\boldsymbol{n}}$ is expanded in spherical harmonics, $\Theta(\hat{\boldsymbol{n}}) = \sum_{\ell m} a_{\ell m} Y_{\ell m}(\hat{\boldsymbol{n}})$. Multipole ℓ corresponds to an angular scale $\theta \approx \pi/\ell$. The primordial fluctuations are a statistically homogeneous and isotropic random field, so the $a_{\ell m}$ are uncorrelated and their variance depends only on ℓ: $\langle a_{\ell m} a^*_{\ell' m'} \rangle = \delta_{\ell\ell'}\delta_{mm'} C_\ell$. All the information in a Gaussian field is in the power spectrum C_ℓ. Since only $2\ell + 1$ values of m are available for each ℓ, C_ℓ can only be measured with a relative precision of $\sqrt{2/(2\ell+1)}$ (cosmic variance), which dominates the error bars at low ℓ.

The primordial curvature perturbation has the power spectrum $\langle \mathcal{R}(\boldsymbol{k})\mathcal{R}^*(\boldsymbol{k}')\rangle = (2\pi)^3\delta^3(\boldsymbol{k}-\boldsymbol{k}')\,2\pi^2\mathcal{P}_\mathcal{R}(k)/k^3$, with $\mathcal{P}_\mathcal{R} = A_s(k/k_{\mathrm{pivot}})^{n_s-1}$. Inflation predicts a nearly scale-invariant spectrum, $n_s \simeq 1$, in which every logarithmic interval of k contributes equally to the curvature fluctuations. Because the evolution is linear and every mode in Milestone III was normalized to $\mathcal{R} = 1$, every observable is a primordial amplitude times a transfer function.

The line-of-sight integral. A direct solution of the photon hierarchy up to $\ell \sim 2500$ would need thousands of coupled equations per mode. Seljak and Zaldarriaga [1] noticed that Eq. (17) is a linear first-order equation in Θ that can be integrated formally. In conformal time it has the form $\dot\Theta + (ik\mu - \dot\tau)\Theta = \tilde S$, where $\tilde S$ collects all terms that do not contain Θ itself. With the integrating factor $e^{ik\mu\eta - \tau}$ and $\tau(\eta_0) = 0$,

$$\Theta(k,\mu,\eta_0) = \int_0^{\eta_0} \tilde S(k,\mu,\eta)\, e^{-\tau(\eta)}\, e^{-ik\mu(\eta_0-\eta)}\, \mathrm{d}\eta. \quad (36)$$

The temperature today is a sum of contributions emitted along the photon path, each attenuated by the probability $e^{-\tau}$ that the photon is not scattered again. The

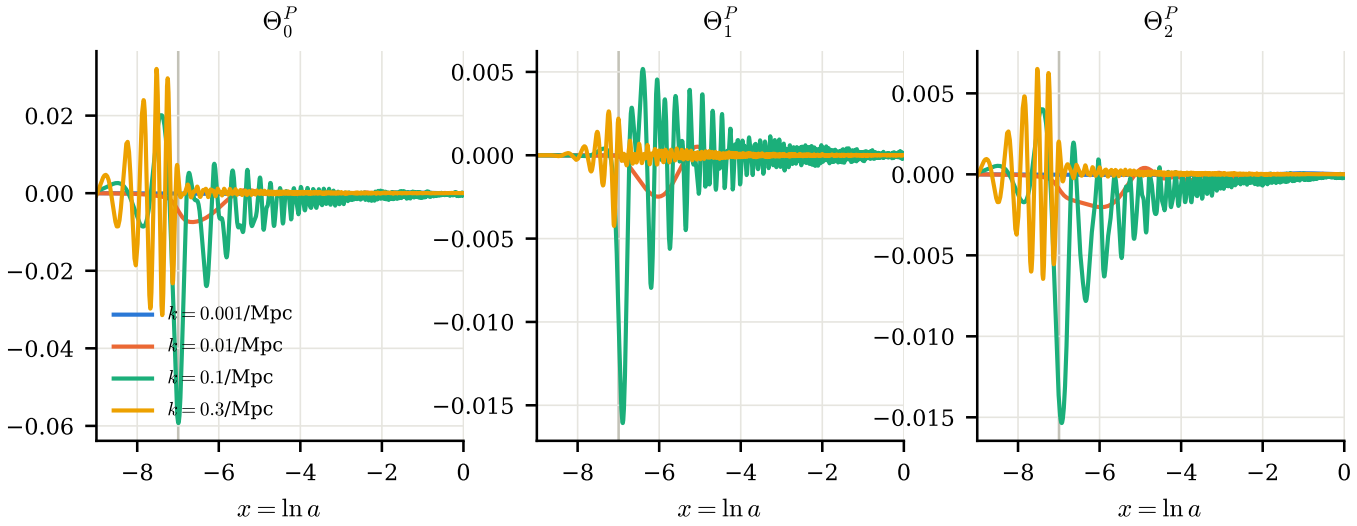


FIG. 14. Polarization multipoles Θ_0^P, Θ_1^P and Θ_2^P from shortly before decoupling (grey line) until today.

phase factor is a plane wave, which the Rayleigh formula $e^{-ik\mu\chi} = \sum_\ell (-i)^\ell (2\ell+1) j_\ell(k\chi) P_\ell(\mu)$ expands into spherical Bessel functions. After integrating the μ-dependent terms of the source by parts we obtain

$$\Theta_\ell(k) = \int_{-\infty}^{0} S_T(k,x)\, j_\ell[k(\eta_0-\eta)]\,\mathrm{d}x, \tag{37}$$

$$\Theta_\ell^E(k) = \sqrt{\frac{(\ell+2)!}{(\ell-2)!}} \int_{-\infty}^{0} S_E(k,x)\, j_\ell[k(\eta_0-\eta)]\,\mathrm{d}x, \tag{38}$$

with the source functions of Eqs. (34) and (35). This separates the problem into physics (the source S, which only needs the low multipoles and is smooth in k) and geometry (the Bessel function, which projects a plane wave at distance $\eta_0-\eta$ onto multipole ℓ). The function $j_\ell(z)$ peaks near $z\approx\ell$, so a perturbation of wavenumber k on the last-scattering surface at distance $\chi_* = \eta_0-\eta_*$ is seen mainly at $\ell \approx k\chi_*$. The angular power spectra are

$$C_\ell^{XY} = 4\pi \int_0^\infty \frac{\mathrm{d}k}{k}\, \mathcal{P}_\mathcal{R}(k)\, \Theta_\ell^X(k)\, \Theta_\ell^Y(k), \tag{39}$$

with $X, Y \in \{T, E\}$. We plot $D_\ell = \ell(\ell+1)C_\ell/2\pi$ in $\mu\mathrm{K}^2$, multiplying by $(10^6 T_\mathrm{CMB})^2$; D_ℓ is the variance of the temperature per logarithmic interval in ℓ.

The Sachs–Wolfe plateau. On scales larger than the sound horizon at decoupling the photons have not oscillated, and only the SW term contributes, $\Theta_\ell \approx (\Theta_0+\Psi)(k,\eta_*)\, j_\ell(k\chi_*)$. In MD the adiabatic initial conditions give $\Theta_0 = -\frac{2}{3}\Psi$, so the observed temperature is $\Theta_0+\Psi = \Psi/3$: the photons from a potential well start out hotter, but lose more energy climbing out. With $\Psi = -\frac{3}{5}\mathcal{R}$, $\Theta_0+\Psi = -\mathcal{R}/5$. Using $\int_0^\infty j_\ell^2(z)\,\mathrm{d}z/z = 1/[2\ell(\ell+1)]$ we find, for $n_s = 1$,

$$D_\ell^\mathrm{SW} = \frac{A_s}{25} T_\mathrm{CMB}^2 \approx 620\,\mu\mathrm{K}^2, \tag{40}$$

independent of ℓ. A scale-invariant primordial spectrum gives a flat D_ℓ on large scales.

Acoustic peaks. On smaller scales the SW term is the acoustic oscillation of Eq. (30), frozen at decoupling, $\Theta_0+\Psi \propto \cos(kr_{s*})$. Modes that are at an extremum at decoupling, $k_n r_{s*} = n\pi$, give peaks in the spectrum. They project to

$$\ell_n \approx n\,\ell_A - \phi_n, \qquad \ell_A = \frac{\pi\chi_*}{r_{s*}}, \tag{41}$$

where ϕ_n is a phase shift caused by the driving potentials and by the projection. Odd peaks ($n = 1, 3, \dots$) are compressions in potential wells and even peaks are rarefactions. The Doppler term $\propto v_b \propto \sin(kr_s)$ is out of phase with the density and fills in the troughs between the peaks. On small scales diffusion damping suppresses the power by $\sim e^{-2(\ell/\ell_D)^2}$, with $\ell_D = k_D\chi_*$.

The matter power spectrum. The matter power spectrum follows from Φ through the Poisson equation. We use the comoving-gauge density contrast, which is the gauge-invariant quantity that reduces to the Newtonian density contrast inside the horizon (in the code we evaluate it directly as $\sum_i \rho_i[\delta_i - 3\mathcal{H}v_i/(ck)]/\sum_i \rho_i$ for CDM and baryons, which is the same up to radiation terms of order 10^{-3} on the largest scales; Sec. VIII),

$$\Delta_M(k,x) = \frac{2}{3}\frac{c^2k^2}{\mathcal{H}^2}\frac{\Phi(k,x)}{\Omega_M(x)}, \tag{42}$$

and $P(k,x) = |\Delta_M(k,x)|^2\, 2\pi^2\mathcal{P}_\mathcal{R}(k)/k^3$. Its shape follows from Milestone III:

- For $k \ll k_\mathrm{eq}$ the mode entered during MD, Φ is constant and the same for all such k. Then $\Delta_M \propto k^2$ and $P \propto k^4\mathcal{P}_\mathcal{R}/k^3 \propto k^{n_s}$.
- For $k \gg k_\mathrm{eq}$ the potential decayed in RD by $\sim (k_\mathrm{eq}/k)^2$, partly compensated by the logarithmic Mészáros growth. This gives $P \propto k^{n_s-4}\ln^2(k/k_\mathrm{eq})$.

The spectrum therefore peaks near $k_{\rm eq}$. Because $k_{\rm eq} \propto \Omega_{M0}h^2$ in units of Mpc^{-1}, the peak position measures the matter density.

B. Implementation and tests

`PowerSpectrum` first tabulates $j_\ell(z)$ for all needed ℓ on a uniform grid with 32 points per 2π, and $j_\ell(z)/z^2$ for the E-mode source of Eq. (35). Uniform grids allow $O(1)$ spline lookups. We evaluate Eq. (37) by the trapezoidal rule on an x-grid that starts at $x = -8.5$, when $e^{-\tau}$ makes the source negligible. The grid has spacing $\Delta x = 0.003$ around recombination, to resolve $\tilde{g}$, and is uniform in η elsewhere, with five points per Bessel oscillation at $k_{\rm max}$. The k-integral uses four points per period $2\pi/\eta_0$ of the transfer functions (and $\Delta \ln k \le 0.02$ at small k), up to $k_{\rm max} = 0.3\,\mathrm{Mpc}^{-1}$, i.e. $k\eta_0 \approx 4300$. We compute C_ℓ at 97 multipoles between 2 and 2500 and spline in ℓ. The full run (background, recombination, perturbations and spectra) takes about 60 s on two cores.

Tests include:

- The SW term alone gives the plateau of Eq. (40) at low ℓ, corrected for the effects discussed below.
- The peak spacing agrees with ℓ_A from the sound horizon and χ_*.
- $C_\ell^{EE} \ge 0$ and $|C_\ell^{TE}| \le \sqrt{C_\ell^{TT} C_\ell^{EE}}$.
- Doubling the LOS and k sampling changes the spectra by less than 5×10^{-5}.
- $P(k)$ has slope n_s at small k and turns over at $k_{\rm eq}$.

C. Results

Transfer functions. Figure 15 shows the projection at work. $\Theta_\ell(k)$ is zero for $k\eta_0 < \ell$, where j_ℓ vanishes: a wave cannot produce structure on angles smaller than its own projected wavelength. It peaks at $k\eta_0 \simeq \ell$ (more precisely $k\chi_* \simeq \ell$), where the wave vector is perpendicular to the line of sight. It then oscillates with a slowly decaying tail from waves whose wave vector is inclined to the line of sight; their crests are farther apart on the sky, so they project onto larger angles. Each multipole therefore receives contributions from a range of k, which smears the acoustic features: the peaks in C_ℓ are much broader than the oscillations of $\Theta_0(k)$ at recombination. The integrand of C_ℓ (bottom panel) is dominated by the first peak at $k\chi_* \approx \ell$. Its height follows the acoustic oscillations: $\ell = 200$ sits on the first peak and has a much larger integrand than $\ell = 100$, while $\ell = 1000$ is suppressed by damping.

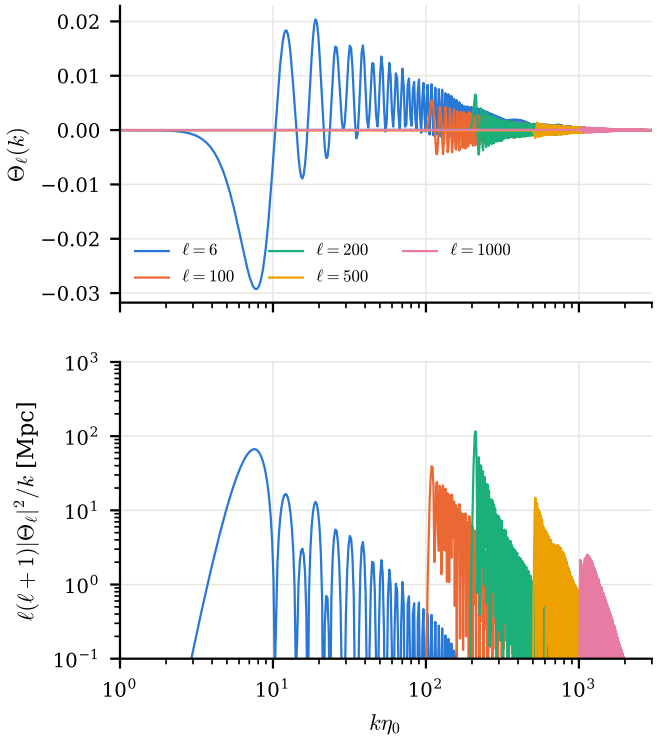


FIG. 15. Photon transfer functions $\Theta_\ell(k)$ (*top*) and the integrand $\ell(\ell+1)\Theta_\ell^2/k$ of C_ℓ (*bottom*) for $\ell = 6$, 100, 200, 500 and 1000, as functions of $k\eta_0$.

The temperature spectrum. Figure 16 compares our TT spectrum with Planck. The model reproduces the Sachs–Wolfe plateau, the positions of the seven acoustic peaks visible in the data, and the damping tail. The main features have the following physical explanation:

- *Plateau.* At $\ell = 10$, $D_\ell = 815\,\mu\mathrm{K}^2$, higher than the estimate $620\,\mu\mathrm{K}^2$ of Eq. (40). Two effects account for most of the difference. First, at decoupling radiation still makes up 24% of the density, so the largest modes have $\Theta_0 + \Psi = -0.236$ instead of the MD value -0.2. Second, the red tilt $n_s < 1$ gives more power at $k \sim \ell/\chi_* \ll k_{\rm pivot}$, a factor 1.16 at $\ell = 10$. Reionization lowers the plateau only marginally, since the scales it affects are smaller.
- *Peak positions.* The first peak is at $\ell = 221$ and the next ones at $\ell = 538$, 817, 1133, 1432, 1745 and 2046. Their spacing, ≈ 300, agrees with $\ell_A = \pi\chi_*/r_{s*} = 303$ for $\chi_* = 13\,911\,\mathrm{Mpc}$ and $r_{s*} = 144.4\,\mathrm{Mpc}$. The angular size of the sound horizon at last scattering is thus about 0.6°. The first peak lies at $\approx 0.73\,\ell_A$ because of the phase shift from the decaying potentials. The peak positions measure the angle r_{s*}/χ_*: in a curved universe the same ruler would appear at a different angle, so the peak positions are the CMB's main constraint on spatial curvature.
- *Peak heights.* The first peak is boosted by radi-

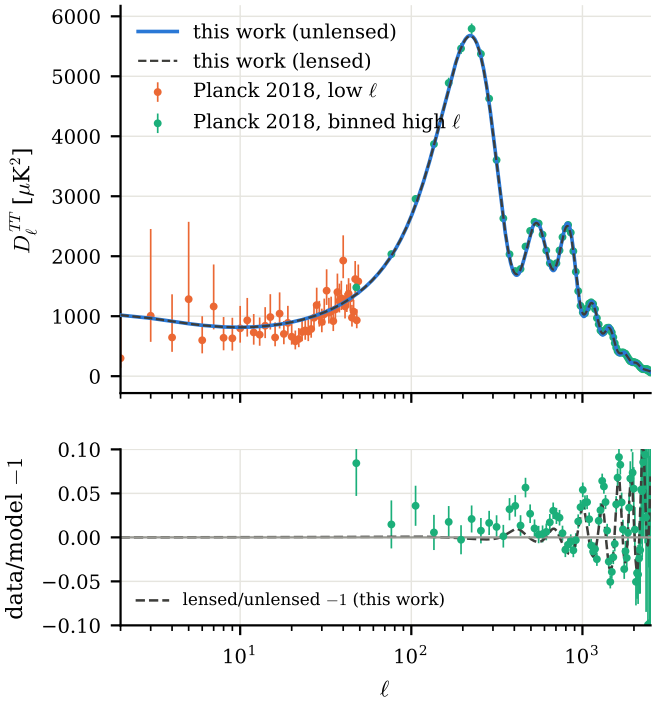


FIG. 16. *Top:* our TT spectrum, unlensed and lensed, compared with Planck 2018 data [6]: the low-ℓ spectrum with asymmetric error bars and the binned high-ℓ spectrum. *Bottom:* residuals of the high-ℓ data relative to our unlensed model. The dashed line is our own lensed/unlensed ratio minus one (Sec. V C), which follows the oscillating residuals.

ation driving. The second peak (a rarefaction) is lower than the first and third relative to a smooth envelope, because the baryons shift the zero point of the oscillation towards compression. The ratio of odd to even peaks measures $\Omega_b h^2$.

- *Damping tail.* Above $\ell \sim 1000$ the power falls steeply. From Eq. (31) we find $k_D = 0.14\,\mathrm{Mpc}^{-1}$ at decoupling, i.e. $\ell_D = k_D\chi_* \approx 2000$. The finite thickness of the last-scattering surface ($\approx 40\,$Mpc, Sec. III) adds to the suppression, since fluctuations smaller than the thickness average out along the line of sight.

The fiducial parameters are not fitted to the data, and the residuals are at the few-percent level. Against the 83 binned high-ℓ points we find $\chi^2 = 780.7$. The residuals oscillate for $\ell \gtrsim 1000$: the data are above the model in the troughs and below it at the peaks. This is the signature of weak gravitational lensing by large-scale structure [15]. The deflections of CMB photons are only $\sim 2'$, but they are coherent over degrees and remap the temperature field. In harmonic space this convolves the spectrum with the lensing kernel, which smooths the acoustic peaks and moves power into the troughs and the damping tail. With lensing included (Sec. V C) we find $\chi^2 = 220.2$ for the same 83 bins. Most of the residual mismatch of the unlensed model is therefore lensing; the remainder comes from the parameters, which are not the best fit, and from the simple recombination model.

Gravitational lensing. Photons from the last-scattering surface are deflected by the gravitational potentials of the intervening structure. To first order the observed temperature in direction $\hat{\mathbf{n}}$ is the unlensed one at $\hat{\mathbf{n}} + \nabla\phi$, where the lensing potential is a projection of the Weyl potential $\Psi - \Phi$ (the combination that bends light, since both potentials contribute to the null geodesic),

$$\phi(\hat{\mathbf{n}}) = -\int_0^{\chi_*} d\chi\, \frac{r(\chi_* - \chi)}{r(\chi_*)\, r(\chi)}\, (\Psi - \Phi)\big(\chi\hat{\mathbf{n}}, \eta_0 - \chi\big), \quad (43)$$

with $r(\chi)$ the comoving angular distance ($r = \chi$ when flat; Sec. VII). The geometric kernel peaks halfway to the source, so lenses at $z \sim 2$ dominate. We treat the last-scattering surface as a single source plane at the peak of the visibility function. $C_\ell^{\phi\phi}$ follows from the same line-of-sight machinery as the temperature, with source (43), for $\ell < 30$; above that we use the Limber approximation $C_\ell^{\phi\phi} \simeq \int d\chi\, W^2(\chi)\, P_{\Psi-\Phi}\big((\ell+\tfrac{1}{2})/r(\chi), \chi\big)/r^2(\chi)$, which avoids integrating to large k (the Weyl potential is extrapolated beyond our $k_{\max}$ as $\ln k/k^2$, its asymptotic form in the radiation-to-matter transfer function). The two agree to 0.2% at $40 \le \ell \le 200$. $[\ell(\ell+1)]^2 C_\ell^{\phi\phi}/2\pi$ peaks at $\ell = 34$ with value 1.41×10^{-7}, and the rms deflection is $2.5'$ (Fig. 17).

Because the deflections are small but coherent over degrees, we lens the spectra with the full-sky correlation-function method of Challinor and Lewis [16], as implemented in CLASS. The lensed correlation function is the unlensed one evaluated with the deflection field averaged over: e.g. for temperature $\tilde{\xi}(\beta) = \sum_\ell \frac{2\ell+1}{4\pi} C_\ell\, e^{-\ell(\ell+1)\sigma^2(\beta)/2}\, d^\ell_{00}(\beta) + \ldots$, where $\sigma^2(\beta)$ is the variance of the relative deflection of two points separated by β and the dots are terms of first and second order in the anisotropic part $C_{\mathrm{gl},2}(\beta)$ of the deflection correlation. The Gaussian factor is exact for Gaussian deflections; only the anisotropic part is expanded. Polarization uses the spin-2 correlation functions ξ_+ and $\xi_\times$ with the Wigner functions d^ℓ_{22}, d^ℓ_{2-2}, d^ℓ_{20}, which we compute by the three-term recurrence in ℓ. The lensed spectra follow by Gauss–Legendre integration over $\cos\beta$. Since lensing couples multipoles over $\Delta\ell \sim$ a few hundred, the unlensed spectra are computed to $\ell_{\max} + 500$.

The effects are those of a remapping of the sky (Fig. 16). The acoustic peaks are smoothed: at $\ell = 1000$ (a trough) the power rises by 3.4% and at the fifth peak ($\ell = 1500$) it drops by 2.8%, while scales larger than the coherence of the deflections ($\ell \lesssim 200$) change by less than 0.1%. In the damping tail lensing transfers power from larger scales, +7% at $\ell = 2500$. The total variance $\sum(2\ell+1)C_\ell$ is almost unchanged. The deflections also distort the E-mode pattern without respecting its parity, creating B modes from pure E: D_ℓ^{BB} rises to 0.094 μK^2 at $\ell \approx 900$, with an almost white C_ℓ^{BB} at low ℓ (Fig. 17). This lensing B-mode is a foreground for primordial grav-

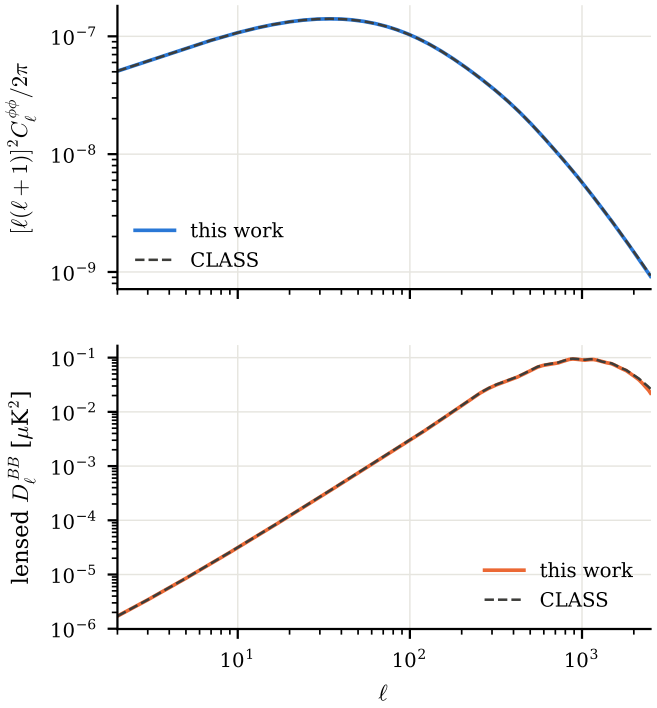


FIG. 17. *Top:* lensing potential power spectrum. *Bottom:* B-mode spectrum generated by lensing of the E modes (no primordial B modes). Dashed: CLASS run with the same parameters.

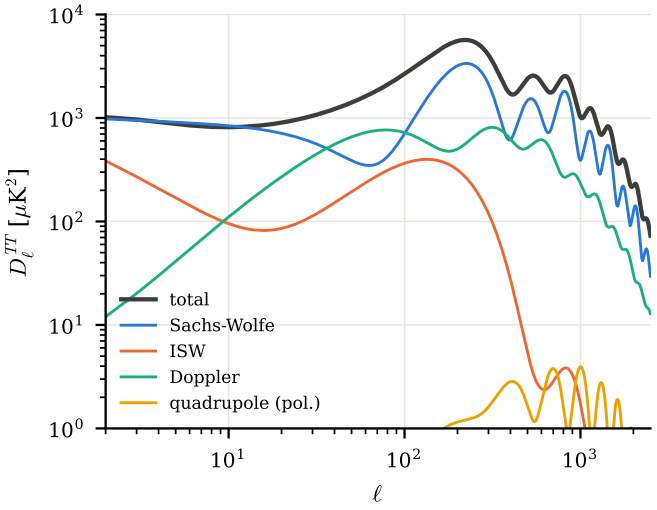


FIG. 18. Contributions to the TT spectrum from the four terms of the source function (34), each computed alone. They add in amplitude, not in power, so the cross terms make the total differ from the sum of the curves.

itational waves and has been measured by several experiments.

Decomposition. Figure 18 splits the TT spectrum into the four terms of Eq. (34):

- The SW term, $\Theta_0 + \Psi$ at last scattering, dominates on all scales. It gives the plateau ($843\,\mu\mathrm{K}^2$ at $\ell = 10$) and the acoustic peaks.
- The Doppler term, from the baryon velocity at last scattering, is out of phase with the density by $\pi/2$. It fills in the troughs: at $\ell = 400$ it is as large as the SW term. Its amplitude is reduced relative to the density term by $\sqrt{3}c_s/c = (1+R_b)^{-1/2}$, so the peaks remain visible. On the largest scales it vanishes, because the velocity is $\propto \kappa$ outside the horizon.
- The ISW term has two parts:
  - The early ISW effect comes from the potentials still decaying just after recombination, because radiation is not yet negligible. It adds power near the first peak ($270\,\mu\mathrm{K}^2$ at $\ell = 220$), and since it adds coherently with the SW term, it raises the first peak considerably.
  - The late ISW effect comes from the decay of Φ in the Λ era. A photon crossing a decaying potential well gains more energy falling in than it loses climbing out. The potentials decay only on very large scales today, so the late ISW dominates the rise at the lowest multipoles ($385\,\mu\mathrm{K}^2$ at $\ell = 2$). It is a direct signature of dark energy.
- The polarization/quadrupole term is very small, $\sim 10^{-3}$ of the total.

Polarization. The EE and TE spectra (Fig. 19) agree well with Planck. Their shapes follow from the fact that polarization is created by the quadrupole during the brief decoupling, $\Pi \propto (\kappa/\tau')\Theta_1$ (Sec. IV):

- The E-mode peaks at $\ell = 395$, 690, 994 and 1304 lie between the temperature peaks, close to the TT troughs, because they trace the velocity $\propto \sin(kr_s)$ rather than the density $\propto \cos(kr_s)$.
- The EE amplitude grows from large to small scales up to $\ell \approx 1000$, because the quadrupole is generated by free streaming over one mean free path, which is more effective for short wavelengths ($\propto k$). It then falls with the damping. The EE peaks are about 100 times lower than the TT peaks.
- At $\ell \lesssim 10$ the reionization bump ($D_\ell \approx 0.04\,\mu\mathrm{K}^2$) comes from rescattering at $z \lesssim 10$. The relevant quadrupole is the large-scale temperature anisotropy seen by the free electrons at reionization, and the horizon at $z = 8$ projects to $\ell \sim$ a few. The amplitude scales as τ_{reion}^2, so this bump is the main handle on reionization.
- The TE spectrum correlates the density and velocity terms, $\propto \cos(kr_s)\sin(kr_s) \propto \sin(2kr_s)$. It therefore changes sign twice as often as the TT peaks

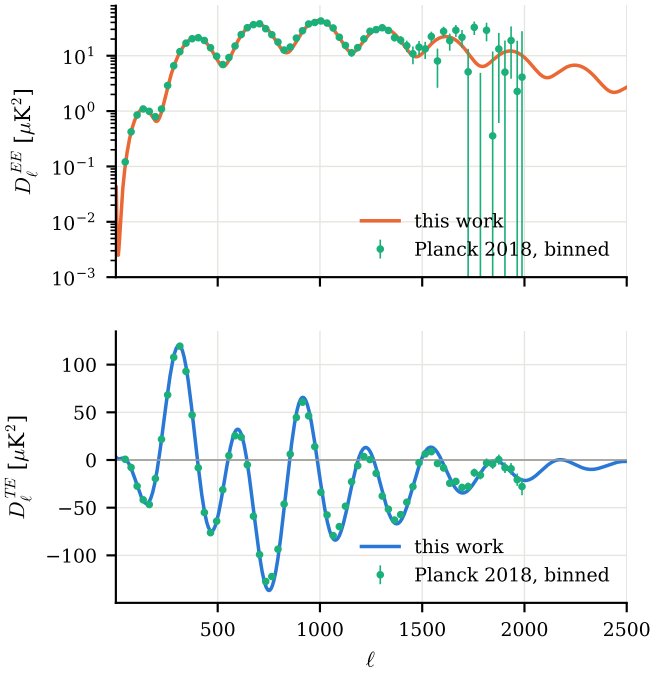


FIG. 19. *E*-mode polarization (*top*) and temperature–polarization cross spectrum (*bottom*) compared with binned Planck 2018 data.

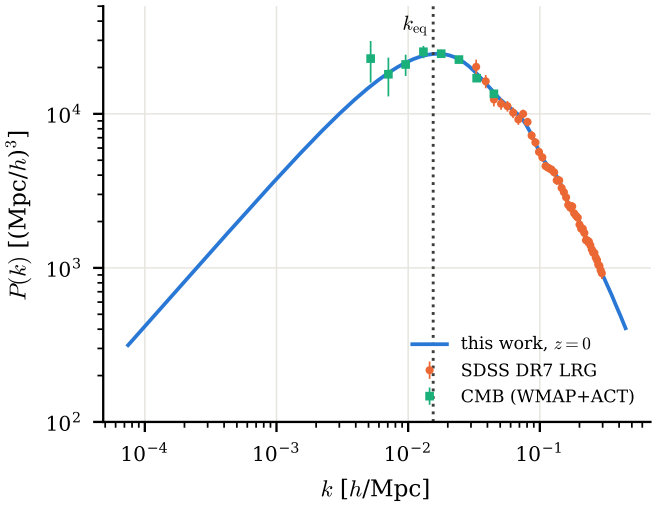


FIG. 20. Matter power spectrum today compared with the SDSS DR7 luminous red galaxy power spectrum [26] and with the estimate from WMAP and ACT CMB data. The vertical dotted line marks $k_{\rm eq}$.

repeat: at $\ell = 52$, 211, 401, 550, ... Its largest values are $+120\,\mu$K^2 at $\ell = 310$ and $-137\,\mu$K^2 at $\ell = 751$. The anti-correlation at $\ell \sim 50$–200, on scales larger than the sound horizon, is a key signature that the fluctuations were adiabatic and present before they entered the horizon, as inflation predicts.

Matter power spectrum. The matter power spectrum (Fig. 20) peaks at $k = 0.0167\,h\,$Mpc^{-1} with $P = 2.46 \times 10^4\,(h^{-1}\,\text{Mpc})^3$, close to $k_{\rm eq} = 0.0155\,h\,$Mpc^{-1} from Milestone I. On large scales the slope is $0.95 \approx n_s$, the primordial spectrum. On small scales the spectrum falls as modes that entered during RD were suppressed. Between 0.1 and $0.4\,h\,$Mpc^{-1} the local slope is -1.7; the asymptotic $k^{n_s-4}\ln^2 k$ is only reached at much larger k. The peak is not sharp, because the transition from RD to MD is gradual.

The baryon acoustic oscillations appear as small wiggles at $k \sim 0.05$–$0.3\,h\,$Mpc^{-1}, with period $2\pi/r_s(z_d) \approx 0.064\,h\,$Mpc^{-1}. They are the same sound waves as in the CMB, frozen into the baryon distribution at the drag epoch, and then imprinted in the dark matter as the baryons fell into the dark matter wells. Their amplitude is small because the baryons are only 16% of the matter.

The model matches the CMB-derived points and the SDSS DR7 LRG spectrum, with $\chi^2 = 12.2$ for the 45 LRG points (compared directly, without a free bias or amplitude parameter). The LRG data cover the turnover only partially, but they clearly favour the decline at $k > k_{\rm eq}$ predicted by a universe dominated by cold dark matter. In a baryon-only universe there would be no turnover near $k_{\rm eq}$ of this form, and the BAO wiggles would be large.

Toy cosmology. As a cross-check with the reference figures of the course, we also ran a matter-heavy toy model (Fig. 21). The TT peaks are at $\ell = 199$, 485 and 685, compared with ≈ 199, 491 and 690 read off the reference figure. The differences from the fiducial model are what the physics above predicts:

- With $\Omega_M = 0.5$ and no neutrinos, equality is at $z \approx 9900$ instead of 3400. The potentials have decayed less by recombination, so radiation driving and the early ISW effect are weaker.
- The baryon loading term then dominates the peak heights, and the third peak is much higher than the second.
- The peaks shift to lower ℓ because the larger matter density reduces the distance to the last-scattering surface ($\chi_* \approx 11\,090\,$Mpc, 20% less) more than the sound horizon (136 Mpc, 6% less), so $\ell_A \approx 256$ instead of 303.
- $P(k)$ peaks at $k = 0.031\,h\,$Mpc^{-1}, as expected from $k_{\rm eq} \propto \Omega_M/\sqrt{\Omega_r}$ in units of h/Mpc, which is larger by a factor 2.1.

The overall amplitude of both C_ℓ and $P(k)$ is a constant factor ≈ 2.1 higher than the reference. A constant factor in both spectra is what a different A_s would give, and the factor is consistent with the reference using $A_s = 10^{-9}$ (this is our inference; the reference does not state A_s).

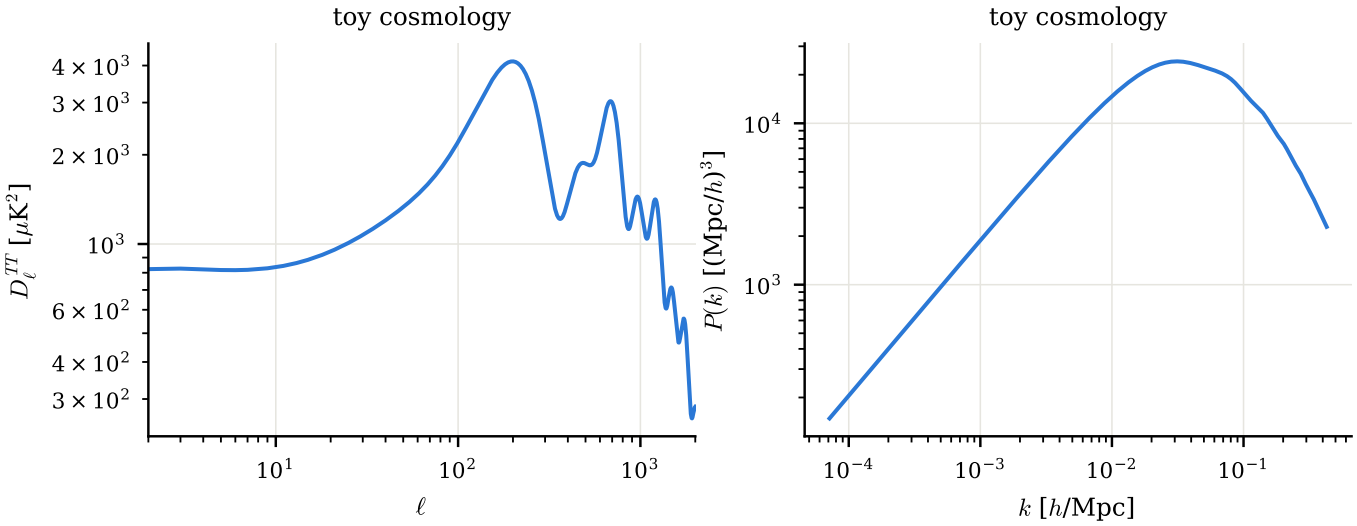


FIG. 21. TT spectrum (*left*) and matter power spectrum (*right*) for the toy cosmology of the course reference figures ($h = 0.7$, $\Omega_b = 0.05$, $\Omega_{\rm CDM} = 0.45$, $\Omega_\Lambda = 0.5$, no neutrinos, no helium, no reionization, no polarization).

VI. VALIDATION AGAINST CLASS

As an end-to-end test, we ran CLASS [3, 4] (through its Python wrapper) with the same parameters and physics: massless neutrinos and the same tanh reionization, unlensed unless stated otherwise. This section uses the Peebles recombination model of Sec. III; with RECFAST (Sec. X), the code's default, the recombination-driven differences below drop by an order of magnitude in TT and by a factor of four in EE. The comparison (Fig. 22) is:

- **Background.** H, t and η agree to $\sim 10^{-7}$. The age (13.8483 Gyr) and η_0 (14 191.6 Mpc) agree in all printed digits.
- **Recombination.** CLASS uses HyRec, which includes the multi-level atom and non-equilibrium helium recombination. Our X_e agrees to $\sim 0.1\%$ during hydrogen recombination. It differs by 3–4% during He^+ recombination at $z \approx 2000$–2500, where our Saha helium recombines too fast, and by +15% in the freeze-out tail. These differences have little effect on the observables:
  - $z_* = 1084.85$ versus 1084.70, and $r_s(z_*) = 144.85$ versus 144.87 Mpc;
  - $z_d = 1060.32$ versus 1060.03, and $r_s(z_d) = 147.03$ versus 147.06 Mpc;
  - $\tau_{\rm reion} = 0.0580$ versus 0.0578.
- **Angular spectra.**
  - TT agrees to 1.1% for $2 \le \ell \le 2000$, and to 0.94% for $\ell \le 1200$.
  - EE agrees to 1% at $\ell = 2$–8 and to 3.3% for $\ell \ge 100$. The largest difference, up to +6%, is in the EE trough at $\ell \approx 15$–20, where the spectrum is tiny.
  - TE differs by at most 1.4% of its peak amplitude.

  When we remove helium from both codes, TT agrees to 0.3–0.9%. The remaining differences are therefore mostly due to the recombination model, which changes the damping tail and the polarization amplitude.
- **Matter power spectrum.** $P(k)$ agrees to 0.11% for all k.
- **Lensing.** To separate the lensing from the unlensed spectra we compare the ratio of lensed to unlensed spectra from each code, with CLASS run with lensing on and $\Delta\ell = 500$ extra multipoles. The TT ratios agree to 1.3×10^{-4} for $\ell \le 1000$, 1.0×10^{-3} for $\ell \le 2000$ and 3.7×10^{-3} for $\ell \le 2500$. $C_\ell^{\phi\phi}$ agrees to 0.11% for $\ell \le 1000$ and 1.1% up to $\ell = 2500$, where our extrapolation of the Weyl potential beyond $k_{\rm max}$ takes over. The lensing B modes agree to 1.5% for $\ell \le 1000$ and 3.4% for $\ell \le 2000$; above that the difference grows to 16% because the unlensed EE spectrum is cut at $\ell_{\rm max} + \Delta\ell$, which matters more for BB, built from a broad range of EE multipoles. Fed with identical input spectra, our lensing module reproduces CLASS's lensed TT to 1.4×10^{-4}.

VII. EXTENSION: SPATIAL CURVATURE

The milestones assume a flat universe. As an extension we added spatial curvature to all modules. Curvature is one of the main targets of the CMB: it changes the distance to the last-scattering surface, and therefore the angle under which the acoustic scale appears. Throughout

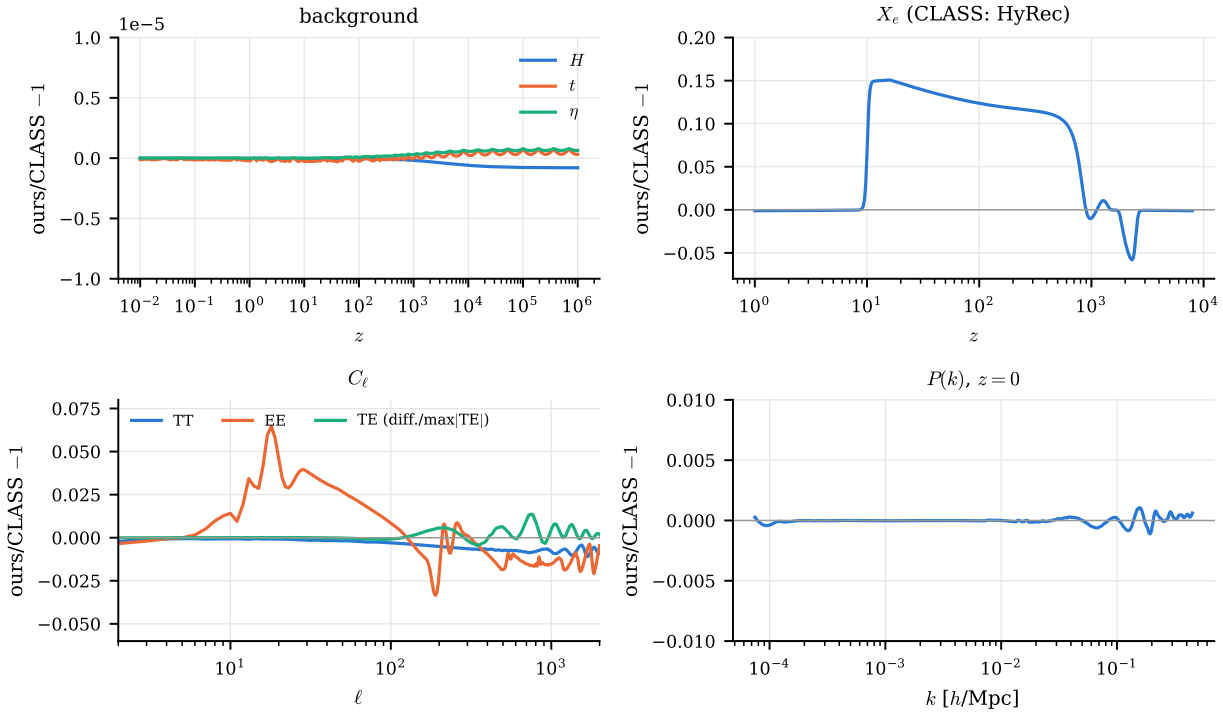


FIG. 22. Relative differences between our code and CLASS run with the same physics. *Top left:* background H, t and η. *Top right:* free electron fraction (CLASS uses HyRec). *Bottom left:* unlensed TT and EE spectra, and the TE difference relative to the maximum of $|C_\ell^{TE}|$. *Bottom right:* matter power spectrum today.

this section the matter, radiation and baryon densities are those of the fiducial model, and $\Omega_{\Lambda 0} = 1 - \Omega_{M0} - \Omega_{r0} - \Omega_{K0}$ absorbs the change.

A. Theory

Background. With $K = -\Omega_{K0} H_0^2/c^2$ ($K < 0$ open, $K > 0$ closed), the Friedmann equation already contains the curvature term $\Omega_{K0} e^{-2x}$. Distances use the comoving transverse distance $r = \sin_K(\chi) \equiv \sin(\sqrt{K}\chi)/\sqrt{K}$, χ, or $\sinh(\sqrt{-K}\chi)/\sqrt{-K}$ for closed, flat and open space. In a closed universe light rays from a point converge again, so objects at fixed χ look larger; in an open universe they diverge and objects look smaller.

Perturbations. In curved space the plane waves $e^{i\boldsymbol{k}\cdot\boldsymbol{x}}$ are replaced by eigenfunctions of the Laplacian, $\nabla^2 G = -k^2 G$. It is convenient to label them by $q^2 = k^2 + K$ and $\nu = q/\sqrt{|K|}$. In an open universe every $q > 0$ is allowed. A closed universe is finite, so the spectrum is discrete: only integer $\nu \geq 3$ describe scalar perturbations ($\nu = 1$ is a homogeneous change of the background and $\nu = 2$ is a pure gauge mode). Three changes follow in the equations of Milestone III [29, 30]:

- The 00 Einstein equation contains $k^2 - 3K$ instead of k^2: curvature adds to the potential energy of a perturbation. The same factor appears in the Poisson equation for Δ_M.
- The free-streaming couplings between neighbouring multipoles of photons, polarization and neutrinos carry the factors $s_\ell = \sqrt{1 - (\ell^2 - 1)K/k^2}$: $\Theta'_\ell = \frac{\kappa}{2\ell+1}[\ell s_\ell \Theta_{\ell-1} - (\ell+1) s_{\ell+1}\Theta_{\ell+1}] + \dots$. Geometrically, the angle subtended by a wavelength changes along the line of sight. In a closed universe $s_\nu = 0$, so the hierarchy of mode ν ends at $\ell = \nu - 1$: a finite space has no structure on angles smaller than its modes allow. In this convention the physical shear is Θ_2/s_2, which enters Ψ.
- The truncation of the hierarchies uses $\cot_K(\eta)$ instead of $1/\eta$.

The initial conditions are those of the flat case, since curvature is negligible at $x = -18$, and we keep the normalization $\mathcal{R} = 1$ and $\mathcal{P}_\mathcal{R} = A_s (k/k_{\rm pivot})^{n_s - 1}$, as in CLASS.

Hyperspherical Bessel functions. The projection of a mode onto the sky now uses the radial eigenfunctions $\Phi_\ell^\nu(\chi)$ (with χ in units of the curvature radius) instead of $j_\ell(kr)$. They obey

$$\Phi'' + 2\cot_K\chi\,\Phi' + \left[\nu^2 - \mathrm{sgn}K - \frac{\ell(\ell+1)}{\sin_K^2\chi}\right]\Phi = 0, \quad (44)$$

with $\Phi_0^\nu = \sin(\nu\chi)/(\nu \sin_K\chi)$ and the three-term recurrence

$$\sqrt{\nu^2 - \mathrm{sgn}K(\ell+1)^2}\,\Phi_{\ell+1} = (2\ell+1)\cot_K\chi\,\Phi_\ell - \sqrt{\nu^2 - \mathrm{sgn}K\,\ell^2}\,\Phi_{\ell-1}. \quad (45)$$

They reduce to $j_\ell(\nu\chi)$ for $\nu \to \infty$. The line-of-sight integral no longer allows the integrations by parts used in Eq. (34), so we project the three parts of the source separately:

$$\Theta_\ell(k) = \int \mathrm{d}x \left[S_{T0}\Phi_\ell + S_{T1}\frac{\sqrt{|K|}}{k}\Phi'_\ell + S_{T2}\frac{3(|K|/k^2)\Phi''_\ell + \Phi_\ell}{2s_2}\right], \quad (46)$$

$$\Theta^E_\ell(k) = \sqrt{\tfrac{(\ell+2)!}{(\ell-2)!}}\int \mathrm{d}x\, \frac{3\tilde{g}\,\Pi}{4}\,\frac{\Phi_\ell}{s_2(kr)^2}, \quad (47)$$

with $S_{T0} = \tilde{g}(\Theta_0 + \Psi) + e^{-\tau}(\Psi' - \Phi')$, $S_{T1} = -\tilde{g}v_b$ and $S_{T2} = \tilde{g}\Pi/2$. In the flat limit these are j_ℓ, j'_ℓ and $(3j''_\ell + j_\ell)/2$, and integrating by parts recovers Eq. (34). In a closed universe the k-integral of Eq. (39) becomes a sum over integer ν.

B. Implementation and tests

Φ^ν_ℓ is computed by upward recurrence [31, 32] below the turning point $\ell(\ell+1) = (\nu^2 - \mathrm{sgn}K)\sin^2_K\chi$, where it oscillates, and by Miller's downward recurrence above it, where it decays exponentially; in a closed universe the downward recurrence starts exactly at $\ell = \nu - 1$. The functions agree with j_ℓ in the flat case to 2×10^{-12}, satisfy the radial equation to the accuracy of the finite differences, match the closed-form Gegenbauer polynomials of a closed universe to 10^{-13}, and approach $j_\ell(\nu\chi)$ as $1/\nu^2$.

The perturbations are sampled in q; in a closed universe only integer ν are used, and in an open universe the grid extends down to $q = 0.01\sqrt{|K|}$, since the modes near the curvature scale contribute to the lowest multipoles. For each mode and each point on the line of sight we compute $\Phi_0 \ldots \Phi_{\ell_\mathrm{cap}}$, where ℓ_cap is just beyond the turning point. One subtlety concerned the closed-universe sum. Rounding a uniform k-grid to integer ν gives an irregular stride of 2–3 in ν, while Θ^2_ℓ oscillates with a period of only a few units of ν. The trapezoidal rule then loses the high accuracy it has for oscillating integrands on a uniform grid, and C_ℓ was off by up to 4%. A uniform integer stride fixes this. With the flat Bessel functions the same code reproduces the integrated-by-parts flat result to 5×10^{-5}, which tests the decomposition of the source. Two speed-ups bring the curved line-of-sight integration from ≈ 48 s to 26–42 s on two cores for $\ell \le 2000$ (7 s with the flat tables). First, modes with smaller q use a proportionally coarser x-grid away from recombination (the radial functions oscillate more slowly), but never coarser than $\Delta x = 0.01$, which resolves reionization; this changes C_ℓ by $< 3\times10^{-5}$. Second, for $\nu \ge 4000$ we use the flat approximation of CLASS: $\Phi^\nu_\ell(\chi)$ is replaced by j_ℓ with the argument rescaled to put its turning point at χ_tp and an empirical amplitude correction [30], read from tables of j_ℓ and j'_ℓ. It is accurate near the turning point but drifts in phase far beyond it. Since those oblique projections carry little weight for Silk-damped modes, the effect on C_ℓ is $\lesssim 6\times10^{-4}$ at $\Omega_K = \pm0.05$, where a third of the modes use it. For $|\Omega_K| = 0.2$ no mode reaches $\nu = 4000$ below $k = 0.3\,\mathrm{Mpc}^{-1}$, and lowering the threshold quickly costs accuracy (0.4% for $\nu \ge 2000$, 1% for $\nu \ge 1000$).

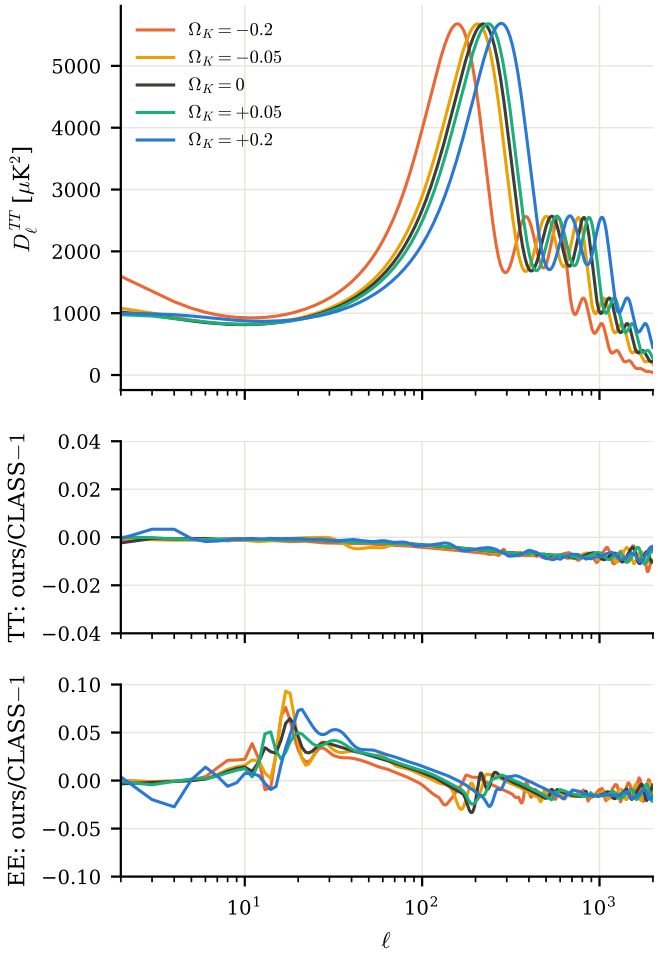


FIG. 23. *Top:* TT spectra for $\Omega_K = -0.2$, -0.05, 0, 0.05 and 0.2 at fixed matter and baryon density. *Middle and bottom:* TT and EE relative to CLASS run with the same physics (unlensed).

C. Results

Figure 23 and Table III show the result. Our spectra agree with CLASS as well in curved as in flat models: TT to 0.15–0.35% for $\ell \le 30$ and 1–1.4% up to $\ell = 2000$, EE to 2–3% for $\ell \ge 100$, and $P(k)$ to 0.12% for $k > 10^{-3}\,\mathrm{Mpc}^{-1}$ ($\Omega_K = \pm0.05$). The background agrees to 10^{-6} and the potentials today to $\sim 10^{-5}$. The remaining differences are the same as in the flat case and come from the recombination model.

The main effect of curvature is geometric. The physics at recombination is unchanged, because curvature is neg-

TABLE III. Curved models: the first TT peak from our code and from CLASS, the acoustic scale $\ell_A = \pi r(\chi_*)/r_s$, and the largest TT difference from CLASS for $30 \le \ell \le 2000$.

Ω_K	Ω_Λ	$\ell_{\rm peak}$	CLASS	ℓ_A	$\ell_{\rm peak}/\ell_A$	TT diff.
−0.20	0.883	158	158	215.7	0.732	1.4%
−0.05	0.733	206	206	280.8	0.734	1.1%
0	0.683	221	221	301.7	0.733	1.1%
+0.05	0.633	236	236	322.2	0.733	1.2%
+0.20	0.483	279	280	381.4	0.731	1.0%

ligible then, so the sound horizon is the same in all models to 0.05% ($r_s = 144.82$–$144.89\,$Mpc; the small spread comes from the optical depth to reionization, which shifts the $\tau = 1$ surface). What changes is the transverse distance to the last-scattering surface, $r(\chi_*)$: it is smaller in a closed universe, where light rays focus, and larger in an open one. The whole acoustic pattern therefore shifts in ℓ. The ratio $\ell_{\rm peak}/\ell_A$ is 0.731–0.734 in all five models, so the first peak follows the geometric prediction, while $\Omega_K = \mp 0.2$ moves it by -29% and $+26\%$. A measurement of the peak positions therefore measures $r(\chi_*)/r_s$, which is why the CMB constrains curvature so well once $\Omega_M h^2$ is fixed by the peak heights.

The low multipoles change too. In the closed model with $\Omega_K = -0.2$ the larger $\Omega_\Lambda = 0.88$ makes the potentials decay earlier and more strongly, and the late ISW effect raises D_2 to $1600\,\mu\mathrm{K}^2$ against $1020\,\mu\mathrm{K}^2$ in the flat model. In the open models the opposite change in Ω_Λ is partly compensated by curvature itself, which also makes the potentials decay once it dominates the expansion. These large-scale effects are limited by cosmic variance, and in practice the geometric degeneracy between Ω_K and Ω_Λ is broken by combining the CMB with lensing or BAO.

VIII. EXTENSION: MASSIVE NEUTRINOS

Neutrino oscillations show that at least two neutrino species are massive, with $\sum m_\nu \gtrsim 0.06\,$eV. For cosmology the masses matter because the neutrinos become non-relativistic in the late universe: before that they behave like radiation, afterwards like a hot component of the dark matter that does not cluster on small scales. We add one massive species of mass m_ν (the minimal normal-ordering case, if the lighter ones are treated as massless), following Refs. [13, 17, 18]. As in CLASS, its temperature today is $T_\nu = 0.71611\,T_{\rm CMB}$, which gives $\Omega_\nu h^2 = m_\nu/93.14\,$eV and contributes 1.0132 to $N_{\rm eff}$; the other $N_{\rm eff} - 1.0132 = 2.0328$ species stay massless. We keep h, ω_b and $\omega_{\rm cdm}$ of the fiducial model, so that Ω_Λ absorbs the neutrino density.

A. Theory

Background. The neutrinos decoupled at $T \sim 1\,$MeV while relativistic, so their distribution keeps the relativistic Fermi–Dirac form $f_0(q) = 1/(e^q + 1)$ in the comoving momentum $q = ap/(k_B T_\nu)$, even after they become non-relativistic. With $M(a) = a m_\nu c^2/(k_B T_\nu)$ the comoving energy is $\epsilon = \sqrt{q^2 + M^2}$, and

$$\rho_\nu = \frac{15}{\pi^4}\Big(\frac{T_\nu}{T_{\rm CMB}}\Big)^4 \frac{\rho_{\gamma 0}}{a^4}\int_0^\infty dq\, q^2 \epsilon f_0, \tag{48}$$

$$p_\nu = \frac{15}{\pi^4}\Big(\frac{T_\nu}{T_{\rm CMB}}\Big)^4 \frac{\rho_{\gamma 0}}{a^4}\int_0^\infty dq\, \frac{q^4}{3\epsilon} f_0. \tag{49}$$

For $M \ll 1$ this is radiation ($p = \rho/3$, $\rho \propto a^{-4}$); for $M \gg 1$, $\rho \to \frac{3\zeta(3)}{2} M$ times the prefactor, i.e. matter with $\rho \propto a^{-3}$. The transition, $M \approx 3$, happens at $1 + z \approx M_0/3$, i.e. $z_{\rm nr} = 118$, 296 and 594 for $m_\nu = 0.06$, 0.15 and 0.3 eV (Fig. 24a). The derivatives needed for $\mathcal{H}'$ and $\mathcal{H}''$ follow from $\rho' = -3(\rho + p)$ and $p' = -(5p - \mathfrak{p})$, where $\mathfrak{p} = \frac{15}{\pi^4}(\dots)\int dq\, q^6 f_0/(3\epsilon^3)$ is the "pseudo-pressure". We evaluate the integrals with 48-point Gauss–Laguerre quadrature, accurate to 10^{-7}; age and conformal time agree with CLASS in all printed digits.

Perturbations. A massive particle's velocity, q/ϵ, depends on its momentum, so the perturbation cannot be described by a single temperature multipole hierarchy. Following Ma and Bertschinger [13] we write $f = f_0(q)[1 + \Psi(\boldsymbol{k}, \hat{n}, q, \eta)]$ and expand Ψ in Legendre multipoles $\Psi_\ell(q)$ for a set of momenta q_i. In our variables ($\kappa = ck/\mathcal{H}$, Newtonian gauge),

$$\Psi_0' = -\kappa \frac{q}{\epsilon}\Psi_1 + \Phi' \frac{d\ln f_0}{d\ln q}, \tag{50}$$

$$\Psi_1' = \frac{\kappa}{3}\frac{q}{\epsilon}\big(\Psi_0 - 2\Psi_2\big) - \frac{\kappa}{3}\frac{\epsilon}{q}\Psi\frac{d\ln f_0}{d\ln q}, \tag{51}$$

$$\Psi_\ell' = \frac{\kappa}{2\ell+1}\frac{q}{\epsilon}\big[\ell\Psi_{\ell-1} - (\ell+1)\Psi_{\ell+1}\big], \quad \ell \ge 2. \tag{52}$$

The free-streaming terms are those of massless neutrinos with c replaced by the particle velocity cq/ϵ, and the metric terms act through $d\ln f_0/d\ln q$: a potential well shifts all momenta, which changes f in proportion to its slope. The neutrinos enter the Einstein equations through their density, velocity and shear,

$$\begin{aligned} \delta\rho_\nu &\propto \int dq\, q^2 \epsilon f_0 \Psi_0, \\ (\rho + p)_\nu \theta_\nu &\propto k \int dq\, q^3 f_0 \Psi_1, \\ (\rho + p)_\nu \sigma_\nu &\propto \frac{2}{3}\int dq\, \frac{q^4}{\epsilon} f_0 \Psi_2, \end{aligned} \tag{53}$$

which replace $4\rho_\nu \mathcal{N}_0$ and the shear of $\mathcal{N}_2$ in Eqs. (25)–(26). For $M \to 0$ and $\Psi_\ell = -\mathcal{N}_\ell\, d\ln f_0/d\ln q$ the system reduces exactly to the massless one, which also gives the

adiabatic initial conditions. We use 8 Gauss–Laguerre momenta and $\ell_{\max} = 17$ (as CLASS); 16 momenta or $\ell_{\max} = 25$ change C_ℓ and $P(k)$ by less than 3×10^{-5}.

Fluid approximation. Deep inside the horizon ($k\eta > 31$) the hierarchy is replaced by the fluid equations of Ref. [17] for δ_ν, θ_ν and σ_ν, with the adiabatic sound speed $c_a^2 = \frac{w}{3(1+w)}(5 - \mathfrak{p}/p)$ and a viscosity closure for the shear, as in CLASS. This reduces the system from 190 to 49 variables. Compared with the full hierarchy it changes $P(k)$ by up to 0.3% at $m_\nu = 0.3\,\mathrm{eV}$; CLASS's fluid approximation has the same error, to 10^{-4}. With 190 variables the dense linear algebra of the implicit (BDF) solver makes it about 20 times slower than an explicit Runge–Kutta method, which gives identical results while the hierarchy is active; we use the explicit method there. A run with massive neutrinos then takes 40–43 s for the perturbations on 2 cores, compared with 37 s for massless neutrinos.

Matter power spectrum. With massive neutrinos the total matter is CDM, baryons and neutrinos. We compute the comoving density contrast directly, $\Delta_m = \sum_i \rho_i[\delta_i - 3(1+w_i)\mathcal{H}v_i/(ck)]/\sum_i \rho_i$, which is the synchronous-gauge δ_m used by CLASS; for massless neutrinos it agrees with the Poisson equation (42) up to radiation terms of order 10^{-3} at $k \sim 10^{-4}\,\mathrm{Mpc}^{-1}$, and reduces the difference from CLASS there from 2×10^{-3} to 10^{-5}. We also output the CDM+baryon spectrum P_{cb}, the relevant one for galaxy clustering.

B. Results

Free streaming. While relativistic, neutrinos stream out of any structure smaller than the horizon. After z_{nr} their thermal velocity $\langle v\rangle \approx 3.15\,k_B T_\nu/(m_\nu a)$ decays, and the free-streaming scale, the distance travelled in a Hubble time, shrinks. Its comoving wavenumber $k_{\mathrm{fs}} \sim \sqrt{3/2}\,\mathcal{H}/\langle v\rangle$ grows as $a^{1/2}$ in matter domination; the smallest value reached, at z_{nr}, is $k_{\mathrm{nr}} \approx 0.018\sqrt{\Omega_m m_\nu/\mathrm{eV}}\,h\,\mathrm{Mpc}^{-1} \approx 0.003$–$0.006\,h\,\mathrm{Mpc}^{-1}$. Figure 24b shows the consequence. Initially $\delta_\nu = \frac{4}{3}\delta_{\mathrm{cdm}}$ (adiabatic). When a mode enters the horizon the neutrino perturbation decays by free streaming. Later, when $k < k_{\mathrm{fs}}(a)$, the neutrinos fall into the CDM potential wells and $\delta_\nu \to \delta_{\mathrm{cdm}}$: at $k = 0.01\,\mathrm{Mpc}^{-1}$ this has largely happened today for 0.3 eV, but at $k = 0.1\,\mathrm{Mpc}^{-1}$ the neutrinos still lag far behind.

Matter power spectrum. On scales above k_{nr} neutrinos behave like CDM, and $P(k)$ changes only by 0.6–3% because Ω_Λ is smaller (Fig. 24c). Below the free-streaming scale the spectrum is suppressed, by 2.9%, 8.0% and 16.1% at $k = 0.5\,h\,\mathrm{Mpc}^{-1}$ for the three masses (CLASS: 3.0, 8.0, 16.2%). There are two reasons [18, 19]. First, a fraction $f_\nu = \Omega_\nu/\Omega_m = 0.0045,\ 0.011,\ 0.022$ of the matter does not cluster, which removes $2f_\nu$ from $P_m \propto [(1-f_\nu)\delta_{cb}]^2$. Second, the CDM and baryons grow more slowly, since only a fraction $1 - f_\nu$ of the matter falls into the wells: $\delta_{cb} \propto a^{1-3f_\nu/5}$ instead of a, which accumulates to a suppression of P_{cb} of several f_ν. Together the suppression is $\approx -8f_\nu$ asymptotically; at $k = 0.5\,h\,\mathrm{Mpc}^{-1}$ we find $-6.5f_\nu$, $-7.1f_\nu$ and $-7.3f_\nu$. The CDM+baryon spectrum shows only the second effect. This step-like suppression is why galaxy surveys are sensitive to $\sum m_\nu$.

CMB. The masses we consider make the neutrinos non-relativistic well after recombination, so the primary anisotropies change only through the background. The extra matter today shortens the distance to the last-scattering surface (η_0 decreases by 118 Mpc for 0.3 eV), which moves the peaks to lower ℓ (the first from 221 to 219) and causes the oscillating residuals in Fig. 24d. The lower Ω_Λ reduces the late ISW effect, and for 0.3 eV the neutrinos are no longer fully relativistic at decoupling, which changes the early ISW effect. The lensing potential, a projection of the matter distribution, is suppressed like $P(k)$: at $\ell = 1000$ by 1.9%, 5.7% and 12.0%. CMB lensing therefore constrains the neutrino mass: through the smoothing of the peaks (Sec. V C) and through direct reconstructions of $C_\ell^{\phi\phi}$.

Validation. We compare the effect of the mass, $X(m_\nu)/X(0) - 1$, with CLASS run with the same parameters (which removes the recombination differences of Sec. VI). For $m_\nu = 0.06\,\mathrm{eV}$, the effects on TT, lensed TT, $C_\ell^{\phi\phi}$ and P_m agree to 3–5 $\times 10^{-4}$ and on EE to 6×10^{-4}; for 0.3 eV to 8×10^{-4}, 1.0×10^{-3}, 3×10^{-4}, 4×10^{-4} and 3.7×10^{-3}, i.e. better than 3% of the effect itself. A nearly massless species ($m_\nu = 10^{-4}\,\mathrm{eV}$) reproduces the massless neutrinos to 4×10^{-7} in Φ at recombination when the same $\ell_{\max}$ is used.

IX. EXTENSION: TWO-POINT FUNCTIONS

Power spectra are the natural output of a Boltzmann code, because each Fourier mode or multipole evolves independently in linear theory. Observers often measure the equivalent real-space two-point functions instead. They show localized features, such as the acoustic scale or the BAO peak, as features at one separation, and they are easier to estimate on a masked sky or in a survey with complicated geometry. We added functions that compute the angular correlation functions of the CMB, the three-dimensional correlation function of the matter, and the angular power spectrum and correlation function of the matter (or galaxy) density projected over a redshift distribution.

A. Theory

Angular correlation functions. For a statistically isotropic field on the sphere, $a_{\ell m}$ are uncorrelated with variance C_ℓ, and the addition theorem

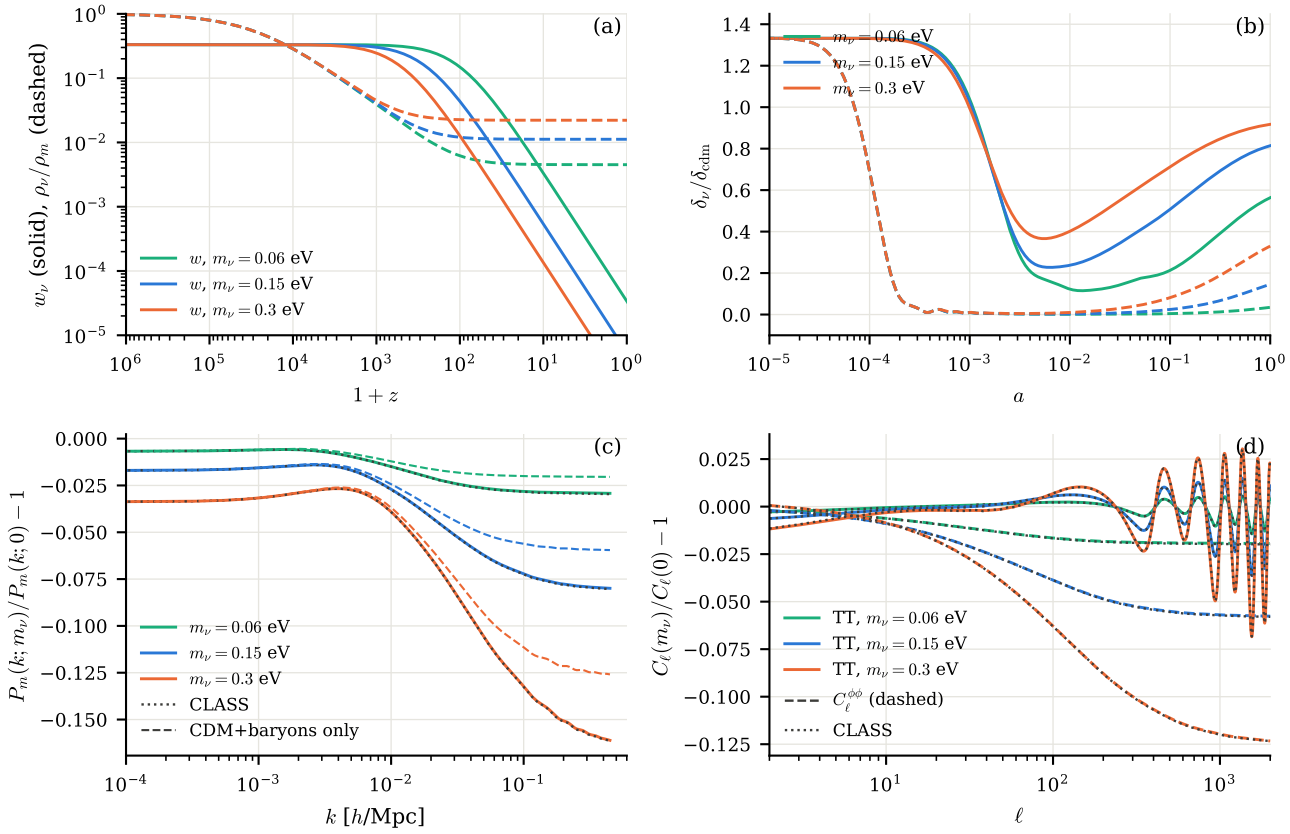


FIG. 24. Massive neutrinos with $m_\nu = 0.06$, 0.15 and 0.3 eV (fixed h, ω_b, $\omega_{\rm cdm}$). *(a)* Equation of state w_ν and density relative to all matter. *(b)* $\delta_\nu/\delta_{\rm cdm}$ for $k = 0.01\,\mathrm{Mpc}^{-1}$ (solid) and $0.1\,\mathrm{Mpc}^{-1}$ (dashed): below the free-streaming scale the neutrinos stop clustering, and fall back into the CDM wells after they slow down. *(c)* Change of the total matter power spectrum (solid) and of the CDM+baryon spectrum (dashed) today, with CLASS (dotted). *(d)* Change of the unlensed TT spectrum (solid) and the lensing potential (dashed), with CLASS (dotted).

$\sum_m Y_{\ell m}(\hat n) Y^*_{\ell m}(\hat n') = \frac{2\ell+1}{4\pi} P_\ell(\hat n\cdot\hat n')$ gives

$$C(\theta) = \langle T(\hat n) T(\hat n')\rangle = \sum_\ell \frac{2\ell+1}{4\pi} C_\ell B_\ell^2 P_\ell(\cos\theta), \quad (54)$$

with an optional Gaussian beam $B_\ell = e^{-\ell(\ell+1)\sigma_b^2/2}$. The polarization is a spin-2 field, so its correlation functions involve the Wigner functions $d^\ell_{mm'}$ of Sec. V C. With Q and U measured relative to the great circle joining the two points [16], $\xi_\pm = \langle QQ\rangle \pm \langle UU\rangle = \sum_\ell \frac{2\ell+1}{4\pi}(C_\ell^{EE} \pm C_\ell^{BB}) d^\ell_{2,\pm2}$ and $\xi_X = \langle TQ\rangle = \sum_\ell \frac{2\ell+1}{4\pi} C_\ell^{TE} d^\ell_{20}$. We sum $\ell = 2$–2500 (the monopole and dipole are not cosmological) with the Legendre and Wigner recurrences.

Matter correlation function. In three dimensions the transform is

$$\xi(r) = \int_0^\infty \frac{k^2 dk}{2\pi^2} P(k) j_0(kr) = \frac{1}{2\pi^2 r}\int_0^\infty dk\, kP(k)\sin(kr). \quad (55)$$

The integrand oscillates many times for $r \sim 100\,\mathrm{Mpc}$, so we use Filon's method [20]: $kP(k)$ is interpolated linearly in k between tabulated points and the product with $\sin(kr)$ is integrated analytically. The grid then only has to resolve $P(k)$, including its BAO wiggles (we use $d\ln k = 0.002$ and $dk \le 5\times10^{-4}\,\mathrm{Mpc}^{-1}$ below $1\,\mathrm{Mpc}^{-1}$), for all r at once. The integral needs $P(k)$ up to $k \sim 10\,\mathrm{Mpc}^{-1}$. Beyond the computed range we extrapolate $\Delta_M \propto k^2 T(k)$ with the shape of the no-wiggle transfer function of Ref. [21] anchored at $k_{\rm max}$ (accurate to 1.5% up to $20\,\mathrm{Mpc}^{-1}$ from $k_{\rm max} = 0.3\,\mathrm{Mpc}^{-1}$), and $\Delta_M \propto k^2$ below $k_{\rm min}$. We also compute σ_R, the rms of Δ_M in top-hat spheres, and σ_8.

Projected density. A galaxy survey with redshift distribution $n(z)$ and linear bias b sees the projected field $\delta_g(\hat n) = \int d\chi\, W(\chi)\, b\, \Delta_M(\chi\hat n, \chi)$ with $W = n(z)\,dz/d\chi$ normalized to unit integral. Expanding the plane waves in spherical harmonics, as for the CMB (Sec. V),

$$C_\ell = 4\pi \int \frac{dk}{k} \mathcal{P}_\mathcal{R}(k) \Delta_\ell(k)^2,$$
$$\Delta_\ell(k) = \int d\chi\, W(\chi)\, b\, \Delta_M(k,\chi)\, j_\ell(k\chi). \quad (56)$$

For $\ell \gg 1$, $j_\ell(k\chi)$ picks out $k\chi \approx \ell + \frac12$, and Eq. (56) reduces to the Limber approximation [22, 23], $C_\ell =$

$\int d\chi\, W^2 b^2 P((\ell+\frac{1}{2})/\chi,\chi)/\chi^2$: the multipole ℓ probes the transverse wavenumber $k = (\ell+\frac{1}{2})/\chi$ at each distance. The Limber approximation drops the radial modes, which matter when the window is not much wider than the wavelength. For $\sigma_z = 0.1$ at $z = 1$ it is 4% wrong at $\ell = 20$ and 0.4% at $\ell = 100$, so we use the exact projection below $\ell = 200$ (with all j_ℓ for one $k\chi$ from a single recurrence) and Limber above. We include only the density term: redshift-space distortions, lensing magnification and the relativistic corrections are left out, and $P(k)$ is linear. The angular correlation function $w(\theta)$ follows from Eq. (54); small angles need ℓ up to $\sim 3\times10^4$ (Limber on a logarithmic grid, with a cosine taper at the end to avoid ringing from the truncation).

B. Results

CMB. The variance of the temperature is $C(0) = 12\,550\,\mu\mathrm{K}^2$, i.e. $112\,\mu$K rms (Fig. 25a). The correlation falls to half at $\theta \approx 0.5°$, the angle subtended by the scales of the first acoustic peak, and shows a shoulder near 1°. This is the real-space counterpart of the acoustic peaks: the sound horizon subtends $\theta_* = r_s/\chi_* = 0.60°$, and a localized initial perturbation produces a ring of that radius around itself. At large angles $C(\theta)$ is set by the lowest multipoles. It crosses zero at 41° and 117°, with $-205\,\mu\mathrm{K}^2$ at 72° and $+259\,\mu\mathrm{K}^2$ at 180°. Because only a few multipoles contribute, the cosmic variance of $C(\theta)$ is large here; the observed sky shows almost no correlation above 60°, one of the known large-angle anomalies. Lensing changes $C(\theta)$ by at most $6\,\mu\mathrm{K}^2$ (at 1.1°, where it smooths the acoustic shoulder), and conserves the variance to 10^{-4}, since it only remaps the sky. A 1° beam reduces the variance to $4\,940\,\mu\mathrm{K}^2$. The polarization has rms $\sqrt{\xi_+(0)} = 6.3\,\mu$K (Fig. 25b). The temperature–polarization correlation ξ_X is $+18\,\mu\mathrm{K}^2$ at 0.57° and $-31\,\mu\mathrm{K}^2$ at 1.22°: around a temperature hot spot the polarization alternates between radial and tangential. This pattern comes from the flow of the photon–baryon fluid into and out of the potential wells at decoupling, and it has been seen by stacking hot spots in the WMAP data [24].

Matter. The matter correlation function (Fig. 25c) has the BAO peak at $r = 100.5\,h^{-1}\,\mathrm{Mpc} = 150\,\mathrm{Mpc}$, close to the sound horizon at the drag epoch, $r_d = 147\,\mathrm{Mpc}$ (the smooth decline of ξ moves the maximum of $r^2\xi$ slightly outward). Each initial point perturbation launches a spherical sound wave in the photon–baryon fluid that stalls at r_d when the baryons are released, so the baryons, and the dark matter that later falls into them, end up with a shell at r_d around every overdensity [25]. In Fourier space this single feature appears as the series of BAO wiggles. We find $\sigma_8 = 0.82071$ (CLASS: 0.82068). Our $\xi(r)$ agrees with the transform of CLASS's $P(k)$ (computed independently with adaptive quadrature) to 3×10^{-4} at $r = 20\,h^{-1}\,\mathrm{Mpc}$, and to 0.9% of the BAO peak height of $r^2\xi$ for $r \geq 60\,h^{-1}\,\mathrm{Mpc}$. The latter is not numerical, as the transform reproduces analytic and quadrature results to 10^{-6} and 7×10^{-4} in the unit tests. It comes from the $\pm7\times10^{-4}$ oscillating difference between the two $P(k)$, which follows the BAO phase (Sec. VI): the BAO peak in ξ is a small residual of a large cancellation, and it is sensitive to the damping and phase of the wiggles.

Projected density. For Gaussian redshift distributions at $z = 0.5$ and 1 ($\sigma_z = 0.1$), our C_ℓ agree with CLASS's number-count spectra (density term only, exact projection) to 7×10^{-4} and 1.7×10^{-3} for $\ell \leq 500$, and to 1.9×10^{-3} for $500 < \ell \leq 2000$ (computed with $k_{\max} = 1\,\mathrm{Mpc}^{-1}$; with the default $0.3\,\mathrm{Mpc}^{-1}$ the extrapolated $P(k)$ misses the residual BAO wiggles, which costs up to 1.5% at $\ell = 1000$–2000). In $w(\theta)$ (Fig. 25d) the BAO shell appears at the angle $r_d/\chi(z)$. For $z = 1$ we find the feature at 2.36°, against $r_d/\chi = 2.47°$; projection smooths it into a shoulder that ends at this angle, since pairs at different distances along the line of sight have transverse separations below r_d. For $z = 0.5$ the expected angle, 4.3°, is close to where w changes sign, and the feature is hard to see. The correlation is weaker at higher redshift because the structure is younger (growth factor) and a fixed angle corresponds to a larger physical scale.

X. EXTENSION: RECFAST RECOMBINATION

The Peebles model of Sec. III treats hydrogen as a three-level atom with the case-B rate of Ref. [7], and helium in Saha equilibrium. In the comparison with CLASS (Sec. VI) this left differences of up to 1% in TT and 3% in EE, mostly through the damping tail and the width of the visibility function. The code now uses RECFAST [12, 33, 34] by default (the default of CAMB, and of CLASS before version 3), which reproduces the multi-level calculations to $\sim 0.1\%$ in X_e with three ordinary differential equations. The Peebles model is kept as an option (`recombination = peebles`) and is used for Secs. III–V.

A. Equations

RECFAST evolves the hydrogen ionization fraction $x_\mathrm{H} = n_\mathrm{HII}/n_\mathrm{H}$, the singly ionized helium fraction $x_\mathrm{He} = n_\mathrm{HeII}/n_\mathrm{He}$ and the baryon temperature T_b, with $X_e = x_\mathrm{H} + f_\mathrm{He}x_\mathrm{He}$ and $f_\mathrm{He} = Y_p/[3.9715(1-Y_p)]$ (the helium-to-hydrogen mass ratio is 3.9715, not 4 as in Sec. III). For hydrogen,

$$\frac{dx_\mathrm{H}}{dz} = \frac{X_e x_\mathrm{H} n_\mathrm{H}\alpha_\mathrm{H} - \beta_\mathrm{H}(1-x_\mathrm{H})e^{-h\nu_{\mathrm{Ly}\alpha}/k_B T_b}}{H(1+z)}\,C_\mathrm{H}, \tag{57}$$

$$C_\mathrm{H} = \frac{1 + K_\mathrm{H}\Lambda_{2s1s}n_{1s}}{F_\mathrm{H}^{-1}\left(1 + K_\mathrm{H}\Lambda_{2s1s}n_{1s}\right) + K_\mathrm{H}\beta_\mathrm{H}n_{1s}}, \tag{58}$$

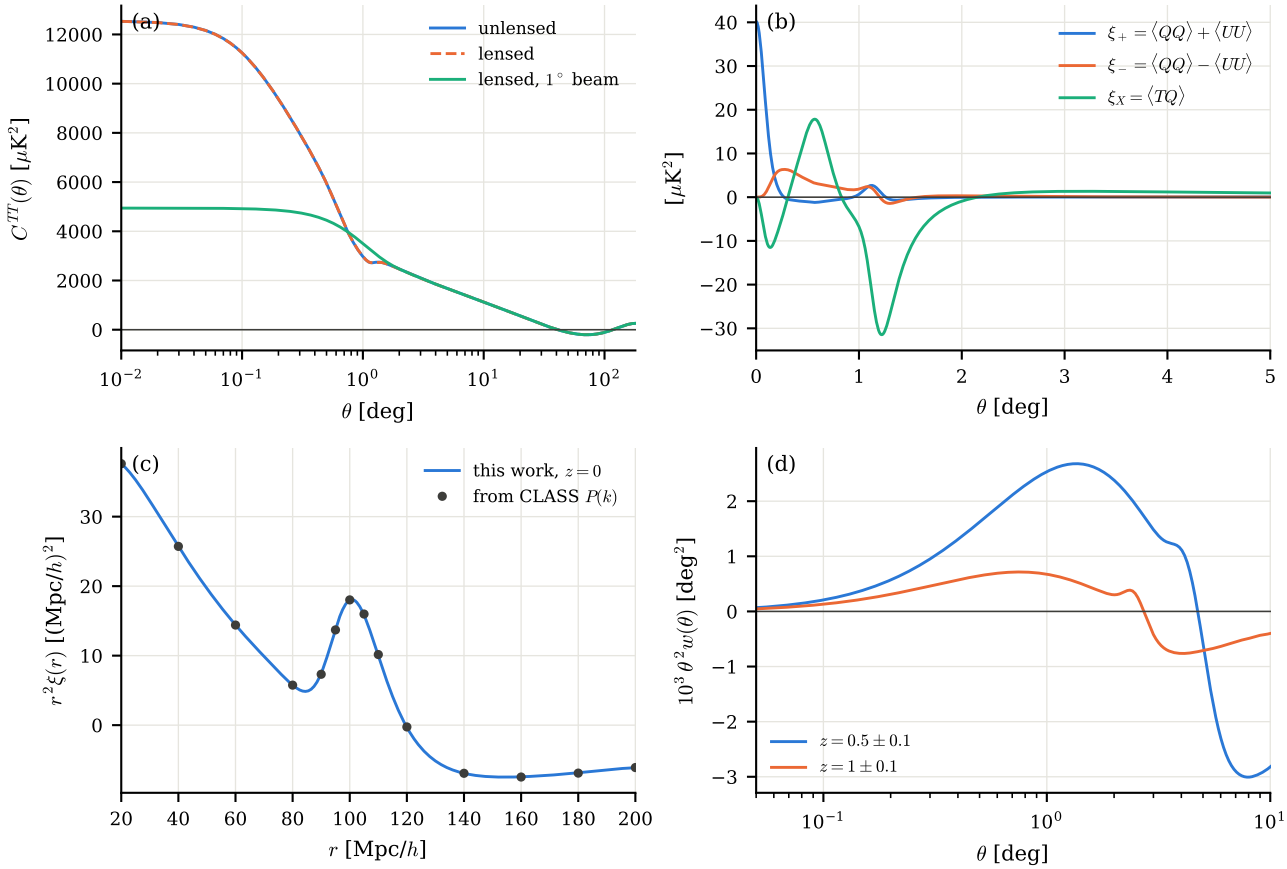


FIG. 25. Two-point functions of the fiducial model. *(a)* CMB temperature correlation function, unlensed, lensed, and lensed with a 1° beam. *(b)* Polarization correlation functions (unlensed). *(c)* Matter correlation function today, with the BAO peak; points are the transform of CLASS's $P(k)$. *(d)* Angular correlation function of the matter projected over Gaussian redshift distributions ($\sigma_z = 0.1$, $b = 1$).

with $n_{1s} = n_{\rm H}(1 - x_{\rm H})$. The case-B recombination coefficient is the fit $\alpha_{\rm H} = 10^{-19}\,a\,t^b/(1 + c\,t^d)\,{\rm m^3\,s^{-1}}$, $t = T_b/10^4\,{\rm K}$, of Ref. [35], and $\beta_{\rm H} = \alpha_{\rm H}(2\pi m_e k_B T_b/h^2)^{3/2} e^{-h\nu_{2s}/k_B T_b}$ is the photoionization rate from $n = 2$. The Peebles factor $C_{\rm H}$ is the probability that an atom in $n = 2$ reaches the ground state before it is ionized: through the two-photon decay ($\Lambda_{2s1s} = 8.22\,{\rm s^{-1}}$) or through the redshifting of Lyman-α photons out of the line, with

$$K_{\rm H} = \frac{\lambda_{{\rm Ly}\alpha}^3}{8\pi H}\left[1 + \sum_{i=1}^{2} A_i e^{-[(\ln(1+z) - z_i)/w_i]^2}\right]. \qquad (59)$$

The two Gaussians in $\ln(1+z)$ (RECFAST 1.5) and the fudge factor $F_{\rm H} = 1.125$ (1.14 in earlier versions) calibrate the three-level atom against the multi-level calculations of Refs. [33, 34]: they absorb the effects of the high-n states, of the two-photon transitions from higher levels and of the Lyman series feedback. For neutral helium RECFAST uses the same form for the He I singlet system (2^1S two-photon rate $51.3\,{\rm s^{-1}}$, the fit of Ref. [36] for the recombination rate), with two additions that make helium recombine much faster than Saha-equilibrium hydrogen-like treatments: the escape of the 2^1P photons is helped by their absorption by the small amount of neutral hydrogen present at $z \sim 2000$ (a continuum-opacity term calibrated by the fudge $F_{\rm He} = 0.86$), and atoms can also recombine through the triplet 2^3P state. The baryon temperature follows Compton heating by the CMB and adiabatic cooling, as before.

B. Implementation

As in RECFAST, the solution has three phases. While both hydrogen and helium are close to equilibrium we use the Saha equations of Sec. III, now for H^+, He^+ and He^{++} together. When the singly ionized helium fraction drops below 0.995 ($z \approx 2900$) we integrate the helium equation, with hydrogen still in Saha equilibrium, and when $x_{\rm H} < 0.995$ ($z \approx 1600$) we integrate the full system $(x_{\rm H}, x_{\rm He}, T_b)$. The equations are stiff, because the rates exceed H by many orders of magnitude when the gas is close to equilibrium, and we use an implicit BDF method. The constants (energy levels from the measured inverse wavelengths, Einstein coefficients, fitting formulae) are

those of RECFAST 1.5 as used in CLASS. The switch to the full hydrogen equation leaves a small kink in dX_e/dx at $z \approx 1600$, where the visibility function is negligible (10^{-13} of its maximum); elsewhere τ'' agrees with a finite difference of τ' to a few 10^{-6}. The whole calculation takes 0.35 s.

C. Validation

Figure 26 compares X_e without reionization and the unlensed spectra with CLASS run with RECFAST.

- X_e agrees with CLASS RECFAST to 3.3×10^{-4} for $0 \le z \le 8000$, including both helium recombinations (the largest difference is at $z \approx 1530$, just after the switch to the full hydrogen equation); the Peebles model differs by up to 3.8% during hydrogen recombination ($800 < z < 1600$). Last scattering is at $z_* = 1089.79$ (CLASS: 1089.79) with $r_s(z_*) = 144.421\,\mathrm{Mpc}$ (CLASS: 144.421). The difference from HyRec, CLASS's default, is up to 1% during helium recombination and in the freeze-out tail, and 0.11% during hydrogen recombination.

- The rms difference of TT from CLASS RECFAST is 0.048% for $30 \le \ell \le 1000$ and 0.29% for $1000 < \ell \le 2000$, against 0.64% and 0.75% with the Peebles model. For EE the rms difference is 0.32% and 0.46%, against 1.4% and 1.4%. The largest single-ℓ EE differences, 2.3% for $\ell \le 30$, are at the zeros of the reionization feature.

- In CLASS, the RECFAST and HyRec spectra differ by less than 10^{-3} rms (dashed in Fig. 26). Our remaining differences from CLASS, which grow to a few 10^{-3} in the damping tail, are larger than this. They therefore come from other numerical approximations in the code, and recombination is no longer the limiting approximation.

XI. EXTENSION: MODELS BEYOND ΛCDM

The code has a library of eleven dark-energy and modified-gravity models beyond ΛCDM (twelve configurations, since DGP has two branches). This section describes how a model is organized in the code, the equations of each model, and how it is validated. All models are selected by name in the parameter file (`model = ...`), and the model parameters are set there too.

A. Species and gravity

A model is a list of *species* and a theory of *gravity*. A species provides its background density and pressure, $\rho(x)$ and $p(x)$, and the first two derivatives of ρ. If it clusters, it also has a perturbation block: its variables, initial conditions and equations. The block is coupled to the metric only through four moments: $\delta\rho$, δp, the momentum $(\rho + p)\theta/k$ and the shear $(\rho + p)\sigma$. It receives Φ, Ψ and Φ' in return. Interacting components, such as dark matter coupled to dark energy, are a single block that holds the variables of both. The gravity gives $E^2 = H^2/H_0^2$ from the species. It also owns its own perturbation variables: Φ in GR, and Φ plus two scalar-field variables in the scalar–tensor theories. From these and the summed moments it computes Φ, Ψ and Φ'. Photons and baryons, with Thomson scattering, tight coupling and the line-of-sight sources, are the fixed core of the perturbation module. All models use the same conventions as Sec. IV: conformal Newtonian gauge, $x = \ln a$ and our sign of Φ. Below, the spatial potential is written with the Planck sign, $\Phi_P = -\Phi$.

Every model is tested in its GR/ΛCDM limit. Where possible we also check the Einstein constraint equations that the code does not integrate. Some models evolve Φ with the energy constraint but not the momentum constraint; in an interacting model the momentum constraint holds only if the exchanged energy and momentum cancel between the species. We test the growth of structure on sub-horizon scales against independent solutions of the quasi-static growth equation, and compare with CLASS, hi_class and CLASS_EDE where possible. Table IV summarizes the models, and Figs. 27 and 28 show their effects for the parameters of the table. All other parameters are those of the fiducial model; Ω_Λ, or the density of the dark-energy field, is set by closure. The runs use RECFAST and include lensing.

B. Dark energy and dark-sector interactions

Quintessence (`quintessence`). A minimally coupled scalar field with potential $V = 3H_0^2 M_{\rm pl}^2 V_0 e^{-\lambda\phi}$ (the field ϕ is in units of $M_{\rm pl}$). It starts at rest in the radiation era, where Hubble friction freezes it, and V_0 is found by shooting so that the field closes the universe. With $\phi_x = d\phi/dx$, the background equations are

$$E^2 = \frac{\rho_o + v}{1 - \phi_x^2/6}, \quad \phi_{xx} = -\Big(3 + \frac{d\ln E}{dx}\Big)\phi_x - \frac{3v_\phi}{E^2}, \quad (60)$$

where $v = V/(3H_0^2M_{\rm pl}^2)$ and ρ_o is the density of the other species. The field perturbation $\delta\phi$ obeys the Klein–Gordon equation of Mukhanov. We evolve $\delta\phi$ and $w = \delta\phi_x - \phi_x\Psi$; the combination w removes Ψ' from the equation. *Validation:* the background agrees with an independent integration in cosmic time, and the Poisson and momentum constraints hold to 10^{-7} with the field's density and momentum.

Coupled quintessence (`coupled_quintessence`). The dark matter particle mass depends on the field, $m \propto e^{-\beta\phi}$ [37, 38]. This gives

$$\rho_c' = -(3 + \beta\phi_x)\rho_c, \quad \phi_{xx} = \cdots - \frac{3v_\phi}{E^2} + \frac{3\beta\rho_c}{E^2}. \quad (61)$$

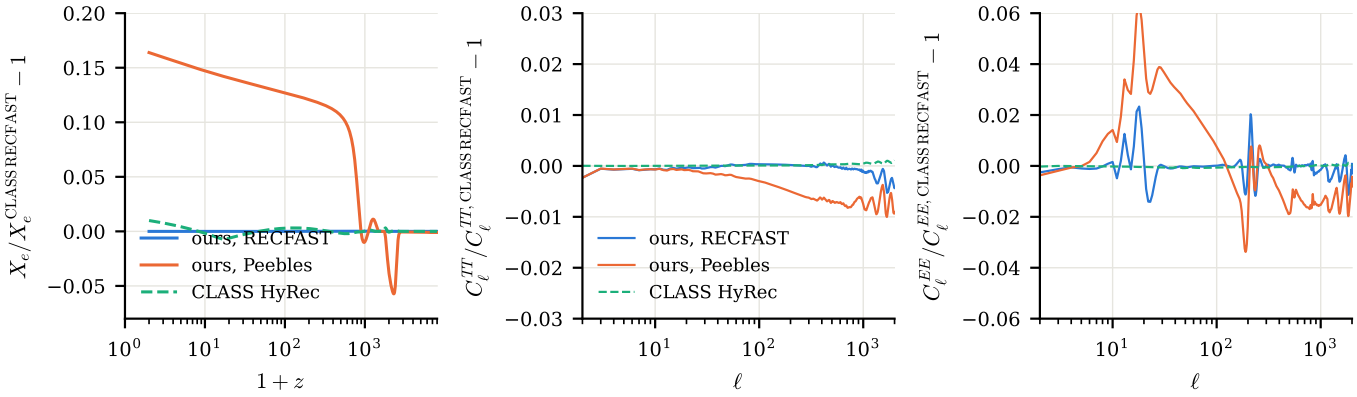


FIG. 26. The RECFAST recombination model. *Left:* $X_e(z)$ without reionization relative to CLASS run with RECFAST, for our RECFAST (blue) and Peebles (orange) models; the dashed line is CLASS with HyRec. *Middle, right:* the unlensed TT and EE spectra (with reionization) relative to CLASS RECFAST.

The dark matter feels a fifth force, which makes its effective gravitational constant $G(1+2\beta^2)$ on small scales. It also feels a drag term $\propto \beta\phi_x v_c$. *Validation:* an independent integration in cosmic time with the coupling; the constraint equations, which hold only if the exchange cancels; and the dark matter growth with fifth force and drag.

Dark scattering (`dark_scattering`). Elastic scattering between dark matter and a dark-energy fluid with constant w and sound speed $\hat{c}_s^2$ [39]. The cross section per unit mass is $\xi = \sigma_D/m_c$. Only momentum is exchanged, with rates $\Gamma_{\rm DE} = ac\xi\rho_c$ and $\Gamma_c = (1+w)(\rho_{\rm DE}/\rho_c)\Gamma_{\rm DE}$. At early times $\Gamma_{\rm DE} \gg \mathcal{H}$, and the dark-energy velocity is locked to that of the dark matter. *Validation:* the constraints with momentum exchange; the drag in the growth against two-fluid growth equations (5×10^{-4}); and, for $\xi = 0$, the wCDM/ΛCDM ratios against CLASS's fluid dark energy (P(k) to 5×10^{-5}, TT to 2×10^{-4} for $\ell \le 800$).

Early dark energy (`ede`). An axion-like field with $V = m^2f^2(1-\cos\phi/f)^n$ [40, 41], plus a cosmological constant. The field is frozen at $\theta_i = \phi_i/f$ until $H \sim m$. It then oscillates, and its energy dilutes as $a^{-6n/(n+1)}$. The model is specified as in CLASS_EDE [42], by the maximum fraction $f_{\rm EDE}$ of the total density, its redshift z_c, θ_i and n. A Newton iteration on $(\ln m^2f^2, \ln f)$ finds the potential, and Ω_Λ follows from closure. For $n \le 2$ the oscillating field does not dilute faster than radiation, so its fraction does not peak and $(f_{\rm EDE}, z_c)$ do not define the model; the code rejects these cases. The background and the perturbations are integrated through all oscillations, with the Klein–Gordon equations of quintessence. Both models share one scalar-field class. A run takes about 90 s instead of 65 s for ΛCDM. *Validation:* the shooting reproduces $f_{\rm EDE}$ and z_c to 10^{-10}, and the oscillation-averaged dilution is $a^{-4.50}$ for $n = 3$. The constraints hold to 5×10^{-6}, and $f_{\rm EDE} \to 0$ is ΛCDM. Compared with CLASS_EDE, the decay constant f agrees to 10^{-3}. The ratios to ΛCDM agree to 3×10^{-5} in $P(k)$ for $k \le 0.05\,h\,\mathrm{Mpc}^{-1}$ (7×10^{-4} at $0.1\,h\,\mathrm{Mpc}^{-1}$) and to 10^{-3} in TT.

Interacting dark energy (`ide`). A dark-energy fluid with constant w that exchanges energy with the dark matter at the rate $Q = \xi H\rho_{\rm DE}$ [43, 44]. The energy is transferred along the dark-matter velocity, so the dark matter feels no drag:

$$\rho_{\rm DE} \propto a^{-3(1+w)-\xi}, \quad \rho_c = \frac{1}{a^3}\left[\Omega_c + \xi\Omega_{\rm DE}\frac{1-a^{-(3w+\xi)}}{3w+\xi}\right]. \tag{62}$$

In the perturbations, H in Q is replaced by the local expansion rate of the dark matter, $\Theta_c/3$. The dark-energy pressure is transformed to the fluid rest frame with $\rho'_{\rm DE}$ including the transfer. Momentum conservation then gives $\theta'_{\rm DE} = \cdots + \xi\mathcal{H}[\theta_{\rm DE} - \theta_c/(1+w)]$. The early-time instability of these models [43] is avoided if the "doom factor" $d = \xi/[3(1+w)]$ is negative; the code rejects $d > 0$ and models with $\rho_c < 0$ before today. *Validation:* ρ_c against an independent integration (10^{-14}); the constraints with energy exchange (2×10^{-6}); and $\xi = 0$ against the separate wCDM implementation of dark scattering (identical). With $\xi = -0.1$ the dark matter loses energy uniformly, so δ_c grows faster: $P(k)$ is 1.254 times that of ΛCDM at $k = 0.1\,h\,\mathrm{Mpc}^{-1}$.

C. Modified gravity

μ–Σ *(`musigma`).* The Planck/DES parameterization [45, 46]. It modifies the Poisson equations for the comoving density $\rho\Delta$ by

$$k^2\Psi = -4\pi G a^2 \mu\,\rho\Delta, \quad k^2(\Phi_P + \Psi) = -8\pi G a^2 \Sigma\,\rho\Delta, \tag{63}$$

with $\mu = 1 + \mu_0\Omega_{\rm DE}(a)/\Omega_{\rm DE,0}$ and similarly for Σ. On super-horizon scales, $\rho\Delta \sim (k/\mathcal{H})^2\delta\rho$, so computing the potentials from $\rho\Delta$ directly would lose all precision there. We therefore evolve the GR potential of $\rho\Delta$ with an equation derived from energy–momentum conservation. This

equation reduces exactly to GR for $\mu = \Sigma = 1$. *Validation:* the potentials against the algebraic relations, and the growth against an independent solution of the growth equation with $\mu(a)$ ($< 10^{-4}$).

DGP (`dgp`). The braneworld model [47]. Its background is $E = \epsilon\sqrt{\Omega_{rc}} + \sqrt{\Omega_{rc} + \rho/\rho_{c0}}$, on the normal branch ($\epsilon = -1$, with Λ) or the self-accelerating branch ($\epsilon = +1$). Optionally the expansion is that of ΛCDM, as in N-body work. The perturbations are quasi-static, with $\mu = 1 + 1/(3\beta)$, $\beta = 1 - 2\epsilon(Hr_c/c)(1 + \dot{H}/3H^2)$, and $\Sigma = 1$ [48]. They use the μ–Σ equations. *Validation:* as for μ–Σ, with $\mu(a)$ from the background.

Jordan–Brans–Dicke (`jbd`). The scalar field ϕ multiplies R, with coupling ω, plus a cosmological constant [49]. ϕ_0 is fixed so that the Newtonian constant measured today is G, and Ω_Λ follows from $E(0) = 1$; both are found by iteration. The perturbations are evolved in full (Φ, $\delta\phi$, $\delta\phi'$). We derived them symbolically. *Validation:* the background equations; the matter-era solution $\phi \propto a^{1/(1+\omega)}$; the momentum constraint (5×10^{-8}); and the growth with $G_{\rm eff} = G\phi_0/\phi$.

Hu–Sawicky $f(R)$ (`fr`). Ref. [50] with $f_R = -|f_{R0}|(R_0/R)^{n+1}$. The expansion is that of ΛCDM to $\mathcal{O}(f_{R0})$. The perturbations are quasi-static, with the scale-dependent

$$\mu(k,a) = \frac{1+4Q}{(1+3Q)(1+f_R)}, \quad \Sigma = \frac{1}{1+f_R}, \qquad Q = \frac{k^2}{a^2}\frac{f_{RR}}{1+f_R} = \frac{k^2\lambda_C^2}{3a^2}. \tag{64}$$

Below the Compton wavelength of the scalaron, gravity is enhanced by 4/3. *Validation:* $f_R(a)$, the Compton wavelength and the limits of μ; the potentials at every k; and the scale-dependent growth at each k (2×10^{-4}).

Cubic Galileon (`cubic_galileon`). The Lagrangian is $c_2X - g_3X\Box\phi$ on its tracker solution [51, 52], with $E^2 = [\rho + \sqrt{\rho^2 + 4\Omega_\phi}]/2$ and no free parameter beyond those of ΛCDM ($w_0 \approx -1.19$). The braided scalar perturbations are evolved in full. We derived them covariantly and checked them with the identity $\nabla^\mu T_{\mu\nu} = (\nabla \cdot J)\phi_\nu$. *Validation:* the tracker relations ($E^4 - \rho E^2 = \Omega_\phi$, $w = -2/(1 + \Omega_\phi)$); the momentum constraint with all braiding terms (3×10^{-8}); and the sub-horizon growth against the quasi-static $G_{\rm eff}$ (7×10^{-5}). The model is ruled out by the ISW–galaxy cross-correlation. Here it shows the strongest effects: $C_\ell^{\phi\phi}$ is 1.177 times the ΛCDM value at $\ell = 100$.

Horndeski in the α basis (`horndeski`). Following Bellini and Sawicki [53] and hi_class [54], the linear theory is described by the kineticity α_K, the braiding α_B, the running of the Planck mass $\alpha_M = d\ln M_*^2/d\ln a$, and the background. We set $\alpha_T = 0$, as required by GW170817. We derived the Newtonian-gauge equations for $L = F(\phi)R/2 + G_2(\phi, X) - G_3(\phi, X)\Box\phi$, which realizes any $\alpha_K, \alpha_B, \alpha_M$. We then eliminated the functions G_i using the definitions of the α's, their time derivatives and the background equations, and reduced the system to first order in $v = -\delta\phi/\phi'$ and $u = v' + \Psi$. The no-ghost and gradient-stability conditions,

$$D = \alpha_K + \tfrac{3}{2}\alpha_B^2 > 0,$$
$$Dc_s^2 = (2 - \alpha_B)\Big[\frac{\alpha_B}{2} + \alpha_M - \frac{\dot{H}}{H^2}\Big] - \frac{\rho_m + p_m}{H^2M_*^2} + \frac{\dot{\alpha}_B}{H} > 0, \tag{65}$$

are checked before a run. The expansion is ΛCDM or w_0w_a. The α's use the hi_class parameterizations $\alpha_X = c_X\Omega_{\rm DE}(a)$, c_Xa or constant. All terms of the scalar equation are proportional to an α or to the dark-energy momentum, so nothing cancels catastrophically at early times when $D \to 0$. *Validation:* the momentum constraint (3×10^{-8}). Three theories rebuilt from their α's reproduce the dedicated implementations: the cubic Galileon ($\alpha_K = 6\Omega_{\rm DE}$, $\alpha_B = 2\Omega_{\rm DE}$, to 5×10^{-9}), quintessence ($\alpha_K = 3(\rho+p)_\phi/H^2$, 2×10^{-11}), and Brans–Dicke on its own expansion ($M_*^2 = \phi$, $\alpha_M = -\alpha_B$, $\alpha_K = \omega\alpha_M^2$, 9×10^{-9}). Against hi_class the ratios to ΛCDM agree to 10^{-4} in $P(k)$ and 8×10^{-4} in TT.

D. Discussion

The models act on the observables in four ways (Figs. 27–28, Table IV).

- *Late-time expansion.* Thawing quintessence raises H at $z \lesssim 2$. It slows the growth (σ_8 lower by 2%) and changes the ISW plateau of TT at $\ell \lesssim 20$.
- *Dark-sector interactions.* These act on the dark matter directly. The fifth force of coupled quintessence and the energy outflow of IDE (with $\xi < 0$) both enhance the growth of sub-horizon modes, by 7% and 25% in $P(0.1\,h\,\mathrm{Mpc}^{-1})$. IDE also has more dark matter in the past, which raises H by about 3% at high redshift and shifts the acoustic peaks. The drag of dark scattering suppresses the growth by 7% (with $w > -1$).
- *Modified gravity.* μ–Σ, nDGP, Brans–Dicke, the Galileon and Horndeski strengthen gravity below the horizon. The result is a scale-independent enhancement of $P(k)$ on small scales, between 4% (Horndeski) and 20% (Galileon). $f(R)$ gives a scale-dependent enhancement that grows towards the Compton scale. The lensing potential also changes, through Σ or through the running Planck mass. The self-accelerating DGP branch weakens gravity and expands faster, so $P(k)$ is 35% lower. The scalar–tensor models with dedicated field equations also change the large-scale ISW signal of TT strongly: by up to +35% for the Galileon (at $\ell \approx 4$) and −27% for Horndeski (at $\ell = 2$).
- *Early times.* Early dark energy is the only model that acts before recombination. It raises H by up

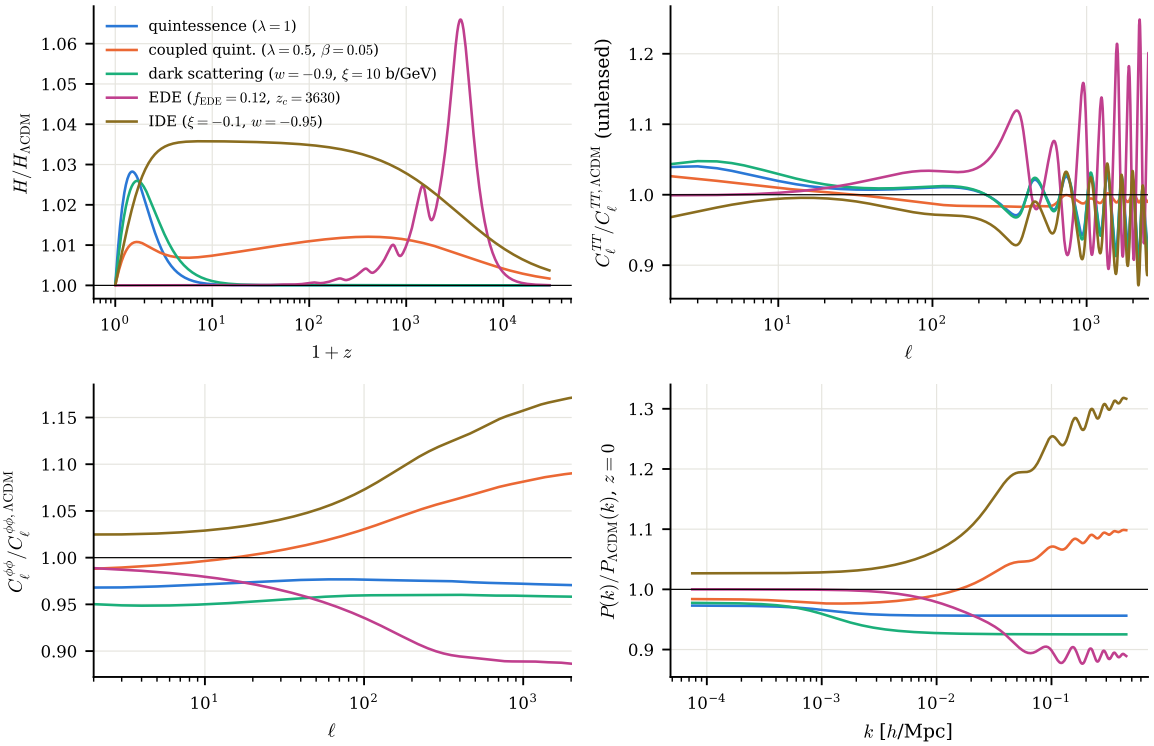


FIG. 27. Dark-energy models and dark-sector interactions relative to ΛCDM: expansion rate, unlensed TT spectrum, lensing potential and matter power spectrum today.

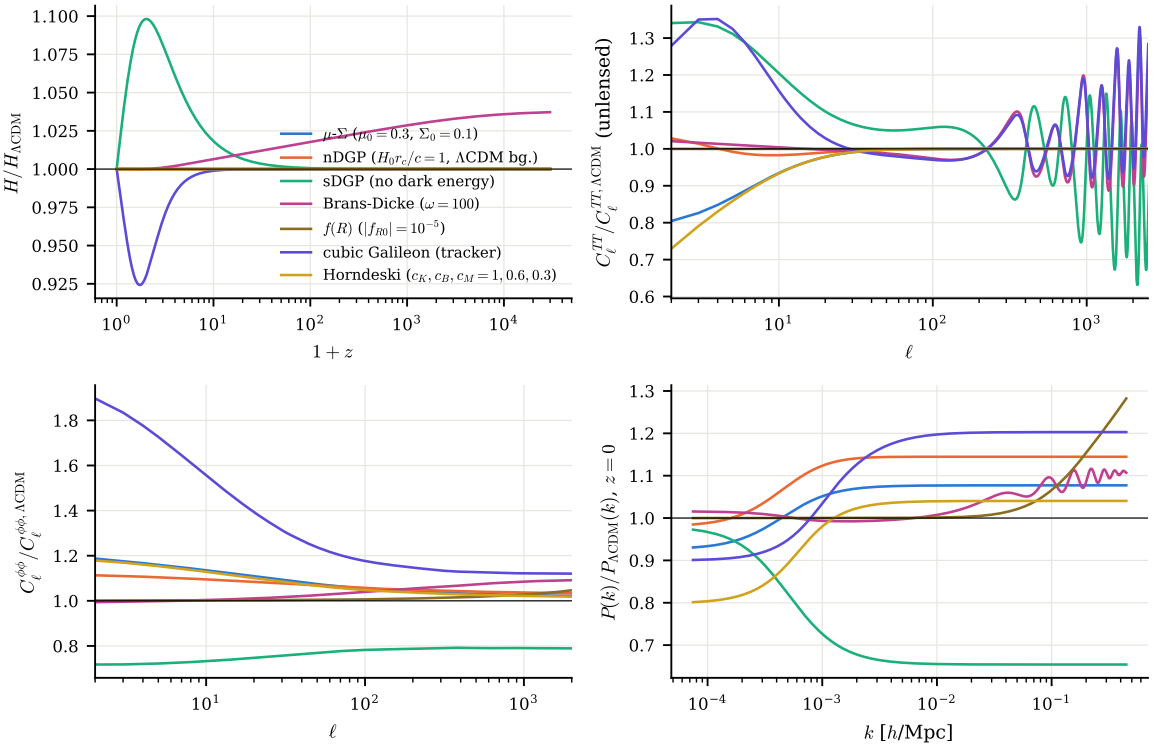


FIG. 28. Modified-gravity models relative to ΛCDM, as in Fig. 27.

TABLE IV. The models beyond ΛCDM, the parameters used in Figs. 27–28, the treatment of the perturbations, and σ_8 and $P(k = 0.1\,h\,\mathrm{Mpc}^{-1})$ relative to ΛCDM (same primordial spectrum, Ω_b, Ω_c and h).

Model	Parameters	Perturbations	Main validation	$\sigma_8/\sigma_8^{\Lambda\mathrm{CDM}}$	$P/P^{\Lambda\mathrm{CDM}}$
quintessence	$\lambda = 1$	full field	constraints, bg.	0.978	0.956
coupled quint.	$\lambda = 0.5$, $\beta = 0.05$	full field + CDM	constraints, growth	1.037	1.071
dark scattering	$w = -0.9$, $\xi = 10\,\mathrm{b/GeV}$	fluid + CDM	CLASS ($\xi = 0$), drag	0.962	0.925
EDE	$f_{\mathrm{EDE}} = 0.12$, $z_c = 3630$, $\theta_i = 2.83$	full field	CLASS_EDE	0.945	0.899
IDE	$\xi = -0.1$, $w = -0.95$	fluid + CDM	constraints, wCDM limit	1.124	1.254
μ–Σ	$\mu_0 = 0.3$, $\Sigma_0 = 0.1$	μ–Σ	growth equation	1.038	1.077
nDGP	$H_0 r_c/c = 1$, ΛCDM bg.	quasi-static	growth equation	1.070	1.145
sDGP	$\Omega_\Lambda = 0$	quasi-static	growth equation	0.809	0.654
Brans–Dicke	$\omega = 100$	full field	constraints, G_{eff}	1.043	1.093
$f(R)$	$\lvert f_{R0}\rvert = 10^{-5}$, $n = 1$	quasi-static $\mu(k, a)$	scale-dep. growth	1.073	1.066
cubic Galileon	—	full field (braiding)	constraints, G_{eff}	1.097	1.203
Horndeski	$c_K = 1$, $c_B = 0.6$, $c_M = 0.3$	full (α basis)	hi_class, 3 theories	1.020	1.040

to 7% around z_c, which reduces the sound horizon. It also slows the growth of the dark matter while the field is significant. At fixed h this shifts the acoustic peaks (the ratio to ΛCDM oscillates by ±9%), and lowers $P(k)$ on small scales by 10%. In fits to data the smaller sound horizon is compensated by a larger H_0, which is why the model has been proposed to resolve the Hubble tension.

For every model the code computes the unlensed and lensed CMB spectra and the matter power spectrum with the same pipeline as for ΛCDM.

XII. CONCLUSIONS

We have written a modular Einstein–Boltzmann solver and used it to compute the CMB temperature and polarization spectra and the matter power spectrum for a flat ΛCDM model. The four milestones gave:

1. The background. Radiation–matter equality is at $z = 3400$ and acceleration starts at $z = 0.63$, before matter–Λ equality at $z = 0.29$, because the negative pressure of Λ counts twice in the acceleration equation. The Universe is 13.85 Gyr old, older than a matter-only universe could be. Supernova distances require an accelerating expansion ($\chi^2 = 31$ versus 317 for a matter-only universe) and prefer $h \approx 0.70$ when the density parameters are held fixed.

2. The recombination history. The huge number of photons per baryon delays recombination until $k_B T \approx \epsilon_0/50$. Photons decouple at $z = 1090$, when the Universe is 372 000 years old (or at $z_* = 1085$ including reionization). The baryons, outnumbered by the photons, follow at $z_d = 1060$, with sound horizon 147 Mpc. Saha equilibrium ignores the $n = 2$ bottleneck and would predict decoupling ~ 200 units of redshift too early and no freeze-out.

3. The perturbations show the main physics of structure formation: acoustic oscillations of the photon–baryon fluid, driven by decaying potentials and damped by photon diffusion; the stagnation of dark matter growth in the radiation era; baryons falling into dark matter wells after decoupling; the decay of the potentials in the Λ era; and the brief generation of polarization during decoupling.

4. The spectra reproduce the Planck TT, EE and TE data and the SDSS galaxy power spectrum without any fitting. The CMB peaks are the acoustic oscillations frozen at decoupling and projected over χ_*, with spacing set by $\ell_A = \pi\chi_*/r_s = 303$. Their heights encode the baryon and matter densities, and the turnover of $P(k)$ at $k_{\rm eq}$ is the memory of radiation–matter equality. The oscillating residuals of the unlensed TT spectrum are explained by gravitational lensing: with our lensing module the χ^2 against the 83 Planck high-ℓ bins falls from 781 to 220, and the lensed spectra and the lensing B modes agree with CLASS.

All modules agree with CLASS at the percent level or better. As an extension, the code also handles open and closed universes (Sec. VII), with hyperspherical Bessel functions and the discrete modes of a closed space; for $-0.2 \le \Omega_K \le 0.2$ it agrees with CLASS as well as in the flat case, and the first acoustic peak follows the geometric prediction $\ell_{\rm peak} \simeq 0.73\,\pi r(\chi_*)/r_s$. With a massive neutrino species (Sec. VIII) the code reproduces the free-streaming suppression of the matter power spectrum, $\approx -7 f_\nu$ on small scales, and of the lensing potential, with the effect of the mass agreeing with CLASS to 5×10^{-4} in $P(k)$ and C_ℓ^{TT}. The two-point functions of Sec. IX show the acoustic physics in real space: a shoulder near 1° in the CMB temperature correlation, alternating radial and tangential polarization around hot spots, and the BAO shell at r_d in the matter correlation function and, projected, at r_d/χ in $w(\theta)$.

The recombination model was the largest remaining physical approximation. With RECFAST (Sec. X), X_e agrees with CLASS's RECFAST to 3.3×10^{-4}, and the TT and EE spectra agree to a few 10^{-3} (rms) up to $\ell = 2000$, compared with 1–3% for the Peebles model. The model library of Sec. XI extends the code to eleven dark-energy and modified-gravity models. Each is validated against its ΛCDM limit, the unused constraint equations and independent calculations, and against CLASS, hi_class and CLASS_EDE where these apply. Natural next steps are:

- Nonlinear corrections to $P(k)$ (halofit or HMcode) for weak-lensing and galaxy-clustering predictions, and their extension to modified gravity.
- Horndeski gravity with $\alpha_T \neq 0$, and models built directly from the Horndeski functions $G_i(\phi, X)$.
- Several massive neutrino species with different masses (the code has one), and B-mode polarization from tensor modes.
- MCMC parameter estimation with the full code; the MCMC fits of the course project are omitted from this report as instructed.

The code takes about one minute to run on a laptop, which is fast enough for such exploratory work.

ACKNOWLEDGMENTS

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